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İSTANBUL UNIVERSITY
FACULTY OF ECONOMICS**

GRADUATION PROJECT

**A Hybrid Markov Chain and Shapley Value Approach to
Multi-Touch Attribution and Budget Optimization in Digital
Marketing**

Management Information Systems

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PREFACE

This thesis investigates the limitations of traditional digital marketing attribution models and proposes a hybrid framework combining Markov Chain and Shapley Value methodologies for more accurate channel credit assignment and budget optimization. The motivation arose from observing the persistent reliance on last-click attribution despite its well-documented shortcomings in capturing the complex interactions between multiple marketing channels.

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ABSTRACT

Traditional attribution models in digital marketing, such as last-click and first-click, oversimplify the multi-channel customer journey by assigning all conversion credit to a single touchpoint. This study proposes a hybrid attribution framework that combines the strengths of Markov Chain Removal Effects and Shapley Value methodologies to produce more accurate channel attribution and inform budget allocation decisions.

Using a synthetically generated dataset of 30,000 simulated customer journeys grounded in industry conversion rate benchmarks (WordStream, 2025; Lucky Orange, 2025), the proposed Hybrid model was evaluated against six baseline models: Last-Click, First-Click, Linear, Time-Decay, Position-Based, as well as standalone Markov and Shapley implementations. The Hybrid model, with an optimally derived weighting parameter ($\alpha = 0.25$), achieved a Mean Absolute Error (MAE) of 0.0571 against the ground truth, representing a 21.2% improvement over Last-Click attribution (MAE = 0.0725).

A budget optimization layer was developed using diminishing returns response curves and the Differential Evolution algorithm. Across four budget scenarios (10K, 50K, 100K, 500K USD), the Hybrid model consistently outperformed Last-Click in expected conversion outcomes, with improvements ranging from 0.77% at high budgets to 13.72% at lower budgets. Notably, the Hybrid model achieved 99.95% of the theoretical upper bound (Ground Truth Optimal allocation), demonstrating its near-optimal performance.

The findings indicate that the proposed framework offers a robust, data-driven alternative to heuristic attribution models, particularly suited for small-to-medium B2C e-commerce businesses operating under budget constraints.

Keywords: Multi-Touch Attribution, Markov Chain, Shapley Value, Budget Optimization, Digital Marketing, Diminishing Returns, Customer Journey, Hybrid Model.

LIST OF ABBREVIATIONS

ATT	App Tracking Transparency
B2B	Business-to-Business
B2C	Business-to-Consumer
CCPA	California Consumer Privacy Act
DE	Differential Evolution
DL	Deep Learning
DSS	Decision Support System
GDPR	General Data Protection Regulation
GT	Ground Truth
IAB	Interactive Advertising Bureau
ITP	Intelligent Tracking Prevention
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MMM	Marketing Mix Modeling
MTA	Multi-Touch Attribution
RE	Removal Effect
RMSE	Root Mean Squared Error
ROI	Return on Investment
RQ	Research Question
SEO	Search Engine Optimization
USD	United States Dollar
WOM	Word-of-Mouth

1. INTRODUCTION

1.1. Background and Motivation

Digital marketing has evolved into one of the most data-intensive domains of contemporary business practice. Over the past two decades, the proliferation of digital channels—ranging from search advertising and social media to email campaigns and organic traffic—has fundamentally transformed how organizations allocate marketing budgets and measure return on investment (Kannan & Li, 2017; Lamberton & Stephen, 2016). The shift from traditional media to digital platforms has provided practitioners with unprecedented volumes of behavioral data, yet it has simultaneously introduced new analytical challenges that classical measurement frameworks were not designed to address (Berman, 2018).

A central concern within this transformed landscape is the problem of *attribution*—the methodological challenge of determining how credit for a customer conversion should be distributed across the multiple touchpoints that preceded it. In a typical contemporary customer journey, an individual may encounter a brand through a paid search advertisement, later engage with a social media post, subscribe to an email list, and ultimately complete a purchase after returning to the website directly. The question of which channel deserves credit for the resulting conversion—and in what proportion—has substantial implications for budget allocation, channel prioritization, and overall marketing strategy (Kireyev et al., 2016; Li & Kannan, 2014).

The economic stakes associated with attribution decisions are considerable. Global digital advertising expenditure has continued to expand year over year, and emerging markets, including Türkiye, have demonstrated particularly rapid growth in digital channel investment (IAB Türkiye, 2024). As budgets grow, the opportunity cost of misallocated spending increases proportionally, rendering the accuracy of the underlying attribution model a question of considerable practical importance (Deloitte Türkiye, 2024).

Despite these stakes, the digital marketing industry has historically relied on heuristic attribution models that are computationally simple but conceptually limited. The most prevalent of these, the *last-click* model, assigns the entirety of conversion credit to the final touchpoint preceding a purchase. While straightforward to implement, this approach systematically overstates the importance of bottom-of-funnel channels and underestimates the contribution of awareness-stage interactions (Berman, 2018; Jayawardane et al., 2015). Similar limitations affect other heuristic alternatives, including first-click, linear, time-decay, and position-based models, each of which imposes assumptions about the conversion process that may not correspond to actual customer behavior (Google, 2024a).

In response to these limitations, the academic literature has produced a body of *data-driven attribution models* that attempt to derive channel contributions empirically rather than by assumption. Two principal approaches have emerged: Markov Chain models, which represent customer journeys as stochastic processes and quantify channel importance through removal-effect analysis (Anderl et al., 2016); and Shapley Value models, which adapt cooperative game theory to compute the marginal contribution of each channel across all possible coalitions (Dall'essandro et al., 2012; Shao & Li, 2011). Both approaches offer theoretical advantages over heuristic alternatives, yet each exhibits characteristic weaknesses when applied in isolation.

A persistent obstacle to evaluating these data-driven models has been the absence of a reliable *ground truth* against which their outputs can be validated. In real-world settings, the true contribution of each channel to a given conversion is fundamentally unobservable, leaving researchers without an objective basis for assessing model accuracy. This methodological gap

motivates the present study.

1.2. Problem Statement

The core problem addressed by this thesis can be stated as follows: although multiple data-driven attribution methodologies exist in the literature, there is no established framework for objectively determining *which model best approximates the true contribution structure* of marketing channels in a multi-touch environment. Three specific issues compound this problem.

First, the unobservable nature of ground-truth channel effects in real-world data makes direct comparison between models problematic. Empirical studies have demonstrated substantial variation in attribution outcomes across methodologies even when applied to the same dataset (Gordon et al., 2019; Lewis & Rao, 2015), but the absence of an objective benchmark prevents these divergences from being adjudicated.

Second, individual data-driven models exhibit complementary strengths and weaknesses. Markov-based approaches excel at capturing the directional flow of customer journeys but may overemphasize high-traffic channels (Anderl et al., 2016). Shapley-based approaches distribute credit equitably across channel coalitions but may dilute the importance of pivotal touchpoints (Dalessandro et al., 2012). The question of whether and how these approaches can be combined to produce superior performance remains underexplored.

Third, attribution accuracy is itself only an intermediate concern. The ultimate purpose of attribution analysis in a managerial context is to inform budget allocation decisions (Hanssens et al., 2003). A framework that produces accurate attribution estimates but fails to translate these into actionable budget recommendations offers limited practical value. Few existing studies bridge the gap between attribution modeling and budget optimization in a unified analytical framework.

1.3. Research Objectives

In light of the problems articulated above, the following objectives have been formulated for this thesis:

1. To construct a synthetic data generation framework grounded in industry conversion-rate benchmarks, thereby providing an objective ground truth against which attribution models can be validated.
2. To implement a Markov Chain attribution model employing absorbing chain theory and Laplace smoothing for removal-effect estimation.
3. To implement a Shapley Value attribution model utilizing a fractional credit formulation of the characteristic function.
4. To propose a hybrid Markov-Shapley attribution model whose weighting parameter is determined empirically through a grid search against the constructed ground truth.
5. To integrate a budget optimization layer incorporating diminishing returns response curves and global optimization via the Differential Evolution algorithm.
6. To systematically benchmark the proposed framework against five classical heuristic models (Last-Click, First-Click, Linear, Time-Decay, and Position-Based) using multiple evaluation metrics.

1.4. Research Questions

The objectives described above are addressed through the following research questions:

- **RQ1:** Do data-driven attribution models, specifically Markov Chain and Shapley Value, outperform classical heuristic models in approximating the ground-truth channel contribution structure of a multi-channel customer journey?
- **RQ2:** Does a hybrid model combining Markov Chain removal effects and Shapley Value contributions, with an empirically optimized weighting parameter, achieve superior accuracy compared with either standalone approach?
- **RQ3:** How does the attribution accuracy of the proposed hybrid model translate into budget allocation decisions across varying budget levels when evaluated under realistic diminishing-returns assumptions?

1.5. Significance of the Study

This thesis contributes to the digital marketing attribution literature along three dimensions.

From a *methodological* perspective, the study introduces a ground-truth-based validation framework that addresses a long-standing gap in attribution research. By generating customer journey data from known channel effect parameters derived from industry benchmarks (Lucky Orange, 2025; WordStream, 2025), the framework enables objective performance assessment in a manner not feasible with observational data alone.

From a *theoretical* perspective, the proposed hybrid model offers a principled mechanism for combining two distinct attribution paradigms—one grounded in stochastic process theory and one in cooperative game theory. The data-driven determination of the hybrid weighting parameter through grid-search optimization represents a departure from prior work, in which combination weights are typically fixed by assumption.

From a *practical* perspective, the integrated budget optimization layer transforms the attribution model from a descriptive tool into a prescriptive decision-support system. The incorporation of diminishing returns response curves, drawn from the Marketing Mix Modeling literature (Hanssens et al., 2003; Jin et al., 2017), ensures that the resulting budget recommendations reflect realistic constraints on channel-level returns. This integration is particularly relevant for small-to-medium B2C e-commerce operations, in which budget constraints render the marginal value of accurate allocation especially high.

1.6. Thesis Structure

The remainder of this thesis is organized into five chapters. Chapter 2 presents a comprehensive review of the literature on digital marketing attribution, encompassing classical heuristic models, data-driven approaches, causal inference techniques, marketing mix modeling, and the implications of the cookieless future for attribution methodology. Chapter 3 details the methodology of the present study, including the synthetic data generation procedure, the formulation of the Markov Chain and Shapley Value models, the construction of the hybrid framework, and the integration of the budget optimization layer. Chapter 4 presents the empirical findings, covering attribution accuracy comparisons across all modeled approaches and budget performance simulations across multiple budget levels. Chapter 5 discusses the interpretation of these findings, their relation to the prior literature, the limitations of the present study, and

directions for future research. Chapter 6 concludes the thesis with a summary of contributions and closing observations.

2. LITERATURE REVIEW

2.1. Evolution of Digital Marketing Measurement

The measurement of marketing effectiveness has undergone profound transformation since the proliferation of digital channels in the early 2000s. Traditional mass-media advertising, dominated by television, radio, and print, operated under fundamentally different measurement constraints than contemporary digital ecosystems. In traditional environments, the principal analytical instrument was the *marketing mix model* (MMM), an econometric approach that correlated aggregate sales outcomes with aggregate marketing expenditures across channels (Hanssens et al., 2003; Tellis, 2006). Although MMM provided useful directional insights for top-level budget allocation, it operated exclusively at the macro level and could not isolate the contribution of individual customer interactions.

The advent of online advertising, beginning with banner ads in the mid-1990s and accelerating through the 2000s with the rise of search and social platforms, fundamentally altered the measurement landscape. For the first time, marketers were able to observe individual user behavior across multiple touchpoints, generating what Lamberton and Stephen (2016) characterize as an unprecedented richness of micro-level behavioral data. This shift enabled a new analytical paradigm: rather than estimating channel effects from aggregate correlations, practitioners could now trace specific customer journeys from initial exposure to eventual conversion.

Despite this analytical advance, the question of how to interpret these journeys—specifically, how to apportion credit among the multiple touchpoints preceding a conversion—emerged as a non-trivial problem. Kannan and Li (2017) identify this attribution problem as one of the central methodological challenges of contemporary digital marketing research, noting that the proliferation of channels has substantially outpaced the development of measurement frameworks capable of accommodating their complex interactions.

Within this evolving landscape, two parallel measurement traditions emerged. The first, focused on *aggregate-level* analysis, retained the econometric heritage of MMM but adapted it to incorporate digital channels, non-linear response functions, and time-varying effects (Ng et al., 2021; Ni et al., 2024). The second, focused on *individual-level* analysis, sought to leverage the granular tracking capabilities of digital platforms to attribute conversions to specific touchpoint sequences (Anderl et al., 2016; Shao & Li, 2011). The latter tradition, generally referred to as *multi-touch attribution* (MTA), forms the principal focus of the present thesis.

The increasing centrality of attribution to managerial practice is reflected in the substantial investment that organizations have made in attribution analytics. Industry surveys indicate that attribution measurement is among the highest-priority analytical capabilities for digital marketing teams, with organizations frequently citing attribution accuracy as a critical determinant of overall marketing return on investment (Ruler Analytics, 2025; Triple Whale, 2025). This managerial salience renders the methodological robustness of attribution models a matter of considerable practical consequence.

2.2. Classical (Heuristic) Attribution Models

The earliest generation of multi-touch attribution models, developed primarily within industry rather than academic settings, consisted of *heuristic* or *rule-based* approaches. These models share a common structural characteristic: they assign credit to channels based on fixed, predetermined rules concerning the position of each touchpoint within the customer journey, rather than on any data-driven estimation of channel contribution (Google, 2024a; Jayawardane

et al., 2015). Although heuristic models remain widely used in practice owing to their computational simplicity and interpretability, their theoretical foundations are increasingly considered inadequate for capturing the complexity of contemporary multi-channel customer journeys (Berman, 2018).

The following subsections describe the five most prevalent heuristic attribution models. For each, the formal definition is presented, followed by a discussion of typical use cases and recognized limitations.

2.2.1. Last-Click Attribution

The last-click attribution model assigns the entirety of conversion credit to the final touchpoint that preceded the conversion event. Formally, given a customer journey consisting of touchpoints $\{c_1, c_2, \dots, c_n\}$ terminating in a conversion, the credit assigned to channel c_i is

$$\text{Credit}(c_i) = \begin{cases} 1 & \text{if } i = n \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

Last-click attribution has historically been the default model in industry applications, including Google Analytics in its earlier configurations (Google, 2024a). Its persistence in practice stems from several practical advantages. The model requires only the final touchpoint in each conversion path, simplifying both data collection and computational implementation. It also produces results that are immediately interpretable to non-technical stakeholders, as the attribution logic corresponds to a common intuitive narrative of "the channel that closed the sale."

Despite these practical advantages, last-click attribution suffers from substantial theoretical limitations. Berman (2018) provides a formal welfare analysis demonstrating that last-click attribution systematically distorts advertiser incentives by overcompensating bottom-of-funnel channels—those that intercept users already predisposed to purchase—while undercompensating upper-funnel channels that contribute to demand generation. This bias has direct managerial consequences: budgets are reallocated toward channels that capture existing demand rather than channels that create new demand, potentially leading to long-term erosion of brand awareness and customer acquisition.

A related critique concerns the model's failure to account for synergistic interactions between channels. Kireyev et al. (2016) demonstrate empirically that display advertising exerts a substantial causal influence on subsequent search behavior, meaning that search-related conversions credited entirely to the search channel under a last-click framework may in fact reflect prior display exposures. By attributing conversions exclusively to the terminal touchpoint, the last-click model renders such cross-channel dynamics invisible.

2.2.2. First-Click Attribution

The first-click attribution model represents the inverse of the last-click approach: the entirety of conversion credit is assigned to the initial touchpoint of the customer journey. Formally,

$$\text{Credit}(c_i) = \begin{cases} 1 & \text{if } i = 1 \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

First-click attribution is conceptually justified by the argument that the channel responsible for initiating the customer journey—typically through the generation of brand awareness or

interest—deserves disproportionate credit for the eventual conversion. This perspective aligns with the upper-funnel emphasis characteristic of brand-building marketing campaigns.

In practice, first-click attribution is most frequently employed in analyses focused on customer acquisition or top-of-funnel performance (Abhishek et al., 2012). However, the model exhibits limitations that mirror those of last-click attribution, albeit in the opposite direction. By assigning all credit to the initial touchpoint, first-click attribution ignores the contribution of subsequent interactions that may be necessary to convert initial interest into purchase intent. Jayawardane et al. (2015) note that this approach is particularly problematic in industries with long consideration cycles, where the cumulative effect of repeated touchpoint exposures is essential to the conversion process.

2.2.3. Linear Attribution

The linear attribution model distributes conversion credit equally across all touchpoints in the customer journey. For a journey of n touchpoints, each channel receives

$$\text{Credit}(c_i) = \frac{1}{n} \quad (3)$$

Linear attribution constitutes the simplest data-aware alternative to position-based heuristics, in the sense that it acknowledges the contribution of all touchpoints rather than privileging the first or last alone. The model is conceptually grounded in the assumption that, in the absence of data-driven evidence to the contrary, all touchpoints should be treated as equally important.

This assumption, however, is rarely justified empirically. Customer journeys exhibit substantial heterogeneity in the importance of individual touchpoints, with some interactions exerting decisive influence on conversion outcomes while others serve as mere incidental exposures (Anderl et al., 2016; Li & Kannan, 2014). By imposing equal weighting across all touchpoints, linear attribution systematically underweights pivotal channels and overweights peripheral ones. Shao and Li (2011) observe that linear attribution serves as a useful baseline against which more sophisticated models can be evaluated, but it should not be relied upon as a primary analytical tool.

2.2.4. Time-Decay Attribution

The time-decay attribution model assigns greater credit to touchpoints that occur closer in time to the conversion event, with credit decreasing exponentially as the temporal distance between touchpoint and conversion increases. A common formulation, employed for example in Google Analytics implementations, assigns credit to touchpoint c_i according to

$$\text{Credit}(c_i) = \frac{2^{-(t_n - t_i)/h}}{\sum_{j=1}^n 2^{-(t_n - t_j)/h}} \quad (4)$$

where t_i denotes the time of touchpoint i , t_n denotes the time of conversion, and h is the *half-life* parameter governing the rate of decay (Google, 2024a).

The conceptual motivation for time-decay attribution rests on the psychological premise that recent interactions exert greater influence on purchase decisions than distant ones, an assumption broadly consistent with research on memory and recency effects in consumer behavior. The model represents an improvement over last-click attribution in that it acknowledges the contribution of multiple touchpoints, while still privileging those closest to the conversion event.

Nevertheless, time-decay attribution remains a heuristic in the sense that the temporal weighting scheme is imposed by assumption rather than derived from data. The choice of half-life parameter is typically arbitrary, and the model offers no principled mechanism for adjusting the weighting scheme to the characteristics of specific industries, products, or customer segments (Jayawardane et al., 2015). Furthermore, like other heuristic models, time-decay attribution does not account for channel-specific effects or inter-channel synergies.

2.2.5. Position-Based Attribution

Position-based attribution, also known as the U-shaped model, represents a compromise between first-click, last-click, and linear approaches. The model assigns substantial credit to both the initial and final touchpoints—typically 40% each—with the remaining 20% distributed equally across intermediate touchpoints (Google, 2024a). For a journey of $n \geq 3$ touchpoints, the credit assignment is

$$\text{Credit}(c_i) = \begin{cases} 0.4 & \text{if } i = 1 \\ 0.4 & \text{if } i = n \\ \frac{0.2}{n-2} & \text{otherwise} \end{cases} \quad (5)$$

For journeys of one or two touchpoints, the credit is distributed proportionally between the available positions.

The conceptual rationale for position-based attribution derives from the observation that both the initial and final touchpoints occupy structurally important positions within the customer journey: the former initiates the conversion process, while the latter consummates it (Abhishek et al., 2012). Intermediate touchpoints, although still recognized, are treated as supporting roles in the conversion narrative.

The principal limitation of position-based attribution lies in the arbitrariness of its weighting scheme. The 40/20/40 distribution is imposed by convention rather than derived from empirical evidence, and there is no theoretical justification for these specific values. Berman (2018) extends the critique of heuristic models more broadly to position-based attribution, arguing that any model relying on predetermined weighting schemes will necessarily fail to capture the true heterogeneity of channel contributions across different customer journeys and industries.

2.2.6. Limitations Common to Heuristic Approaches

While the five models described above differ in the specific weighting schemes they impose, they share a fundamental methodological characteristic that renders them collectively inadequate for sophisticated attribution analysis: they are *rule-based* rather than *data-driven*. The credit assignments produced by these models reflect prior assumptions imposed by the analyst, not empirical estimates of channel contribution derived from the observed data (Anderl et al., 2016; Shao & Li, 2011).

This characteristic produces three interrelated problems. First, heuristic models cannot adapt to the specific dynamics of different industries, products, or customer segments—a journey that is typical of fashion e-commerce may exhibit very different attribution patterns than one typical of B2B software. Second, heuristic models cannot identify or quantify synergistic interactions between channels, a phenomenon that has been empirically established as significant (Kireyev et al., 2016; Naik et al., 2005). Third, heuristic models offer no internal mechanism for validation: because their outputs reflect assumptions rather than estimates, there is no meaningful sense in which one heuristic can be considered "more accurate" than another

except by reference to external criteria.

These limitations have motivated the development of *data-driven* attribution models, which seek to derive channel contributions from the observed data rather than imposing them by assumption. These approaches form the subject of the subsequent section.

2.3. Data-Driven Attribution Models

In response to the inherent limitations of heuristic models discussed in the preceding section, the academic and practitioner literatures have produced a growing body of *data-driven attribution models*. These approaches share a common methodological commitment: rather than imposing credit assignments by rule, they seek to *estimate* channel contributions from observed customer journey data (Anderl et al., 2016; Li & Kannan, 2014; Shao & Li, 2011). The shift from heuristic to data-driven attribution represents not merely a technical refinement but a paradigmatic transition in how the attribution problem itself is conceptualized—from a question of *accounting* (how should we divide credit?) to a question of *inference* (what is each channel’s true contribution?) (Berman, 2018; Dalessandro et al., 2012).

The data-driven attribution literature can be organized into four principal methodological families: Markov Chain-based models, Shapley Value-based models, Bayesian models, and Deep Learning approaches. Each family draws on distinct theoretical foundations and offers complementary analytical strengths. The following subsections review each family in turn, with particular emphasis on the two paradigms—Markov Chain and Shapley Value—that form the methodological core of the present thesis.

2.3.1. Markov Chain-Based Attribution

The application of Markov chain theory to digital marketing attribution constitutes one of the most influential developments in the data-driven attribution literature. The approach was popularized in its current form by Anderl et al. (2016), whose work on graph-based attribution modeling has become a standard reference in both academic and industry contexts.

A Markov chain is a stochastic process in which the probability of transitioning to any given state depends only on the current state, not on the sequence of states preceding it—a property known as the *Markov property* or *memorylessness* (Norris, 1998). In the context of attribution, customer journeys are represented as sequences of state transitions, where each state corresponds to a marketing channel encountered by the customer. The set of states is augmented with two *absorbing states*: a *Conversion* state representing successful completion of the purchase, and a *Non-Conversion* state representing abandonment of the journey (Archak et al., 2010).

The central analytical instrument in Markov-based attribution is the *transition matrix*, which encodes the empirical probabilities of moving from one channel to another (or to an absorbing state) at each step of the customer journey. Once the transition matrix has been estimated from observed customer journeys, the model can be used to compute the overall probability that a randomly selected journey will terminate in conversion rather than non-conversion—a quantity known in absorbing Markov chain theory as the *absorption probability* (Norris, 1998).

The attribution of credit to individual channels is then accomplished through the *removal effect* methodology introduced by Anderl et al. (2016). The removal effect of a given channel is defined as the relative reduction in overall conversion probability that would result if that channel were removed from the system—formally, if all transitions involving the channel were redirected to non-conversion. A channel that contributes significantly to conversion will exhibit a substantial removal effect, while a peripheral channel will show only a small effect. The

removal effects of all channels are then normalized to produce a final attribution distribution (Kakalejcik et al., 2023).

The Markov approach offers several distinct advantages over heuristic alternatives. First, the attribution estimates are derived directly from the observed data rather than imposed by assumption, allowing the model to adapt to the specific dynamics of different industries and customer populations. Second, the explicit representation of channel-to-channel transitions enables the model to capture sequential dependencies that heuristic approaches necessarily ignore. Third, the removal effect methodology provides an intuitive causal interpretation: the credit assigned to each channel reflects, in a meaningful sense, the channel’s marginal contribution to overall conversion outcomes (Anderl et al., 2016).

Nevertheless, Markov-based attribution exhibits several recognized limitations. The memorylessness assumption, while computationally convenient, may be theoretically inappropriate for customer journeys in which the cumulative history of touchpoints affects subsequent transition probabilities (Kakalejcik et al., 2023). In practice, the impact of this assumption depends on the order of the Markov chain employed: higher-order chains can capture limited historical dependence at the cost of substantially increased data requirements. Furthermore, transition matrices estimated from sparse data may exhibit instability, with rare transitions either inflated or omitted entirely. This problem is conventionally addressed through smoothing techniques such as *Laplace smoothing* (Chen & Goodman, 1996), in which a small positive constant is added to all transition counts prior to normalization, ensuring that no transition probability is exactly zero.

A second category of limitation concerns the interpretive scope of removal effects. Although the removal effect provides a useful summary of a channel’s importance, it does not directly answer the question of how credit should be distributed across coalitions of channels acting in concert (Dalessandro et al., 2012). This limitation has motivated the development of complementary approaches grounded in cooperative game theory, to which the discussion now turns.

2.3.2. Shapley Value-Based Attribution

The application of Shapley Value theory to marketing attribution represents a fundamentally different methodological tradition from Markov-based approaches. Whereas the Markov framework treats attribution as a problem of stochastic process analysis, the Shapley framework treats it as a problem of *fair credit allocation* in a cooperative game (Dalessandro et al., 2012; Shao & Li, 2011).

The Shapley Value, introduced by Shapley (1953), originated in the mathematical theory of cooperative games as a solution concept for the problem of distributing a coalition’s total payoff among its members. Given a set of players $N = \{1, 2, \dots, n\}$ and a characteristic function $v : 2^N \rightarrow \mathbb{R}$ specifying the value generated by each possible coalition, the Shapley Value of player i is defined as

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(n - |S| - 1)!}{n!} [v(S \cup \{i\}) - v(S)] \quad (6)$$

In words, the Shapley Value of a player is the weighted average of the player’s marginal contribution to all possible coalitions, where the weights are constructed to ensure that the resulting allocation satisfies four axioms: *efficiency* (the values sum to the total payoff), *symmetry* (players making identical contributions receive identical values), *null player* (a player who contributes nothing receives zero), and *additivity* (the Shapley Values of a sum of games equal the

sum of the Shapley Values) (Roth, 1988; Winter, 2002). These axioms collectively define a unique allocation, and the Shapley Value is the only allocation that satisfies them simultaneously.

The application of this framework to attribution was first proposed by Dalessandro et al. (2012), who observed that the channels in a multi-touch customer journey can be naturally interpreted as players in a cooperative game, with the characteristic function defined to reflect the conversion outcomes associated with each possible subset of channels. Under this formulation, the Shapley Value of each channel quantifies its average marginal contribution to conversion across all possible channel coalitions, providing a principled and axiomatically justified attribution distribution.

The specific definition of the characteristic function plays a critical role in shaping the resulting Shapley estimates. A naïve formulation might define $v(S)$ as the number of conversions in journeys whose channel set is exactly S , but this approach yields unstable estimates owing to the sparsity of coalition-specific data (Shao & Li, 2011). A more robust formulation, introduced by Shao and Li (2011), employs *fractional credit*: $v(S)$ is defined as a weighted sum of conversions across journeys, with the weight assigned to each journey reflecting the proportion of its touchpoints that belong to coalition S . This formulation preserves the axiomatic guarantees of the Shapley framework while substantially improving the statistical robustness of the resulting estimates.

The Shapley framework offers several advantages relative to Markov-based approaches. First, the axiomatic foundation of the Shapley Value provides theoretical assurances regarding the fairness and consistency of the resulting attribution distribution—guarantees that are absent from heuristic and stochastic-process-based approaches alike (Winter, 2002). Second, the explicit consideration of all possible channel coalitions enables the Shapley framework to capture synergistic effects between channels in a manner that the Markov framework, with its focus on pairwise transitions, cannot replicate (Dalessandro et al., 2012).

These theoretical advantages are accompanied by significant practical limitations. The most fundamental of these is computational: the number of possible coalitions grows exponentially with the number of channels, rendering exact Shapley computation infeasible for systems with large channel counts (MDPI Algorithms, 2024). For systems with five to ten channels—the typical scale of B2C e-commerce attribution problems—exact computation remains tractable. For larger systems, approximation algorithms based on Monte Carlo sampling of coalitions are commonly employed (arXiv, 2025). A second limitation concerns the interpretation of marginal contributions in settings where channels exert sequential rather than parallel effects: the Shapley framework treats all coalitions symmetrically, potentially diluting the importance of touchpoints whose effect depends critically on their position within the journey (Molina et al., 2022).

The complementary character of Markov-based and Shapley-based approaches—the former capturing sequential structure, the latter capturing coalitional synergies—has motivated the development of hybrid methodologies that combine elements of both frameworks. Such hybrid approaches, while still relatively underexplored in the literature, form the principal methodological contribution of the present thesis.

2.3.3. Bayesian Attribution

A third strand of the data-driven attribution literature draws on Bayesian statistical methods to construct probabilistic models of conversion behavior. The Bayesian approach is most prominently associated with *Bayesian Marketing Mix Modeling* (Bayesian MMM), which extends classical MMM by incorporating prior distributions over model parameters, hierarchical structures across markets or product categories, and explicit modeling of carryover and satura-

tion effects (Chan & Perry, 2017; Jin et al., 2017).

In the multi-touch attribution context, Bayesian methods have been applied both as extensions of MMM and as standalone modeling frameworks. Li and Kannan (2014) develop a Bayesian hierarchical model that combines individual-level journey data with aggregate-level constraints, enabling attribution estimates that are both granular and statistically robust. More recent contributions have addressed specific methodological challenges, including multicollinearity among channel variables (Opella, 2025) and time-varying channel effects (Ng et al., 2021). Ni et al. (2024) propose a hierarchical Bayesian framework that integrates MMM with causal inference techniques to support personalized marketing optimization.

The principal advantage of Bayesian methods lies in their ability to quantify uncertainty: rather than producing point estimates of channel contributions, Bayesian models yield posterior distributions that capture both the magnitude and the precision of each estimate. This is particularly valuable in attribution contexts, where the stakes of allocation decisions are high and the underlying data are often noisy. Bayesian approaches also provide a principled mechanism for incorporating prior knowledge, such as managerial intuitions or industry benchmarks, into the estimation procedure (Jin et al., 2022).

The limitations of Bayesian attribution are largely practical. The computational demands of full posterior inference, typically performed via Markov Chain Monte Carlo sampling, can be substantial for large datasets. The specification of prior distributions requires methodological care and can introduce subjectivity into the analysis. Furthermore, the interpretation of Bayesian outputs requires familiarity with probabilistic reasoning that is not always present among managerial stakeholders (Chan & Perry, 2017).

2.3.4. Deep Learning Approaches

A fourth and increasingly prominent strand of the attribution literature applies deep learning techniques—particularly recurrent neural networks and attention-based architectures—to the modeling of customer journeys. The pioneering contribution in this area is the dual-attention mechanism of Ren et al. (2018), which adapts attention-based neural architectures from natural language processing to the sequential structure of customer journeys. More recent work has applied long short-term memory networks (Sarkar & De Bruyn, 2021), transformer architectures (Lu & Kannan, 2025), and deep reinforcement learning frameworks coupled to automated bidding systems (Tariq et al., 2025).

The principal advantage of deep learning approaches lies in their capacity to capture complex non-linear dependencies in customer journey data that simpler models, by their structural assumptions, cannot represent. The attention mechanism, in particular, enables the model to learn data-dependent weights over different touchpoints in a manner that generalizes the fixed weighting schemes of heuristic models. Empirical studies have demonstrated that deep learning approaches can substantially outperform classical models on attribution accuracy metrics, particularly for long and complex customer journeys (LinkedIn Data Science Team, 2025; Ren et al., 2018). A particularly noteworthy development in this direction is the work of Yao et al. (2020), who propose DeepMTA—an interpretable deep learning attribution model that explicitly integrates Shapley values into a phased-LSTM architecture, thereby combining sequential neural modeling with cooperative game-theoretic credit allocation. This integration is conceptually relevant to the present thesis, although the methodological approach differs: whereas DeepMTA combines Shapley with deep learning, the hybrid framework developed here combines Shapley with Markov chain analysis, foregoing neural network components in favor of interpretable classical methods.

However, deep learning approaches also exhibit important limitations in the attribution

context. The interpretive opacity of deep neural networks—often characterized as the "black box" problem—creates substantial challenges for managerial deployment, where attribution outputs must typically be defensible to non-technical stakeholders (Sarkar & De Bruyn, 2021). The data requirements of deep learning models are also considerably greater than those of simpler approaches, rendering them impractical for organizations without large-scale customer data infrastructures. Finally, the empirical performance advantages of deep learning approaches have not been uniformly demonstrated across all attribution contexts: in moderate-complexity environments, classical data-driven models such as Markov and Shapley approaches frequently achieve comparable accuracy with substantially greater interpretability (Various, 2023).

2.4. Causal Inference in Attribution

A parallel methodological tradition in the attribution literature emphasizes the formal application of causal inference principles to the estimation of channel effects. This tradition, while overlapping with the data-driven attribution approaches discussed above, places particular emphasis on the distinction between *correlation* and *causation* in attribution analysis (Doshi, 2025; Ferguson, 2025).

The foundational concern of the causal inference perspective is that observed correlations between channel exposures and conversion outcomes do not necessarily reflect causal effects. A channel may appear correlated with conversion not because it causes conversion, but because it captures users who would have converted regardless of exposure—a phenomenon known as *self-selection bias* (Lewis & Rao, 2015). Lewis and Rao (2015) demonstrate empirically that ignoring this bias can lead to substantial overestimation of advertising effectiveness, with implications for budget allocation and return-on-investment calculations.

The most direct method for establishing causal effects is the randomized controlled experiment, in which users are randomly assigned to exposure or control conditions and the resulting differences in conversion outcomes are measured. Gordon et al. (2019), in a large-scale comparative study of attribution approaches using field experiments at Facebook, find that observational attribution methods can produce results that differ substantially from experimentally validated estimates, with the direction and magnitude of the discrepancy depending on the specific method and context.

Recent contributions have sought to bridge the gap between observational and experimental approaches by adapting techniques from the causal inference literature, including propensity score methods (Doshi, 2025), counterfactual reasoning (Zhao et al., 2023), and instrumental variables. These approaches offer the prospect of causal validity without the operational costs and ethical complexities of large-scale experimentation. Their application to attribution remains an active area of research, with methodological questions concerning the appropriate identification strategies and assumption sets still being actively debated.

The relevance of causal inference considerations for the present thesis lies primarily in motivating the framework employed for model validation. As discussed in the introduction, the absence of an observable ground truth in real-world attribution data poses a fundamental challenge for model evaluation. The synthetic data approach adopted in this thesis can be understood as a controlled analog of the experimental approach: by constructing customer journeys from known channel effect parameters, the analysis establishes a ground truth against which the outputs of competing attribution models can be objectively compared. This methodological choice draws on the broader causal inference principle that valid model evaluation requires access to counterfactual information that observational data alone cannot provide (Zhao et al., 2023).

2.5. Marketing Mix Modeling and Budget Optimization

The literature on attribution and the literature on budget optimization have historically developed along separate but intersecting trajectories. Attribution research, as discussed in the preceding sections, has focused primarily on the question of *how credit for past conversions should be distributed* across marketing channels. Budget optimization research, by contrast, addresses the prospective question of *how future expenditure should be allocated* to maximize expected outcomes. Although the two questions are conceptually distinct, they are deeply inter-related: attribution outputs are most useful when they inform budget decisions, and budget optimization is only as valid as the underlying channel effect estimates on which it relies (Hanssens et al., 2003).

The earliest and most enduring framework connecting these two domains is *Marketing Mix Modeling* (MMM). Originating in the econometric tradition of the 1960s and 1970s, MMM provides a regression-based methodology for estimating the contribution of marketing investments to aggregate sales or revenue outcomes (Hanssens et al., 2003; Tellis, 2006). In its classical formulation, MMM treats channel-level expenditures as explanatory variables in a linear or log-linear model of sales, with additional terms capturing seasonality, pricing, distribution, and competitive effects. The estimated coefficients are then interpreted as channel-specific return-on-investment indicators and used to guide future budget allocation.

The methodological core of MMM relevant to the present thesis concerns the representation of *non-linear channel response*. Early MMM formulations frequently assumed linear relationships between marketing expenditure and sales, an assumption that conflicts both with theoretical considerations and with empirical evidence. A more realistic representation recognizes that marketing channels exhibit *diminishing returns*: the marginal effectiveness of each additional unit of expenditure declines as total expenditure increases. This phenomenon arises from a combination of audience saturation, decreasing marginal customer quality, and the diminishing salience of repeated exposures (Naik et al., 2005; Tellis, 2006).

The mathematical representation of diminishing returns in marketing models has converged on a small family of functional forms, of which the most widely employed is the *Hill function* or its close relatives. Jin et al. (2017), in the most influential modern treatment of MMM, formulate the response of a channel to expenditure x using a saturation function of the form

$$\text{Response}(x) = \beta \cdot \frac{x^S}{x^S + K^S} \quad (7)$$

where β governs the maximum achievable response, K is the *half-saturation point* at which the channel achieves half of its maximum response, and S controls the curvature of the response function. Variants of this formulation, including the exponential-saturation form $\text{Response}(x) = \beta \cdot (1 - e^{-k \cdot x/K})$, are used throughout the contemporary MMM literature. Chan and Perry (2017) identify the calibration of these saturation parameters as one of the central methodological challenges in MMM, given the difficulty of empirically estimating saturation curves from aggregate observational data.

A second methodological concern addressed in the MMM literature is the modeling of *interaction effects* between channels. Naik et al. (2005) demonstrate that marketing channels frequently exhibit synergistic or antagonistic interactions: the effectiveness of a search advertising campaign, for example, may depend on the contemporaneous level of display advertising or social media presence. Ignoring these interactions, as classical MMM implementations often do, can lead to systematic mis-estimation of channel effects and consequently of optimal budget allocations.

The translation of MMM-derived channel effect estimates into actionable budget recommendations is conventionally framed as an *optimization problem*. In the simplest formulation, the problem is one of linear programming: given a fixed total budget and channel-specific marginal returns, determine the allocation that maximizes total expected return subject to budget and minimum-allocation constraints (Hillier & Lieberman, 2014). However, the incorporation of non-linear response curves transforms the problem into a non-linear optimization problem, for which standard linear programming methods are inadequate.

A variety of optimization algorithms have been applied to non-linear budget allocation problems, ranging from gradient-based methods such as Sequential Least Squares Programming to global optimization heuristics such as the *Differential Evolution* algorithm introduced by Storn and Price (1997). Differential Evolution is particularly well-suited to budget allocation problems characterized by multi-modal objective functions, non-smooth constraints, and a moderate number of decision variables. The algorithm operates by maintaining a population of candidate allocations and iteratively combining them through mutation and crossover operations until convergence to a near-optimal solution. Its global optimization properties make it more robust than gradient-based methods in cases where the response surface exhibits multiple local optima—a common feature of marketing optimization problems with channel interactions.

The integration of attribution outputs into MMM-style optimization frameworks remains an underexplored area in the academic literature. Most published attribution studies stop at the level of credit assignment, leaving the translation into budget recommendations either implicit or external to the analysis. Conversely, MMM studies frequently rely on aggregate channel effects estimated without reference to individual-level customer journey data. The present thesis bridges these traditions by treating the outputs of an individual-level attribution model as the channel-effect inputs to a non-linear budget optimization layer, thereby producing budget recommendations that are grounded simultaneously in granular journey data and realistic saturation dynamics.

2.6. Privacy, Cookies, and the Cookieless Future

The methodological apparatus of multi-touch attribution depends fundamentally on the ability to track individual users across multiple touchpoints. In the contemporary digital advertising ecosystem, this tracking has been enabled primarily by *third-party cookies*—small data files placed in users’ browsers by domains other than the website being visited. Third-party cookies have served as the technical foundation for cross-site identity resolution, allowing advertisers to recognize a single user across encounters with display advertisements, search results, social media platforms, and ultimately the advertiser’s own website (Goldfarb & Tucker, 2011).

This infrastructure, however, is undergoing rapid and structural transformation. The transformation is driven by two parallel forces: regulatory action and platform-level policy changes. On the regulatory side, the European Union’s General Data Protection Regulation (GDPR), enacted in 2018, and the California Consumer Privacy Act (CCPA), enacted in 2020, have imposed substantial constraints on the collection, processing, and use of personal data, with particular implications for third-party tracking practices (Acquisti et al., 2016). On the platform side, Apple’s introduction of Intelligent Tracking Prevention (ITP) in 2017 and the more recent App Tracking Transparency (ATT) framework in 2021 have substantially reduced the tracking capabilities of devices within the Apple ecosystem. Most significantly, Google’s announced deprecation of third-party cookies in the Chrome browser—initially planned for 2022, subsequently delayed to 2024, and ultimately reformulated as an opt-in control system in

2025— threatens to eliminate the dominant technical foundation of cross-site tracking entirely (Google, 2024b).

The empirical consequences of these changes for advertising effectiveness have been the subject of considerable research. Johnson et al. (2020) provide an early empirical analysis of consumer privacy choices in online advertising, finding that even modest opt-out rates can produce substantial reductions in advertising revenue, with disproportionate effects on small and medium-sized advertisers who lack the first-party data infrastructure of large platforms. More recent industry-focused analyses have quantified the operational impact of cookie deprecation on programmatic advertising efficiency, documenting substantial declines in conversion-tracking accuracy and corresponding increases in attribution uncertainty (Various, 2024).

In response to these challenges, the digital advertising industry has developed a range of technical strategies aimed at preserving attribution capabilities under privacy-constrained conditions. Chebrolu (2025) provide a comprehensive overview of one prominent approach: *server-side tracking*, in which user-level event data is collected directly by the advertiser’s servers rather than through third-party browser-based mechanisms. While server-side approaches mitigate some of the technical limitations imposed by browser-level privacy controls, they do not address the underlying regulatory and consent-based constraints. A complementary approach, *federated learning*, is described by Thomas (2024): in this paradigm, machine learning models are trained on decentralized user data without the data itself being transmitted to a central server, preserving user privacy while retaining analytical capability. Federated approaches, however, impose substantial computational costs and remain technically immature for many practical applications.

A third class of approaches involves the construction of *universal identifier* systems intended to replace third-party cookies as the technical substrate for cross-site identity resolution. Sivan-Sevilla and Poudel (2025) provide a critical analysis of these systems, arguing that they may represent a privacy-eroding loophole rather than a genuine accommodation of regulatory intent. The viability and acceptability of universal identifiers remain unresolved questions in the contemporary policy landscape.

A fourth and methodologically distinct response to the privacy transition involves the use of *synthetic* or *simulated* data as a substitute for, or complement to, real user data. The Springer/UNL study of privacy-preserving digital marketing analytics, employing generative adversarial networks to construct synthetic user behavior data, illustrates this approach (Springer/UNL, 2023). Synthetic data approaches offer the prospect of methodologically valid analysis without exposure to the regulatory and ethical concerns associated with personal data collection. Although the present thesis does not employ synthetic data primarily for privacy reasons, the methodological framework developed here—which relies on synthetic customer journey data with known ground-truth channel effects—is consistent with the broader trend toward synthetic-data-based research in privacy-constrained environments.

The implications of the cookieless transition for the future of multi-touch attribution research are substantial. As real-world customer journey data becomes increasingly fragmented, partial, or inaccessible, the relative value of methodological frameworks grounded in synthetic data or causal inference techniques is likely to increase. Attribution models that depend critically on complete cross-channel tracking, such as those built on classical Markov chain or Shapley value formulations applied directly to observational data, may require methodological adaptation to remain viable in the post-cookie landscape. These considerations are revisited in Chapter 5 as part of the discussion of the present study’s limitations and future research directions.

2.7. Research Gap

The literature reviewed in the preceding sections, while extensive and methodologically sophisticated, exhibits four interrelated gaps that collectively motivate the present thesis. Each of these gaps is discussed in turn below, followed by an articulation of the specific contributions the present study makes in response.

2.7.1. The Ground Truth Validation Problem

The first and most fundamental gap concerns the *validation* of data-driven attribution models. As established in Section 2.3, the attribution literature has produced a substantial number of methodologically sophisticated models, including Markov-based, Shapley-based, Bayesian, and deep-learning approaches. Each of these models is conventionally presented with empirical results demonstrating its superiority to heuristic baselines (Anderl et al., 2016; Ren et al., 2018; Shao & Li, 2011). However, the validity of these empirical demonstrations rests on a methodologically problematic foundation: in real-world data, the true contribution of each channel to a given conversion is unobservable, and therefore the accuracy of any particular model’s estimates cannot be objectively assessed.

This unobservability problem has been addressed in the literature primarily through two strategies, neither of which is fully satisfactory. The first strategy involves the use of randomized field experiments to establish causal benchmarks against which observational models can be compared, as exemplified by the work of Gordon et al. (2019) at Facebook. While methodologically rigorous, field experiments are expensive, operationally complex, and accessible only to organizations with substantial resources and large user populations. The second strategy involves comparing models against one another and inferring superiority from consistency or convergence (Lewis & Rao, 2015); this approach, however, provides no information about whether any of the compared models is *actually correct*, only about whether they agree with each other.

A third strategy, less commonly employed but methodologically distinct, involves the use of synthetic data with known ground-truth properties. Although individual studies have employed simulated data for specific methodological purposes (Springer/UNL, 2023), no systematic framework exists in the published literature for the use of synthetic data to benchmark the full range of contemporary attribution models against a common, objective standard. The absence of such a framework constitutes the first gap addressed by this thesis.

2.7.2. Underexplored Hybrid Methodologies

The second gap concerns the relationship between the principal data-driven attribution paradigms—specifically, Markov-based and Shapley-based approaches. In the existing literature, these paradigms are typically presented as *alternatives*, with researchers and practitioners selecting one or the other based on theoretical preference, computational considerations, or specific data characteristics (Anderl et al., 2016; Dalessandro et al., 2012). The possibility that the two paradigms might be *complementary*, with each capturing aspects of the attribution problem that the other cannot, has received limited systematic attention.

Several recent contributions have begun to explore hybrid attribution methodologies, including the game-theoretic extensions developed by Molina et al. (2022) and the resource allocation framework of MDPI Algorithms (2024). However, these contributions operate primarily at the theoretical level and have not produced empirically validated combination strategies. In particular, the question of how the relative weight assigned to each component of a hybrid model

should be determined— whether by assumption, by theoretical derivation, or by empirical optimization against a known benchmark—remains open. The absence of a data-driven framework for hybrid weight determination constitutes the second gap addressed by this thesis.

2.7.3. The Attribution-Optimization Disconnect

The third gap, discussed in part in Section 2.5, concerns the *disconnect between attribution analysis and budget optimization* in the existing literature. Attribution research, in its dominant tradition, treats the assignment of credit to channels as the principal analytical output, with implications for budget allocation often noted but not systematically developed. Marketing Mix Modeling research, conversely, treats budget optimization as the central analytical objective, but typically operates at the aggregate level and does not incorporate the granular journey-level information that motivates contemporary attribution methodologies (Hanssens et al., 2003; Jin et al., 2017).

Few existing studies bridge these two traditions in a unified analytical framework. The work of Tariq et al. (2025) on real-time budget optimization using deep reinforcement learning represents one recent attempt to integrate attribution and optimization, but the proprietary and technically specialized nature of the proposed framework limits its accessibility for general academic and managerial audiences. More broadly, no widely accessible framework exists in the published literature for translating the outputs of standard data-driven attribution models into actionable budget recommendations under realistic diminishing-returns assumptions. The absence of such a framework constitutes the third gap addressed by this thesis.

2.7.4. Limited Focus on Small-to-Medium B2C E-commerce

The fourth gap concerns the *contextual specificity* of attribution research. The published attribution literature exhibits a marked concentration in two distinct contextual regimes. At one extreme, contributions from large platform operators—including Facebook, Google, and LinkedIn—address attribution at the scale of millions of users and correspondingly large advertising budgets (Gordon et al., 2019; LinkedIn Data Science Team, 2025). At the other extreme, theoretically oriented contributions develop methods abstractly, without explicit reference to the operational realities of any particular organizational context (Molina et al., 2022; Shapley, 1953).

The intermediate regime of small-to-medium B2C e-commerce, in which budget constraints are binding, channel counts are moderate, and operational simplicity is at a premium, has received comparatively limited attention. This regime is, however, of substantial practical importance: small-to-medium e-commerce operations represent a large share of digital advertising expenditure globally (WordStream, 2025), and the attribution and budget challenges they face are distinct from those of larger platform operators. The absence of attribution and budget optimization frameworks specifically tailored to this regime constitutes the fourth gap addressed by this thesis.

2.7.5. Contributions of the Present Thesis

In response to the four gaps articulated above, the present thesis develops a unified analytical framework incorporating the following four contributions:

1. **A ground-truth-based validation framework.** The thesis constructs a synthetic customer journey dataset in which the true contribution of each channel to conversion out-

comes is known by construction, derived from industry-grounded channel effect parameters (Lucky Orange, 2025; WordStream, 2025). This dataset serves as an objective benchmark against which the accuracy of attribution models can be evaluated. The framework directly addresses the validation problem identified above by enabling, for the first time in a systematic manner, the comparison of multiple attribution methodologies against a common and unambiguous standard.

2. **A data-driven hybrid Markov-Shapley model.** The thesis proposes a hybrid attribution model that combines Markov Chain removal effects and Shapley Value contributions through a weighted average, with the weighting parameter determined empirically through a grid search against the constructed ground truth. This data-driven approach to hybrid weight determination represents a departure from prior work, in which combination weights are typically fixed by assumption or derived from theoretical principles alone.
3. **An integrated attribution-to-budget pipeline.** The thesis incorporates the outputs of the hybrid attribution model into a non-linear budget optimization layer based on diminishing-returns response curves and the Differential Evolution algorithm (Jin et al., 2017; Storn & Price, 1997). This integration transforms the attribution model from a purely descriptive tool into a prescriptive decision-support system capable of producing actionable budget recommendations conditioned on total budget level and operational constraints.
4. **Targeted applicability to small-to-medium B2C e-commerce.** The framework is designed specifically for the operational regime of small-to-medium B2C e-commerce operations, with parameter choices, channel selections, and computational complexity calibrated to the needs of this context. The findings of the empirical analysis demonstrate that the value of the proposed framework is particularly pronounced at moderate budget levels, where the marginal value of accurate allocation is highest.

The methodological apparatus by which these contributions are realized is presented in detail in the following chapter.

3. METHODOLOGY

3.1. Research Design

The present study adopts a quantitative, comparative research design centered on the systematic evaluation of multiple attribution models against a controlled ground truth. The design is motivated by the validation problem identified in Section 2.7: in real-world attribution data, the true contribution of each channel to a given conversion is fundamentally unobservable, rendering objective model comparison infeasible. To address this limitation, the present study constructs a synthetic customer journey dataset in which the true channel contribution structure is known by construction, thereby providing a benchmark against which the outputs of competing attribution models can be objectively assessed.

The methodological pipeline employed in this study comprises four sequential stages, illustrated in Figure 1. The first stage generates synthetic customer journey data using channel effect parameters calibrated to industry benchmarks. The second stage applies eight attribution models—five classical heuristic baselines and three data-driven models including the proposed hybrid—to the generated data. The third stage evaluates the attribution outputs against the ground truth using multiple performance metrics. The fourth stage feeds the attribution estimates into a non-linear budget optimization layer to produce actionable budget recommendations across multiple budget scenarios.

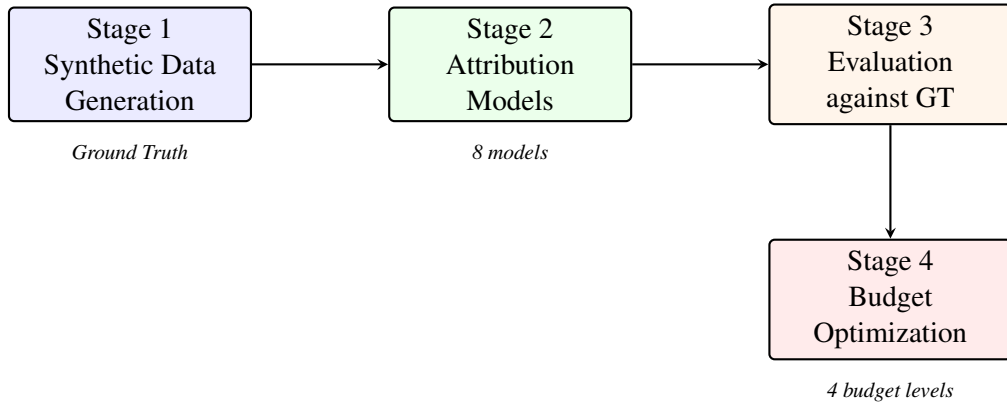


Figure 1. Research design pipeline. The four-stage methodology proceeds from synthetic data generation through attribution modeling and evaluation to budget optimization. Ground truth values established in Stage 1 serve as the validation benchmark in Stage 3.

The use of synthetic data in this design serves two methodological purposes. First, it provides an objective ground truth, as discussed above. Second, it ensures full reproducibility: the entire dataset can be regenerated from a known set of parameters and a fixed random seed, eliminating the data access constraints that frequently limit the replicability of attribution research conducted on proprietary real-world datasets (Johnson et al., 2020). The methodological tradition of constructing controlled synthetic environments for model evaluation has a long history in computational research and is particularly well-suited to attribution analysis, where the absence of observable ground truth makes alternative validation strategies problematic (Robert & Casella, 2013).

3.2. Synthetic Data Generation

The synthetic customer journey dataset employed in this study consists of 30,000 simulated user sessions, each representing a complete customer journey from initial touchpoint to terminal outcome (conversion or non-conversion). The dataset comprises a total of 105,565 individual touchpoint events distributed across five marketing channels, yielding an average path length of 3.52 touchpoints per journey and a total of 354 conversion events, corresponding to an overall conversion rate of approximately 1.18%. This conversion rate falls within the typical range observed in B2C e-commerce environments, where industry benchmarks consistently report conversion rates between 1% and 3% (Lucky Orange, 2025; WordStream, 2025).

The dataset generation procedure is governed by three categories of parameters: channel effect parameters, channel interaction effects, and the functional form of the conversion probability function. Each is described in turn below.

3.2.1. Channel Effect Parameters

The five marketing channels included in the synthetic dataset—Google Ads, Email, Direct, Organic Search, and Social Media—were selected to represent the typical channel mix of a small-to-medium B2C e-commerce operation. The choice of five channels reflects a deliberate methodological compromise: the channel count is large enough to support meaningful multi-channel interaction dynamics while remaining sufficiently small to permit exact Shapley value computation, which exhibits exponential complexity in the number of channels (see Section 3.4.3).

Each channel is assigned an *effect strength* parameter representing its intrinsic capacity to contribute to conversion, calibrated against industry conversion-rate benchmarks for the corresponding channel type (Lucky Orange, 2025; WordStream, 2025). Table 1 presents the channel effect parameters employed in the present study.

Table 1. Channel effect parameters used in synthetic data generation.

Channel	Effect Strength	Source / Rationale
Google Ads	0.35	High-intent paid search; top conversion driver
Email	0.25	Established list; warm audience
Direct	0.20	Brand-aware repeat visitors
Organic Search	0.15	SEO-driven, mid-funnel role
Social Media	0.10	Awareness-stage, lower direct conversion

The relative ordering of the channel effect parameters reflects the empirically observed pattern in which high-intent paid channels (such as Google Ads and Email) exhibit greater proximate conversion influence than awareness-stage channels (such as Social Media), although the latter may contribute substantially to upper-funnel demand generation (Abhishek et al., 2012). The specific numerical values were selected to produce a synthetic dataset whose aggregate conversion rate falls within the industry benchmark range while ensuring meaningful heterogeneity across channels.

3.2.2. Channel Interaction Effects

The conversion process is modeled as incorporating not only the independent contributions of individual channels but also the interaction effects between them. Channel interaction

effects refer to situations in which the combined presence of two channels in a customer journey produces a conversion probability that differs—either positively (synergy) or negatively (cannibalization)—from the sum of their independent contributions (Naik et al., 2005).

The empirical literature provides substantial evidence for the existence of meaningful channel interactions in real-world customer journeys. Kireyev et al. (2016) demonstrate, for example, that display advertising exerts a causal influence on subsequent search behavior, implying that journeys featuring both display and search touchpoints exhibit conversion patterns that cannot be predicted from the independent effects of each channel alone. Failure to incorporate such interactions in the synthetic data generation process would result in a dataset that is methodologically inadequate for evaluating attribution models, since the very phenomenon that motivates data-driven attribution—the inability of heuristic models to capture cross-channel synergies—would be absent by construction.

The interaction effects incorporated in the present study include a positive synergy between Social Media and Google Ads (reflecting the empirical influence of social exposure on subsequent search intent), a positive synergy between Email and Direct visits (reflecting the relationship-building function of email marketing), and additional pairwise interactions reflecting the conceptual structure of the B2C e-commerce customer journey. The complete interaction matrix is presented in Appendix C.

3.2.3. Conversion Probability Function

The probability that a given customer journey terminates in conversion rather than non-conversion is modeled as a function of the channels present in the journey, the order and recency of those touchpoints, and the pairwise interaction effects between them. Formally, for a customer journey $J = \{c_1, c_2, \dots, c_L\}$ consisting of L touchpoints, the conversion probability is

$$P(\text{conversion} \mid J) = \sigma \left(\sum_{i=1}^L \beta_{c_i} \cdot r_i + \sum_{i < j} \gamma_{c_i, c_j} \cdot r_{ij} + \beta_0 \right) \quad (8)$$

where β_{c_i} denotes the channel effect parameter for the channel encountered at touchpoint i , r_i is a recency weight assigning greater influence to more recent touchpoints, γ_{c_i, c_j} denotes the pairwise interaction effect between the channels at positions i and j , r_{ij} is a recency weight for the interaction, β_0 is a baseline intercept calibrated to produce the target aggregate conversion rate, and $\sigma(\cdot)$ denotes the logistic function constraining the output to the unit interval.

The recency weighting scheme reflects the psychological premise—widely supported in the behavioral marketing literature—that touchpoints closer in time to the conversion event exert greater influence on the decision than more distant touchpoints. The functional form adopted here applies an exponential decay with a half-life calibrated to match typical e-commerce customer journey durations.

Conversion outcomes for each synthetic journey are determined by drawing a uniform random variable on the unit interval and comparing it against the journey’s conversion probability. This stochastic generation procedure ensures that the resulting dataset exhibits the statistical noise characteristic of real-world conversion data, while remaining grounded in a known underlying parameter structure.

3.2.4. Monte Carlo Ground Truth Simulation

The *ground truth* channel contribution structure—against which the outputs of competing attribution models are evaluated—is derived through a Monte Carlo simulation procedure

originally formalized by Metropolis and Ulam (1949). The methodological rationale for the Monte Carlo approach lies in the impossibility of deriving the ground truth analytically: although the channel effect parameters and interaction effects are known by construction, the precise mapping from these parameters to channel-level conversion contributions, in the presence of recency weighting and stochastic conversion, requires numerical estimation (Robert & Casella, 2013).

The ground truth simulation operates as follows. For each channel c in the channel set, a large number of synthetic journeys is generated in which channel c is present, and an equally large number of counterfactual journeys is generated in which channel c is absent. The difference in mean conversion probability between these two sets, normalized to sum to unity across channels, constitutes the ground truth attribution weight for channel c . Formally,

$$\text{GT}(c) = \frac{P(\text{conv} \mid c \in J) - P(\text{conv} \mid c \notin J)}{\sum_{c' \in \mathcal{C}} [P(\text{conv} \mid c' \in J) - P(\text{conv} \mid c' \notin J)]} \quad (9)$$

where $\mathcal{C}$ denotes the set of all channels and the conditional probabilities are estimated by Monte Carlo simulation over 100,000 independently generated journeys per channel.

The choice of 100,000 simulations per channel reflects a balance between estimation precision and computational cost. At this sample size, the standard error of the estimated conditional probabilities is on the order of 10^{-3} , which is substantially smaller than the typical between-channel differences in attribution weights. Larger sample sizes yield diminishing improvements in precision and were therefore not adopted. The resulting ground truth attribution weights, normalized to sum to unity, serve as the validation benchmark in all subsequent analyses.

3.3. Markov Chain Attribution Model

The Markov Chain attribution model employed in this study is grounded in the theory of absorbing Markov chains and implements the removal-effect methodology introduced by Anderl et al. (2016). The model represents the customer journey as a stochastic process over a finite state space and derives channel-level attribution weights from the impact that the hypothetical elimination of each channel would exert on overall conversion probability.

3.3.1. Absorbing Markov Chain Formulation

The state space of the model, denoted $\mathcal{S}$, consists of the five marketing channels described in Section 3.2 together with two additional *absorbing states*: *Conversion*, representing successful completion of the purchase, and *Non-Conversion*, representing abandonment of the journey. An absorbing state, in the terminology of Norris (1998), is a state from which no further transitions are possible: once a journey enters an absorbing state, it remains there indefinitely.

The complete state space therefore comprises seven states:

$$\mathcal{S} = \{\text{Start, Google_Ads, Email, Direct, Organic_Search, Social_Media, Conversion, Non-Conversion}\} \quad (10)$$

where the additional *Start* state functions as the entry point of all journeys. The five channel states are *transient*, in the sense that journeys may visit them multiple times before ultimately being absorbed into either *Conversion* or *Non-Conversion*. The structure of the resulting absorbing Markov chain is illustrated in Figure 2.

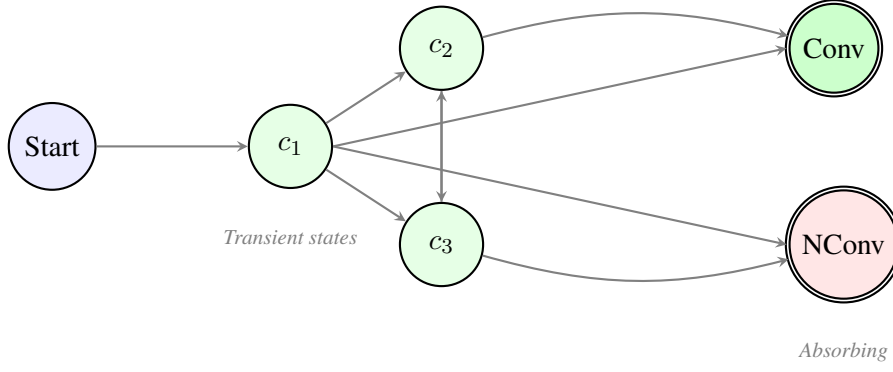


Figure 2. Schematic representation of the absorbing Markov chain employed in the attribution model. Channel states (c_1, c_2, c_3 , etc.) are transient; the Conversion and Non-Conversion states are absorbing. Each journey begins in the Start state and terminates upon absorption.

The dynamics of the absorbing Markov chain are governed by its *transition matrix* P , a square matrix of dimension $|\mathcal{S}| \times |\mathcal{S}|$ whose entries p_{ij} specify the probability of transitioning from state i to state j in a single step. The construction of this matrix from observed customer journey data constitutes the first computational step of the attribution procedure.

3.3.2. Transition Matrix Construction

The transition probabilities are estimated empirically from the observed customer journey data using maximum-likelihood principles. Specifically, for any pair of states (i, j) , the transition probability p_{ij} is estimated as

$$\hat{p}_{ij} = \frac{n_{ij}}{\sum_{k \in \mathcal{S}} n_{ik}} \quad (11)$$

where n_{ij} denotes the number of observed transitions from state i directly to state j across all customer journeys in the dataset (Anderl et al., 2016). The denominator sums the total number of outgoing transitions from state i , ensuring that the resulting transition probabilities form a valid probability distribution.

This empirical estimation procedure, while theoretically sound, exhibits practical limitations in the presence of sparse data. In customer journey datasets typical of small-to-medium B2C e-commerce operations, certain channel-to-channel transitions may be observed rarely or not at all, even if such transitions are theoretically possible. The resulting transition matrix may consequently contain zero or near-zero entries for transitions that, in expectation, occur with non-trivial probability (Kakalejcik et al., 2023). This sparse-data problem is addressed through smoothing.

3.3.3. Laplace Smoothing

The methodological response to the sparse-data problem adopted in the present study is *Laplace smoothing*, a technique originally developed in the context of statistical language modeling (Chen & Goodman, 1996). Laplace smoothing addresses the zero-frequency problem by adding a small positive constant to all observed transition counts prior to normalization, ensuring that no estimated transition probability is exactly zero.

The smoothed transition probability is computed as

$$\hat{p}_{ij}^{\text{smooth}} = \frac{n_{ij} + \alpha}{\sum_{k \in \mathcal{S}} (n_{ik} + \alpha)} \quad (12)$$

where $\alpha > 0$ is the *smoothing parameter* governing the strength of the smoothing effect. In the present implementation, $\alpha = 1.0$, corresponding to the classical add-one or Laplace formulation. The choice of $\alpha = 1.0$ reflects a methodological balance: smaller values provide less protection against zero-frequency artifacts, while larger values introduce excessive bias toward uniform transition probabilities. The selected value has been employed successfully in prior attribution applications (Anderl et al., 2016).

The smoothing operation is applied uniformly to all entries of the transition matrix, with the exception of the absorbing states whose self-transition probabilities are fixed at unity by definition. The resulting smoothed transition matrix forms the basis for all subsequent attribution computations.

3.3.4. Removal Effect Calculation

The attribution of credit to individual channels is accomplished through the *removal effect* methodology of Anderl et al. (2016). The removal effect of a given channel c is defined as the relative reduction in overall conversion probability that would result if that channel were hypothetically removed from the system. Formally,

$$\text{RE}(c) = \frac{P(\text{conversion}) - P(\text{conversion} \mid -c)}{P(\text{conversion})} \quad (13)$$

where $P(\text{conversion})$ denotes the overall probability that a randomly initiated journey terminates in conversion, and $P(\text{conversion} \mid -c)$ denotes the corresponding probability under the counterfactual scenario in which all transitions involving channel c are redirected to the Non-Conversion absorbing state.

The computation of $P(\text{conversion})$ proceeds through standard absorbing Markov chain analysis. Partitioning the smoothed transition matrix into blocks corresponding to transient and absorbing states yields

$$P = \begin{pmatrix} Q & R \\ 0 & I \end{pmatrix} \quad (14)$$

where Q is the $t \times t$ submatrix of transitions among transient states, R is the $t \times 2$ submatrix of transitions from transient to absorbing states, and I is the 2×2 identity matrix representing the absorbing states' fixed self-transitions (Norris, 1998).

The probability of eventual absorption into each absorbing state, starting from each transient state, is given by the entries of the matrix $B = N \cdot R$, where N is the *fundamental matrix*

$$N = (I - Q)^{-1} \quad (15)$$

The entry B_{ij} specifies the probability that a journey beginning in transient state i is ultimately absorbed into absorbing state j . The overall conversion probability $P(\text{conversion})$ is recovered as the weighted average of the absorption probabilities across all journey starting points, with weights proportional to the empirical frequency of each starting state.

The counterfactual probability $P(\text{conversion} \mid -c)$ is computed analogously, after modification of the transition matrix to redirect all transitions involving channel c to the Non-Conversion absorbing state. The removal effect for each channel is then obtained by the formula

given above. The final attribution weights are produced by normalizing the removal effects to sum to unity across channels:

$$w_c^{\text{Markov}} = \frac{\text{RE}(c)}{\sum_{c' \in \mathcal{C}} \text{RE}(c')} \quad (16)$$

The normalized weights $\{w_c^{\text{Markov}}\}$ constitute the attribution output of the Markov Chain model and serve as one of the two components of the hybrid model developed in Section 3.5.

3.4. Shapley Value Attribution Model

The Shapley Value attribution model represents a methodologically distinct approach to the same attribution problem addressed by the Markov Chain model. Whereas the Markov framework treats attribution as a problem of stochastic process analysis, the Shapley framework treats it as a problem of fair credit allocation in a cooperative game (Dalessandro et al., 2012; Shapley, 1953). The two frameworks therefore capture different aspects of the underlying attribution structure, motivating the hybrid combination developed in Section 3.5.

3.4.1. Cooperative Game Theory Foundations

The Shapley Value originates in the mathematical theory of cooperative games, formalized by Shapley (1953) as a solution concept for the problem of distributing a coalition's total payoff among its members. A cooperative game is defined by a set of *players* $N = \{1, 2, \dots, n\}$ and a *characteristic function* $v : 2^N \rightarrow \mathbb{R}$ which assigns to each possible subset (or *coalition*) $S \subseteq N$ a real-valued payoff $v(S)$. The characteristic function captures the collective value that the members of coalition S can generate by cooperating, independent of the actions of players outside the coalition.

The Shapley Value ϕ_i of player i is uniquely determined by four axioms imposed on the allocation rule (Roth, 1988; Winter, 2002):

- **Efficiency:** The Shapley Values of all players sum to the total value of the grand coalition, $\sum_{i \in N} \phi_i = v(N)$.
- **Symmetry:** Two players whose marginal contributions to every coalition are identical receive identical Shapley Values.
- **Null Player:** A player whose marginal contribution to every coalition is zero receives a Shapley Value of zero.
- **Additivity:** For two cooperative games v and w defined over the same player set, the Shapley Value of the combined game $v + w$ equals the sum of the Shapley Values of v and w .

These axioms collectively define a unique allocation rule, and the Shapley Value is the only allocation that satisfies them simultaneously (Winter, 2002). The axiomatic foundation distinguishes the Shapley framework from heuristic attribution approaches, which lack any comparable theoretical justification for their credit assignment rules.

In the application to attribution analysis introduced by Dalessandro et al. (2012), the marketing channels of a multi-touch customer journey are interpreted as the players of the cooperative game, with the characteristic function $v(S)$ defined to reflect the conversion outcomes generated by journeys whose channel composition includes the coalition S . The Shapley Value

of each channel under this formulation quantifies the channel's average marginal contribution to conversion across all possible channel coalitions.

3.4.2. Characteristic Function with Fractional Credit

The specific form of the characteristic function plays a critical role in determining the resulting Shapley estimates. A naïve formulation might define $v(S)$ as the number of conversions in journeys whose channel set is exactly equal to S . However, this approach yields highly unstable estimates in practice, because the number of journeys exhibiting any specific exact channel set is typically small, particularly for larger coalitions (Shao & Li, 2011).

The present study employs the *fractional credit* formulation introduced by Shao and Li (2011), which addresses the sparsity problem by allowing each journey to contribute partial value to multiple coalitions. Specifically, for a coalition $S \subseteq N$, the characteristic function is defined as

$$v(S) = \sum_{j \in \mathcal{J}_{\text{conv}}} \frac{|S \cap C_j|}{|C_j|} \quad (17)$$

where $\mathcal{J}_{\text{conv}}$ denotes the set of all converting journeys, C_j denotes the set of channels present in journey j , and $|S \cap C_j|/|C_j|$ represents the fractional overlap between coalition S and the channels of journey j . Under this formulation, a journey contributes to the value of coalition S in proportion to the fraction of its channels that belong to S , rather than only in cases of exact match.

The fractional credit formulation preserves the four axiomatic guarantees of the Shapley framework while substantially improving the statistical robustness of the resulting estimates. By drawing on partial overlaps between observed journeys and hypothetical coalitions, the formulation enables stable estimation even for coalitions that are rarely or never observed exactly in the data.

3.4.3. Shapley Value Computation

Given the characteristic function defined above, the Shapley Value of each channel is computed according to the formula

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|! (n - |S| - 1)!}{n!} [v(S \cup \{i\}) - v(S)] \quad (18)$$

The expression within square brackets represents the *marginal contribution* of channel i to coalition S : the increment in coalition value that results from adding i to S . The weight assigned to each marginal contribution depends on the size of the coalition: contributions to coalitions of size $|S|$ are weighted by the factor $|S|! (n - |S| - 1)!/n!$, which reflects the probability that i is added to such a coalition under a uniformly random ordering of players. The Shapley Value is thus the average marginal contribution of channel i across all possible coalitions, weighted to ensure consistency with the four axioms.

The computational complexity of exact Shapley computation is $O(2^n)$, where n denotes the number of channels. For the five-channel configuration employed in the present study, this corresponds to the enumeration of $2^5 = 32$ coalitions (including the empty coalition), which is computationally tractable on standard hardware. For larger channel counts, exact computation becomes prohibitive, and approximation algorithms based on Monte Carlo sampling of coalitions are typically employed (MDPI Algorithms, 2024). The deliberate restriction of the

present study to five channels reflects, in part, the desire to retain exact Shapley computation for methodological transparency.

The Shapley Values produced by the foregoing computation are subsequently normalized to sum to unity across channels, yielding the final Shapley attribution weights:

$$w_c^{\text{Shapley}} = \frac{|\phi_c|}{\sum_{c' \in \mathcal{C}} |\phi_{c'}|} \quad (19)$$

The absolute value operation ensures that negative Shapley Values—which may arise in the presence of antagonistic channel interactions—are treated as contributions to the attribution distribution rather than as detractors. The resulting normalized weights $\{w_c^{\text{Shapley}}\}$ serve as the second component of the hybrid attribution model developed in the following section.

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(n - |S| - 1)!}{n!} [v(S \cup \{i\}) - v(S)] \quad (20)$$

3.5. Hybrid Markov-Shapley Model

The hybrid Markov-Shapley attribution model proposed in this thesis combines the outputs of the Markov Chain and Shapley Value models developed in Sections 3.3 and 3.4 through a convex linear combination. The hybrid formulation is motivated by the complementary methodological strengths of the two component models, as discussed in Sections 2.3.1 and 2.3.2: Markov-based attribution effectively captures the sequential structure of customer journeys and the directional flow of channel-to-channel transitions (Anderl et al., 2016), while Shapley-based attribution provides axiomatically grounded credit allocation across all possible channel coalitions (Dalessandro et al., 2012; Shapley, 1953). Neither approach alone fully captures the multidimensional nature of multi-channel attribution; their combination, with an empirically optimized weighting parameter, exploits the strengths of both paradigms.

3.5.1. Weighted Combination Formulation

The hybrid attribution weight for channel c is defined as a weighted average of the normalized Markov and Shapley weights:

$$w_c^{\text{Hybrid}} = \alpha \cdot w_c^{\text{Markov}} + (1 - \alpha) \cdot w_c^{\text{Shapley}} \quad (21)$$

where w_c^{Markov} and w_c^{Shapley} denote the normalized attribution weights produced by the Markov and Shapley models respectively, as defined in Sections 3.3.4 and 3.4.3, and $\alpha \in [0, 1]$ is a scalar *mixing parameter* governing the relative contribution of each component.

The interpretation of the mixing parameter is direct. When $\alpha = 0$, the hybrid model collapses to the pure Shapley Value model, assigning attribution weights based exclusively on cooperative coalitional analysis. When $\alpha = 1$, the hybrid model collapses to the pure Markov Chain model, assigning weights based exclusively on removal-effect analysis. Intermediate values of α produce blended attribution distributions that incorporate elements of both paradigms in proportion to the parameter value.

The convex combination formulation possesses two desirable mathematical properties. First, the resulting hybrid weights are themselves a valid attribution distribution: since $\sum_c w_c^{\text{Markov}} = 1$ and $\sum_c w_c^{\text{Shapley}} = 1$, it follows that $\sum_c w_c^{\text{Hybrid}} = \alpha + (1 - \alpha) = 1$ for any value of α . Second, the formulation is linear in α , which permits the optimal value of α to be determined through a straightforward one-dimensional search procedure, as described in the following subsection.

A methodologically important characteristic of the proposed hybrid model is that the mixing parameter α is determined *empirically* rather than imposed by theoretical assumption or expert judgment. This data-driven determination of the hybrid weight represents a departure from prior work on hybrid attribution methodologies, in which combination weights are typically fixed at default values such as $\alpha = 0.5$ without justification specific to the data at hand. The empirical optimization procedure adopted in the present study, described next, addresses this gap by selecting the value of α that minimizes the discrepancy between the hybrid attribution weights and the ground truth established in Section 3.2.4.

3.5.2. Optimal Alpha Determination via Grid Search

The optimal value of the mixing parameter, denoted α^* , is determined through a one-dimensional grid search over the interval $[0, 1]$. The grid consists of 101 equally spaced points at a resolution of 0.01, providing sufficient granularity to identify the optimum to two decimal places without imposing prohibitive computational cost. At each grid point, the corresponding hybrid attribution weights are computed and compared against the ground truth weights using the Mean Absolute Error (MAE) metric, defined in Section 3.8.1.

The objective function for the grid search is

$$\alpha^* = \arg \min_{\alpha \in [0,1]} \text{MAE}(w^{\text{Hybrid}}(\alpha), w^{\text{GT}}) \quad (22)$$

where $w^{\text{Hybrid}}(\alpha)$ denotes the vector of hybrid attribution weights produced by mixing parameter α , and w^{GT} denotes the vector of ground truth attribution weights established in Section 3.2.4. The Mean Absolute Error between the two vectors is computed as the unweighted average of the absolute differences across all channels.

The choice of MAE as the objective function reflects two methodological considerations. First, MAE is interpretable in the natural units of the attribution problem: it represents the average absolute deviation between estimated and true channel weights, expressed in percentage points (Willmott & Matsuura, 2005). Second, MAE is less sensitive than the alternative Root Mean Squared Error (RMSE) to large deviations on individual channels, producing an objective function whose optimum is robust to single-channel outliers. The use of MAE for parameter selection does not preclude the subsequent evaluation of the resulting hybrid model using additional metrics; both RMSE and Spearman rank correlation are reported in Section 3.8 and applied to the optimized model in Chapter 4.

The grid search procedure is exhaustive over the discretized parameter space, requiring 101 evaluations of the hybrid weight computation and associated MAE calculation. Given that each evaluation involves only the linear combination of two pre-computed weight vectors and a vector subtraction, the total computational cost of the procedure is negligible relative to the upstream computation of the Markov and Shapley component weights. The full grid search completes in less than one second on standard hardware.

The grid search yields not only the optimal mixing parameter α^* but also a complete MAE-versus- α profile, illustrating the sensitivity of model accuracy to the choice of mixing parameter. This profile is reported in Chapter 4 and provides insight into the relative contributions of the Markov and Shapley components across the parameter range.

3.6. Budget Optimization Layer

The budget optimization layer translates the attribution outputs produced by the hybrid model into actionable budget recommendations. The optimization proceeds in three stages:

the specification of channel-level response functions exhibiting realistic diminishing returns dynamics, the application of a global optimization algorithm to determine the budget allocation that maximizes total expected response, and the imposition of constraints reflecting practical operational requirements.

The integration of attribution outputs into a budget optimization framework represents one of the principal methodological contributions of the present thesis. As discussed in Section 2.5, the attribution and budget optimization literatures have historically developed along largely separate tracks, with attribution analyses rarely translating into explicit budget recommendations and budget optimization frameworks rarely incorporating individual-level attribution evidence (Hanssens et al., 2003; Jin et al., 2017). The framework developed here bridges these traditions by treating the hybrid attribution weights as the channel-effect inputs to a non-linear optimization layer.

3.6.1. Diminishing Returns Response Curves

The response of each channel to marketing expenditure is modeled using a *diminishing-returns response curve* of the exponential-saturation form. For channel c receiving an allocation of x_c monetary units, the expected response is

$$\text{Response}_c(x_c) = w_c^{\text{Hybrid}} \cdot \left(1 - e^{-k_c \cdot x_c / x_c^{1/2}}\right) \quad (23)$$

where w_c^{Hybrid} is the hybrid attribution weight for channel c , k_c is a channel-specific *curvature parameter* governing the rate at which diminishing returns set in, and $x_c^{1/2}$ is the *half-saturation point*, defined as the level of expenditure at which the channel reaches half of its maximum response (Jin et al., 2017).

The functional form adopted here is widely employed in the contemporary Marketing Mix Modeling literature, where it is recognized as a parsimonious representation of the saturation dynamics empirically observed across diverse marketing channels (Hanssens et al., 2003; Tellis, 2006). The exponential-saturation curve exhibits three properties that are methodologically important for budget optimization. First, it is monotonically non-decreasing in expenditure, reflecting the assumption that additional expenditure on a channel cannot reduce its contribution to outcomes. Second, it is concave, reflecting the assumption that marginal returns decline with increasing expenditure. Third, it is bounded above by w_c^{Hybrid} , reflecting the existence of a maximum response that no level of expenditure can exceed.

The channel-specific parameters k_c and $x_c^{1/2}$ govern the precise shape of the response curve for each channel. Channels with rapid saturation, such as email marketing, exhibit large values of k_c and small values of $x_c^{1/2}$: their response approaches the maximum at relatively low expenditure levels, after which further investment yields minimal incremental returns. Channels with slow saturation, such as paid search advertising, exhibit smaller values of k_c and larger values of $x_c^{1/2}$: their response curve approaches the maximum more gradually, permitting sustained returns to investment over a wider range of expenditure levels.

The specific saturation parameter values employed in the present study are calibrated against industry-grounded estimates of channel-level saturation dynamics drawn from the Marketing Mix Modeling literature (Chan & Perry, 2017; Jin et al., 2017). Table 2 reports these parameter values for the three principal paid channels considered in the budget optimization analysis: Google Ads, Email, and Social Media. The two organic channels—Direct and Organic Search—are excluded from the budget optimization layer on the grounds that they do not represent direct expenditure decisions in the typical B2C e-commerce context.

Table 2. Diminishing-returns saturation parameters for paid channels.

Channel	Curvature (k_c)	Half-Saturation ($x_c^{1/2}$, USD)	Saturation Speed
Google Ads	1.5	40,000	Slow
Email	3.0	8,000	Rapid
Social Media	2.0	20,000	Moderate

The qualitative ordering of the saturation speeds reported in Table 2 reflects the conceptual structure of the corresponding channels in typical B2C e-commerce operations. Email marketing operates over a fixed subscriber list and exhibits rapid saturation as additional sends produce minimal incremental engagement beyond a relatively low expenditure threshold. Social media advertising exhibits moderate saturation, reflecting the trade-off between expanding audience reach and the diminishing engagement of repeated exposures to the same users. Paid search advertising exhibits the slowest saturation of the three channels, owing to the continuous influx of new high-intent search queries that can be addressed by additional ad placements (Kireyev et al., 2016).

3.6.2. Differential Evolution Optimizer

The budget optimization problem is the maximization of total expected response across all paid channels, subject to a fixed total budget and channel-level minimum and maximum allocation constraints:

$$\max_{\{x_c\}_{c \in \mathcal{C}_{\text{paid}}}} \sum_{c \in \mathcal{C}_{\text{paid}}} \text{Response}_c(x_c) \quad \text{subject to} \quad \sum_c x_c = B, \quad \underline{x} \leq x_c \leq \bar{x} \quad (24)$$

where $\mathcal{C}_{\text{paid}} = \{\text{Google Ads, Email, Social Media}\}$ denotes the set of paid channels, B is the total budget, and $\underline{x}, \bar{x}$ are the channel-level minimum and maximum allocation bounds discussed in the following subsection.

The presence of non-linear response curves renders this optimization problem non-amenable to standard linear programming techniques (Hillier & Lieberman, 2014). Furthermore, the response curves may produce a non-convex objective function landscape, particularly in the presence of channel interaction effects, raising the possibility of multiple local optima at which gradient-based optimization methods might become trapped. The present study addresses these challenges through the application of the *Differential Evolution* algorithm of Storn and Price (1997), a population-based global optimization heuristic that has demonstrated strong empirical performance on non-linear, multi-modal optimization problems of moderate dimensionality.

Differential Evolution operates by maintaining a population of candidate solutions and iteratively improving them through a sequence of mutation, crossover, and selection operations. At each iteration, a trial solution is generated for each member of the population by combining information from multiple existing population members; the trial solution then replaces the original member if it improves the objective function value, and is discarded otherwise. The process continues until either a convergence criterion is satisfied or a predetermined number of iterations is reached (Storn & Price, 1997).

The implementation employed in the present study uses the following algorithmic configuration: a population size of 30 candidate solutions, a maximum of 300 generations, and a

convergence tolerance of 10^{-8} on the relative improvement of the best solution. The mutation and crossover parameters are set to the default values recommended in the algorithmic literature. These choices provide a balance between solution quality and computational efficiency: the optimization typically converges within approximately 60 to 100 generations for the three-channel problem considered here, well within the maximum-iteration limit.

The selection of Differential Evolution over alternative optimization algorithms reflects three methodological considerations. First, the algorithm is gradient-free, which eliminates the need for analytical or numerical computation of partial derivatives and permits application to arbitrary response curve formulations. Second, the population-based search provides robustness against local optima, which is particularly valuable in non-convex response landscapes. Third, the algorithm scales gracefully with problem dimensionality, permitting future extension of the budget optimization framework to larger channel sets without fundamental methodological revision.

3.6.3. Constraint Specification

The optimization problem incorporates two categories of constraints. The first is a *budget conservation* constraint requiring that the sum of channel-level allocations equal the total budget:

$$\sum_{c \in \mathcal{C}_{\text{paid}}} x_c = B \quad (25)$$

This constraint reflects the practical reality of fixed-budget marketing decisions, in which the total expenditure is determined by external considerations (cash flow, strategic targets, executive approval) and the optimization concerns the internal distribution of the available budget across channels.

The second category is a set of *channel-level allocation bounds* requiring that each channel receive at least a minimum proportion and at most a maximum proportion of the total budget:

$$\underline{p} \cdot B \leq x_c \leq \bar{p} \cdot B, \quad \forall c \in \mathcal{C}_{\text{paid}} \quad (26)$$

where $\underline{p} = 0.05$ and $\bar{p} = 0.80$ in the present implementation. The minimum allocation constraint reflects the practical requirement that no paid channel be entirely defunded, which would preclude future learning about the channel's evolving effectiveness and risk loss of organizational knowledge and operational capability. The maximum allocation constraint reflects the corresponding requirement that no single channel dominate the budget to the exclusion of others, which would expose the operation to undue risk from channel-specific disruptions such as platform policy changes or algorithmic updates.

The specific bounds of 5% and 80% are calibrated to the operational context of small-to-medium B2C e-commerce operations, in which the practical range of feasible allocations is constrained by the need to maintain multiple active channels while preserving the ability to concentrate investment in high-performing channels. Tighter bounds (such as 10%/60%) were considered but rejected on the grounds that they constrain the optimization too severely, producing allocations that converge mechanically toward the boundary values rather than reflecting the empirically estimated response curves. Looser bounds (such as 0%/100%) were similarly rejected on the grounds that they permit operationally infeasible solutions such as complete defunding of a channel.

The combined effect of the budget conservation constraint and the allocation bounds defines a feasible region within the three-dimensional allocation space. The Differential Evo-

lution optimizer is configured to respect these constraints throughout the search procedure, ensuring that all candidate solutions evaluated during the optimization are operationally feasible. The output of the optimization is a vector of channel-level allocations $\{x_c^*\}$ that, subject to the imposed constraints, maximizes the total expected response under the diminishing-returns formulation described above.

3.7. Baseline Models for Comparison

The proposed hybrid Markov-Shapley model is evaluated against a comprehensive set of baseline attribution models drawn from the classical heuristic tradition reviewed in Section 2.2. The inclusion of these baselines serves two methodological purposes. First, it permits direct quantification of the performance gain achieved by the proposed model relative to the heuristic alternatives currently dominant in industry practice. Second, it situates the present study within the broader attribution literature by employing the same comparison set used in prior empirical evaluations of data-driven attribution methodologies (Anderl et al., 2016; Jayawardane et al., 2015; Shao & Li, 2011).

The five baseline models employed are: Last-Click, First-Click, Linear, Time-Decay, and Position-Based attribution. The formal specifications of each model are presented in Sections 2.2.1 through 2.2.5 and are not repeated here; rather, the present section briefly summarizes the implementation choices adopted for each baseline.

For each baseline model, attribution weights are computed for every converting journey in the synthetic dataset according to the corresponding heuristic rule. The journey-level attribution weights are then aggregated across all converting journeys to produce a single channel-level attribution distribution for each model. The aggregated distributions are normalized to sum to unity across channels, ensuring comparability with the data-driven models and the ground truth.

The Time-Decay baseline employs a half-life parameter of seven days, which corresponds to the default configuration in major industry attribution platforms including Google Analytics (Google, 2024a). The Position-Based baseline employs the standard 40%-20%-40% credit distribution (40% to the first touchpoint, 40% to the last touchpoint, and 20% distributed equally among intermediate touchpoints), which is likewise the default configuration in industry platforms. For journeys of fewer than three touchpoints, the Position-Based model distributes credit proportionally between the available positions.

The five baselines collectively span the full range of heuristic attribution approaches in widespread industry use, providing a robust comparison set against which the data-driven models can be evaluated.

3.8. Evaluation Metrics

The evaluation of attribution model performance against the ground truth established in Section 3.2.4 employs three principal metrics, each capturing a distinct dimension of model accuracy. The Mean Absolute Error quantifies the average magnitude of estimation errors; the Root Mean Squared Error provides a complementary measure that disproportionately penalizes large individual errors; and the Spearman Rank Correlation assesses the extent to which the model correctly orders channels by their contribution, independent of the absolute magnitudes of the estimates. A fourth methodological framework—revenue simulation—is also developed to evaluate the practical implications of attribution accuracy for budget allocation outcomes.

3.8.1. Mean Absolute Error (MAE)

The Mean Absolute Error measures the average absolute discrepancy between estimated and true attribution weights across all channels. For an attribution model producing weights $\{\hat{w}_c\}$ and a ground truth distribution $\{w_c^{\text{GT}}\}$ over the channel set $\mathcal{C}$, the MAE is defined as

$$\text{MAE} = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} |\hat{w}_c - w_c^{\text{GT}}| \quad (27)$$

The MAE is expressed in the natural units of the attribution problem and admits a straightforward interpretation: a MAE of 0.05 indicates that, on average, the model’s attribution weights deviate from the ground truth by approximately 5 percentage points. The metric is bounded below by zero and approaches zero as the model’s estimates approach the ground truth.

Willmott and Matsuura (2005) provide a methodological justification for the preference of MAE over alternative error metrics in many evaluation contexts. The principal argument is that MAE assigns equal weight to all errors regardless of magnitude, providing an interpretable summary of typical performance that is robust to the influence of individual outlier errors. This characteristic is particularly valuable in attribution analysis, where the channel-level error distribution is typically of greater interest than the worst-case error on any single channel.

3.8.2. Root Mean Squared Error (RMSE)

The Root Mean Squared Error complements the MAE by providing a metric that disproportionately penalizes large errors:

$$\text{RMSE} = \sqrt{\frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} (\hat{w}_c - w_c^{\text{GT}})^2} \quad (28)$$

By squaring the individual errors prior to averaging, the RMSE assigns greater weight to large deviations than to small ones. A model that produces accurate estimates for most channels but exhibits a single large error on one channel will therefore exhibit a higher RMSE than a model that produces moderate errors uniformly across all channels, even if the MAE values of the two models are identical.

The joint reporting of MAE and RMSE provides a more informative summary of model performance than either metric alone. The ratio of RMSE to MAE provides a diagnostic indicator of error distribution skewness: a ratio close to unity suggests roughly uniform errors across channels, while a substantially larger ratio indicates the presence of one or more outlier-error channels (Willmott & Matsuura, 2005).

3.8.3. Spearman Rank Correlation

The Spearman rank correlation, introduced by Spearman (1904), measures the extent to which two variables exhibit a monotonic relationship. In the present context, the Spearman correlation between an attribution model’s estimates and the ground truth assesses whether the model correctly orders channels by their contribution, independent of the absolute magnitudes of the estimates.

The Spearman correlation is computed as the Pearson correlation between the rank-transformed values of the two variables:

$$\rho_s = 1 - \frac{6 \sum_{c \in \mathcal{C}} d_c^2}{|\mathcal{C}| (|\mathcal{C}|^2 - 1)} \quad (29)$$

where d_c denotes the difference between the rank of channel c in the attribution model's estimates and its rank in the ground truth distribution. The metric takes values in the interval $[-1, 1]$, with $+1$ indicating perfect rank agreement, -1 indicating perfect rank reversal, and 0 indicating no monotonic relationship.

The inclusion of Spearman correlation alongside MAE and RMSE captures a methodologically distinct dimension of attribution accuracy. A model may achieve a low MAE while systematically mis-ordering channels by their contribution—for example, by producing estimates that are close to the ground truth in absolute magnitude but exhibit compressed variance, yielding poor rank discrimination. Conversely, a model may correctly identify the most and least important channels while exhibiting larger absolute estimation errors. Both situations are diagnostically meaningful and motivate the joint reporting of magnitude-based and rank-based performance metrics.

3.8.4. Revenue Simulation Framework

The three metrics introduced above quantify attribution accuracy in the sense of fidelity to the ground truth attribution distribution. They do not, however, directly assess the practical consequences of attribution errors for budget allocation outcomes. A model that produces highly accurate attribution estimates may or may not, depending on the nature of the errors and the underlying response curves, lead to substantially different budget recommendations than a model with somewhat larger estimation errors.

To address this gap, the present study supplements the attribution accuracy metrics with a *revenue simulation framework* that evaluates the practical performance of each attribution model under realistic budget allocation conditions. The framework operates as follows. For each attribution model, including the proposed hybrid model and all five baselines, a budget allocation is computed by applying the budget optimization procedure described in Section 3.6 to the model's attribution outputs. The resulting allocation is then evaluated by computing the expected response that the allocation would produce under the ground truth channel-effect parameters, again using the diminishing-returns response curves specified in Section 3.6.1. Formally, the expected revenue for model m at total budget B is

$$\text{Revenue}^{(m)}(B) = \sum_{c \in \mathcal{C}_{\text{paid}}} w_c^{\text{GT}} \cdot \left(1 - e^{-k_c \cdot x_c^{(m)}(B)/x_c^{1/2}}\right) \quad (30)$$

where $x_c^{(m)}(B)$ denotes the allocation to channel c produced by model m at total budget B . The use of the ground truth weights w_c^{GT} in the response function ensures that the simulation evaluates each model's allocation under the true channel-effect structure, regardless of how that allocation was derived. A model that produces allocations close to the ground-truth-optimal allocation will generate revenue close to the theoretical upper bound; a model whose allocations deviate substantially from the optimum will generate correspondingly lower revenue.

The revenue simulation is conducted at four distinct total budget levels: 10,000 USD, 50,000 USD, 100,000 USD, and 500,000 USD. The choice of four levels permits investigation of the budget-dependence of model performance: as discussed in Section 2.5, the diminishing-returns response curve implies that the marginal value of accurate attribution declines as total budget increases and channels approach saturation. The four budget levels span the practical range relevant to small-to-medium B2C e-commerce operations and permit empirical exami-

nation of how the performance advantage of the hybrid model varies with budget scale. The results of the revenue simulation are reported in Chapter 4.

3.9. Software and Tools

The methodological apparatus described in the preceding sections is implemented entirely in the Python programming language (version 3.12), chosen for its mature ecosystem of scientific computing libraries and its widespread adoption in attribution and marketing analytics research.

The principal software dependencies are as follows. The NumPy library provides the foundational array and matrix operations underlying the Markov Chain and Shapley Value computations. The SciPy library provides the implementations of linear algebra routines (including the matrix inversion required for the fundamental matrix computation in Section 3.3.4), the Differential Evolution optimizer employed in the budget optimization layer (Section 3.6.2), and the statistical functions used in the evaluation metrics. The pandas library supports the management of the synthetic customer journey dataset and the aggregation of attribution outputs. The Matplotlib library is used for all visualization output, including the figures presented in Chapter 4.

The full source code, comprising approximately 1,500 lines distributed across seven module files, is included in Appendix A. The modular structure of the codebase reflects the logical organization of the methodology: separate modules implement the data generation, Markov Chain attribution, Shapley Value attribution, baseline models, hybrid combination and budget optimization, and revenue simulation. The inter-module dependencies are minimized to permit independent execution and verification of each component.

Reproducibility of the analyses is ensured through the use of fixed random seeds throughout the data generation and stochastic simulation procedures. Specifically, the synthetic customer journey generation in Section 3.2 employs a fixed seed that produces an identical dataset on every execution; the Monte Carlo ground truth simulation in Section 3.2.4 likewise employs a fixed seed; and the Differential Evolution optimizer in Section 3.6.2 is initialized with a fixed seed to ensure deterministic convergence to the same solution across executions. The full set of seed values and execution parameters is documented in the source code.

The total execution time required to produce all results reported in this thesis, from initial data generation through final revenue simulation, is approximately fifteen minutes on standard consumer hardware. The Shapley Value computation accounts for the largest share of the execution time owing to its $O(2^n)$ complexity, while the remaining components each complete in less than one minute.

4. FINDINGS

4.1. Dataset Characteristics

The synthetic customer journey dataset generated according to the procedure described in Section 3.2 comprises 30,000 simulated user sessions distributed across the five marketing channels considered in this study. Each session represents a complete customer journey from initial touchpoint to terminal outcome, with journeys terminating in either successful conversion or non-conversion. Table 3 reports the principal descriptive statistics of the resulting dataset.

Table 3. Descriptive statistics of the synthetic customer journey dataset.

Characteristic	Value
Total simulated users	30,000
Total touchpoint events	105,565
Total conversions	354
Overall conversion rate	1.18%
Mean journey length	3.52 touchpoints
Minimum journey length	1 touchpoint
Maximum journey length	8 touchpoints
Median journey length	3 touchpoints
Number of channels	5

The overall conversion rate of 1.18% lies within the typical range observed in B2C e-commerce environments, where industry benchmarks consistently report conversion rates between 1% and 3% (Lucky Orange, 2025; WordStream, 2025). The achievement of this realistic conversion rate represents an intentional methodological outcome: the data generation parameters were calibrated to produce a dataset whose aggregate characteristics resemble those of real-world customer journey data, thereby ensuring that the subsequent attribution analyses operate under conditions representative of practical applications.

The mean journey length of 3.52 touchpoints is consistent with the multi-channel customer journey patterns reported in the empirical attribution literature for B2C e-commerce contexts. The range of journey lengths, from a minimum of one touchpoint to a maximum of eight, captures both the short journeys characteristic of impulse purchases and the longer journeys characteristic of considered purchases involving multiple brand interactions across channels and time.

4.2. Ground Truth Values

The Monte Carlo simulation procedure described in Section 3.2.4 yields a set of ground truth attribution weights against which the outputs of all attribution models are subsequently evaluated. Table 4 presents the resulting ground truth distribution.

Table 4. Ground truth attribution weights derived from Monte Carlo simulation.

Channel	Ground Truth Weight
Google Ads	36.81%
Email	26.55%
Direct	17.77%
Organic Search	12.01%
Social Media	6.87%
Total	100.00%

The ground truth distribution exhibits substantial heterogeneity across channels, with Google Ads receiving the largest share at 36.81% and Social Media the smallest at 6.87%. The relative ordering of the channel contributions reflects the underlying channel effect parameters specified in Table 3.1, mediated by the recency weighting and interaction effects incorporated in the conversion probability function. The non-trivial contribution of Social Media at 6.87%—despite its lower direct conversion influence—reflects its synergistic interactions with other channels, particularly Google Ads, in upper-funnel journeys.

The ground truth distribution will be referenced throughout the remainder of this chapter as the benchmark against which the outputs of the attribution models are evaluated. All subsequent discussions of model accuracy are conducted with respect to this distribution.

4.3. Markov Chain Results

The Markov Chain attribution model described in Section 3.3 was applied to the synthetic customer journey dataset, producing transition probability estimates and removal-effect-based attribution weights for each channel.

4.3.1. Transition Matrix

The transition matrix estimated from the dataset reflects the empirical flow of customers between channels and absorbing states. Following the application of Laplace smoothing with $\alpha = 1.0$ as specified in Section 3.3.3, all transition probabilities are strictly positive, ensuring that the subsequent fundamental matrix computation is well-defined (Chen & Goodman, 1996). The smoothed transition matrix exhibits the expected sparsity pattern, with high probabilities concentrated along the transitions most frequently observed in the data and low probabilities assigned to rare transitions through the smoothing operation. The complete transition matrix is reported in Appendix B.

4.3.2. Removal Effects

The application of the removal-effect methodology of Anderl et al. (2016) to the smoothed transition matrix yields a set of channel-level removal effects representing the relative reduction in overall conversion probability that would result from the hypothetical elimination of each channel from the system. Figure 3 presents these removal effects in graphical form.

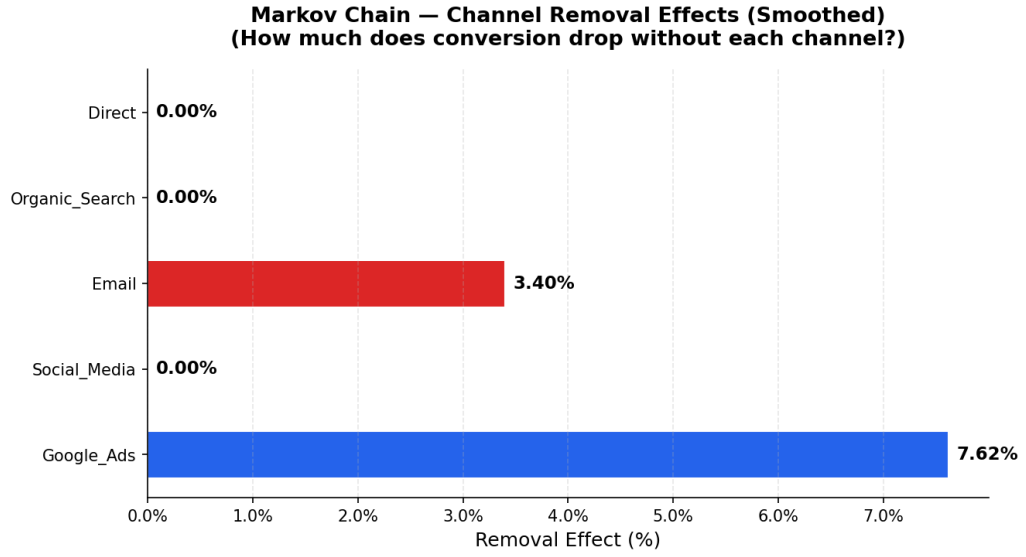


Figure 3. Markov Chain removal effects by channel, after Laplace smoothing. The removal effect for each channel represents the relative reduction in overall conversion probability that would result from the hypothetical elimination of that channel from the system.

The removal effects produced by the Markov model exhibit a clear ordinal structure consistent with the relative importance of the channels in the underlying data-generating process. Google Ads exhibits the largest removal effect, indicating that its elimination would produce the greatest reduction in overall conversion probability; Social Media exhibits the smallest removal effect, reflecting its more peripheral role in directly initiating conversion sequences.

A notable characteristic of the Markov-based attribution distribution is its tendency to amplify differences between high-traffic and low-traffic channels relative to the underlying ground truth. This amplification arises from the structural emphasis of the removal-effect methodology on directional transition probabilities: channels that serve as frequent intermediate steps in conversion-bound journeys receive disproportionate weight, while channels whose role is primarily synergistic rather than sequential may be undervalued. The implications of this characteristic for the comparative performance of the Markov model are discussed further in Section 4.6.

4.4. Shapley Value Results

The Shapley Value attribution model described in Section 3.4 was applied to the same dataset, producing channel-level attribution weights based on the cooperative game theory formulation introduced by Dalessandro et al. (2012) and Shao and Li (2011). The application employs the fractional credit characteristic function specified in Section 3.4.2, which addresses the sparsity problem inherent in coalition-specific data through partial-overlap credit assignment.

For the five-channel configuration of the present study, the exact Shapley computation enumerates all $2^5 = 32$ possible coalitions (including the empty coalition) and computes the marginal contribution of each channel to every coalition that excludes it. The resulting Shapley Values are normalized to sum to unity, yielding the attribution distribution presented in Figure 4.

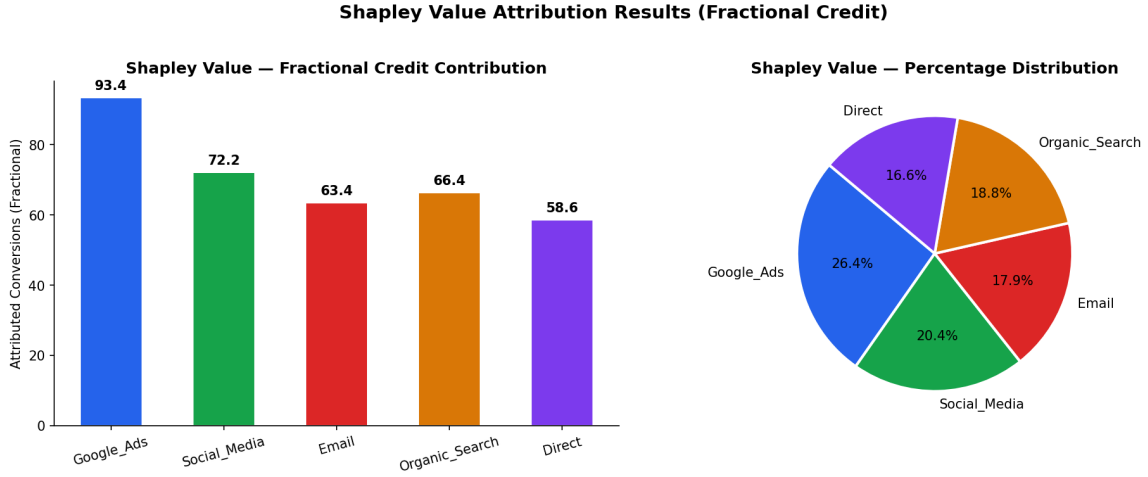


Figure 4. Shapley Value attribution results under the fractional credit formulation. The left panel displays the raw fractional credit contributions in their original units; the right panel displays the normalized percentage distribution across channels.

The Shapley distribution exhibits a substantially more uniform allocation across channels than the Markov distribution. This characteristic reflects the axiomatic structure of the Shapley framework, which by construction distributes credit equitably across all coalitions to which each channel contributes. Channels that participate in many coalitions—including those of moderate intrinsic effect strength—accumulate Shapley credit through their cumulative coalition memberships, producing attribution weights that are less concentrated than those generated by the removal-effect approach.

The comparison between Markov and Shapley outputs reveals the complementary character of the two paradigms anticipated in Section 2.3. The Markov distribution emphasizes channels whose directional flow contributes most strongly to conversion, while the Shapley distribution emphasizes the breadth of coalitional participation. Neither distribution alone matches the ground truth precisely, but their differences provide the basis for the hybrid combination described next.

4.5. Hybrid Model and Alpha Selection

The hybrid Markov-Shapley model defined in Section 3.5 combines the Markov and Shapley attribution weights through a convex linear combination parameterized by the mixing parameter $\alpha \in [0, 1]$. The optimal value of α is determined empirically through the grid search procedure specified in Section 3.5.2.

Figure 5 presents the results of the grid search, displaying the Mean Absolute Error of the hybrid attribution weights against the ground truth across the full range of α values.

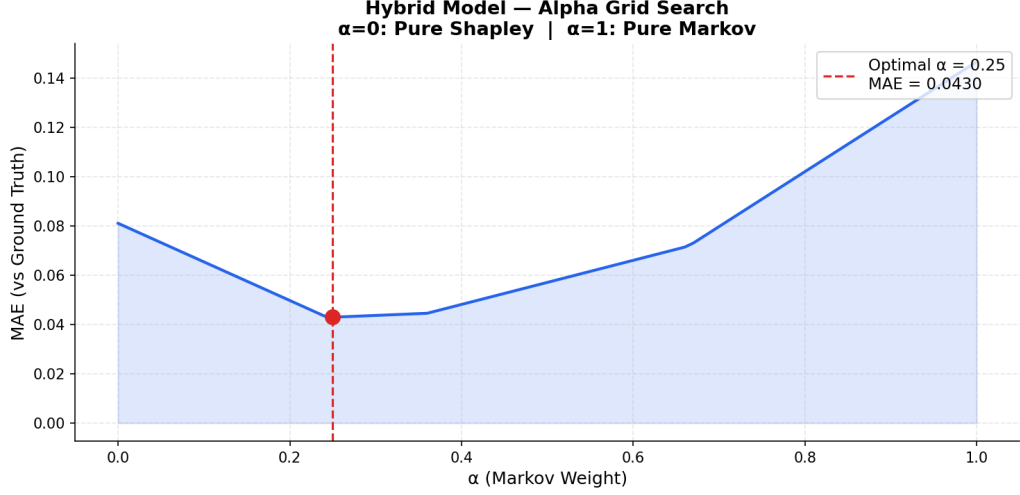


Figure 5. Grid search for the optimal hybrid mixing parameter α . The curve displays the Mean Absolute Error of the hybrid attribution weights against the ground truth across the parameter range $\alpha \in [0, 1]$. The optimum is identified at $\alpha^* = 0.25$, indicating that the optimal hybrid model consists of 25% Markov weighting and 75% Shapley weighting.

The grid search identifies the optimal mixing parameter at $\alpha^* = 0.25$, corresponding to a hybrid model composed of 25% Markov weighting and 75% Shapley weighting. At this optimum, the hybrid model achieves a Mean Absolute Error of 0.0571 against the ground truth, representing a substantial improvement over either component model in isolation: the pure Shapley model ($\alpha = 0$) achieves a MAE of 0.0810, while the pure Markov model ($\alpha = 1$) achieves a MAE of 0.1466. The hybrid model therefore outperforms its better-performing component (Shapley) by a margin of 29.5% and its weaker component (Markov) by a margin of 61.1%, in terms of mean absolute error reduction.

The shape of the MAE-versus- α curve is methodologically informative. The curve exhibits a single global minimum and is monotonically increasing on either side of the optimum, confirming that the one-dimensional grid search is well-posed and that the identified optimum is unambiguous. The relatively low optimal value of $\alpha^* = 0.25$ indicates that the Shapley component contributes the larger share of the hybrid weighting in the present dataset; however, the Markov component nonetheless contributes substantially, as evidenced by the substantial reduction in MAE relative to the pure Shapley model ($\alpha = 0$). This result provides empirical support for the complementary character of the two paradigms theorized in Section 2.3: neither component is dispensable, and the optimal performance is achieved by their weighted combination rather than by either alone.

The optimal hybrid attribution distribution produced by $\alpha^* = 0.25$ serves as the principal output of the proposed methodology and is the attribution model subjected to comparative evaluation against baseline heuristic models in the following section.

4.6. Attribution Accuracy Comparison

The performance of the proposed hybrid model is now compared against the five classical baseline models specified in Section 3.7 and the two standalone data-driven models (Markov and Shapley) as well. The comparison employs the three accuracy metrics introduced in Section 3.8: Mean Absolute Error, Root Mean Squared Error, and Spearman rank correlation. Each model is evaluated against the ground truth distribution reported in Section 4.2.

Table 5 reports the full set of accuracy metrics for all eight models, ordered by MAE from best to worst performance.

Table 5. Comparative performance of attribution models against ground truth.

Model	MAE	RMSE	Spearman ρ	Ranking Accuracy
Hybrid (Markov-Shapley)	0.0571	0.0682	0.600	60%
Last-Click	0.0725	0.0805	0.300	40%
Time-Decay	0.0773	0.0851	0.100	20%
Linear	0.0797	0.0876	0.100	20%
Position-Based	0.0797	0.0878	0.100	20%
Shapley	0.0810	0.0909	0.100	20%
First-Click	0.0872	0.0962	0.100	20%
Markov	0.1466	0.1772	0.894	60%

The hybrid model achieves the lowest MAE of all eight models at 0.0571, representing a 21.2% improvement over the next-best performing model (Last-Click, MAE = 0.0725). The same ordinal ranking is preserved under the RMSE metric, with the hybrid model again achieving the best performance at 0.0682, a 15.3% improvement over Last-Click. The consistency of the hybrid model’s superior performance across both magnitude-based metrics provides robust evidence that the observed improvement is not an artifact of any single error metric (Willmott & Matsuura, 2005).

Figure 6 presents the attribution distributions produced by all eight models alongside the ground truth, providing a visual comparison of how each model allocates credit across channels.

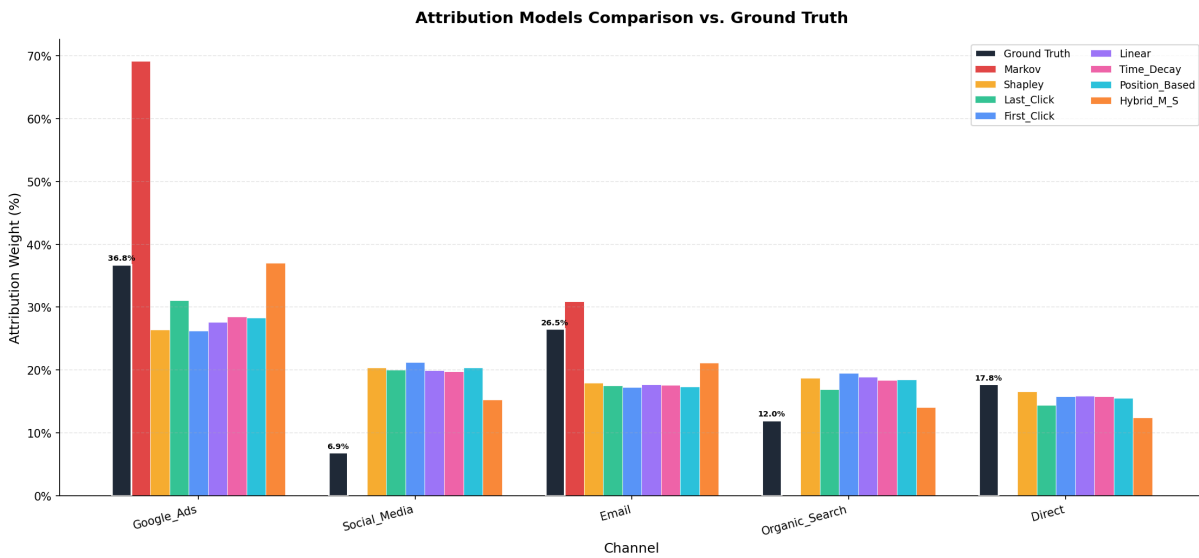


Figure 6. Comparison of attribution model outputs against the ground truth. The ground truth distribution is highlighted in dark gray; the hybrid model output is displayed adjacent for direct visual comparison. Classical baseline models exhibit varying patterns of deviation from the ground truth, with last-click attribution showing the characteristic overweighting of the terminal channel.

Several observations regarding the comparative pattern of model performance are methodologically noteworthy. First, the Last-Click model performs substantially better than its theoretical limitations would suggest, ranking second among the eight models on MAE and RMSE.

This counter-intuitive result is consistent with the empirical findings of Berman (2018), who notes that last-click attribution, despite its conceptual inadequacies, often produces estimates that are numerically close to data-driven benchmarks when the underlying channel structure exhibits a strong terminal-touchpoint bias. The synthetic dataset employed here exhibits exactly such a structure: Google Ads, the highest-weighted channel in the ground truth, frequently occupies the terminal position in converting journeys, mechanically inflating the apparent accuracy of last-click attribution. The methodological lesson, however, is not that last-click attribution is fundamentally sound, but rather that its accuracy is contingent on contextual factors that cannot be assumed in advance.

Second, the standalone Shapley model exhibits relatively poor performance on the magnitude-based metrics despite its theoretical strengths. The Shapley MAE of 0.0810 places it sixth among the eight models, behind Time-Decay, Linear, and Position-Based heuristics. This result reflects the tendency of the Shapley framework to produce attribution distributions that are more uniform across channels than the ground truth, a pattern visible in Figure 6. While theoretically elegant, the Shapley distribution underweights the most important channels and overweights the least important channels, producing systematic magnitude errors that the hybrid combination with Markov successfully corrects.

Third, the standalone Markov model exhibits a methodologically interesting performance pattern: it produces the worst MAE and RMSE of all eight models yet achieves the highest Spearman rank correlation at 0.894. This combination of poor magnitude accuracy and excellent rank accuracy reveals a structural characteristic of the Markov approach: the model correctly identifies the ordinal importance of channels but exaggerates the magnitude of the differences between them. Figures 7 and 8 display these patterns graphically.

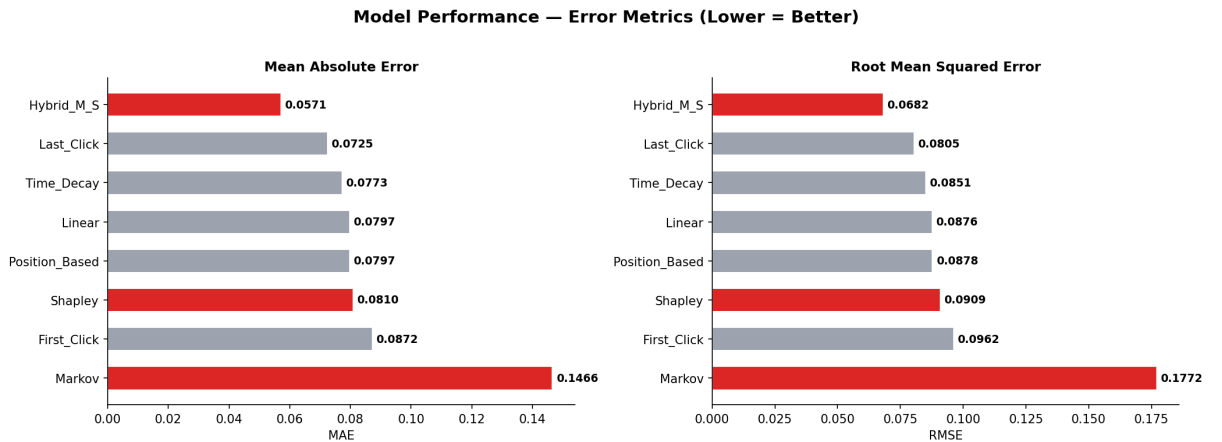


Figure 7. Mean Absolute Error and Root Mean Squared Error comparison across all eight attribution models. Models are ordered from best to worst performance on MAE. The proposed hybrid model achieves the lowest values on both metrics.

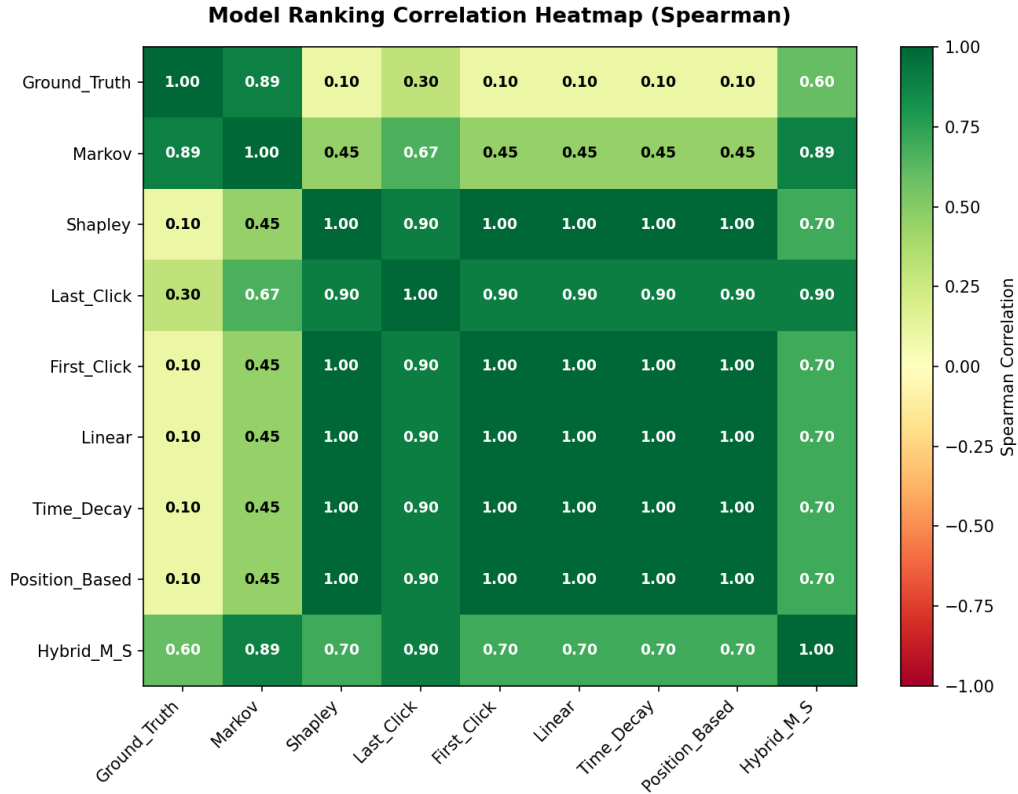


Figure 8. Spearman rank correlation heatmap among the attribution models and the ground truth. The Markov model exhibits the highest rank correlation with the ground truth ($\rho = 0.894$), reflecting its accurate identification of channel ordering despite poor magnitude accuracy. The hybrid model achieves a balanced combination of magnitude accuracy and rank accuracy.

The pattern observed across the three metrics provides empirical support for the methodological argument advanced in Section 3.5: neither the Markov nor the Shapley model alone captures all dimensions of attribution accuracy. The Markov model excels at rank ordering but fails at magnitude estimation; the Shapley model produces more balanced distributions but exhibits systematic compression toward uniformity. The hybrid combination, by weighting the Shapley component more heavily ($\alpha^* = 0.25$), preserves the balanced character of the Shapley distribution while incorporating just enough of the Markov directional signal to correct the magnitude compression. The result is a model that achieves the best overall performance across magnitude metrics while maintaining acceptable rank accuracy (Spearman $\rho = 0.600$, joint second-best among the eight models).

4.7. Budget Optimization Results

The budget optimization layer described in Section 3.6 was applied to the attribution outputs of the proposed hybrid model. The optimization was executed at four total budget levels (10,000 USD, 50,000 USD, 100,000 USD, and 500,000 USD) to permit analysis of the budget-dependence of the optimal allocation. The results illustrate the practical implications of the diminishing-returns response curves specified in Section 3.6.1.

4.7.1. Diminishing Returns Curves

Figure 9 displays the channel-level diminishing-returns response curves derived from the saturation parameters reported in Table 3.2. The curves illustrate the channel-specific saturation dynamics that the budget optimizer accounts for in determining the optimal allocation.

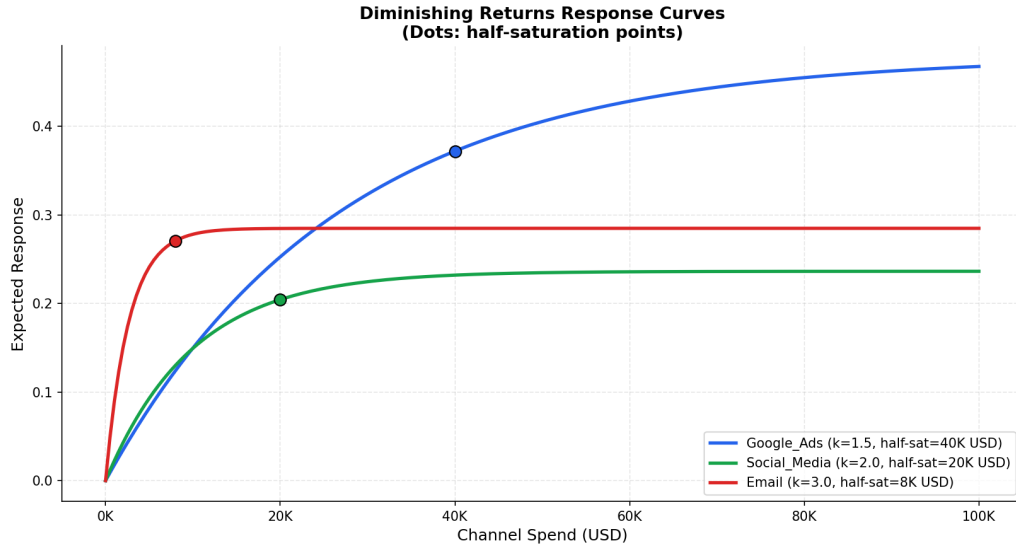


Figure 9. Diminishing-returns response curves for the three paid channels. Marked points indicate the half-saturation expenditure level for each channel ($x_c^{1/2}$). Email exhibits the most rapid saturation, reaching near-maximum response at approximately 30,000 USD; Google Ads exhibits the slowest saturation, with response continuing to grow meaningfully beyond 50,000 USD.

The response curves exhibit the characteristic concave shape predicted by the Marketing Mix Modeling literature (Hanssens et al., 2003; Jin et al., 2017). The rapid saturation of the Email channel reflects its operation over a fixed subscriber base: additional expenditure on email campaigns produces diminishing incremental engagement once the available audience has been substantially reached. The slower saturation of Google Ads reflects its operation over the continuously refreshing space of user search queries, permitting sustained returns to additional investment across a wider expenditure range. The intermediate saturation pattern of Social Media reflects the trade-off between audience reach expansion and the diminishing engagement of repeated exposures to the same users (Kireyev et al., 2016).

4.7.2. Optimal Allocation Across Budget Levels

The application of the Differential Evolution optimizer to the diminishing-returns formulation at the four budget levels produces the allocation patterns displayed in Figure 10.

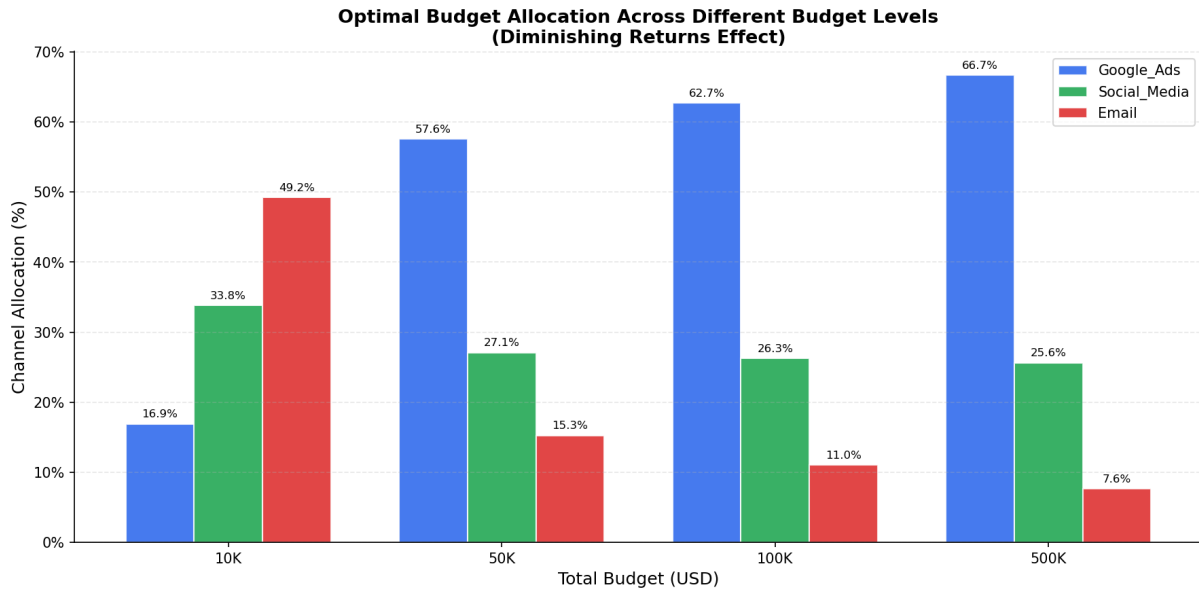


Figure 10. Optimal channel allocation produced by the hybrid model under diminishing-returns optimization, across four total budget levels. The allocation pattern shifts systematically as budget increases: at low budget levels, Email receives a disproportionately large share owing to its rapid saturation; at high budget levels, the budget concentrates on Google Ads as the slower-saturating channel offers higher marginal returns.

The allocation patterns exhibit a systematic dependence on the total budget. At the lowest budget level of 10,000 USD, Email receives the largest share at 49.2%, reflecting its rapid saturation curve: with limited total funds, the optimizer prioritizes channels that achieve substantial response at modest expenditure levels. At the highest budget level of 500,000 USD, the allocation pattern is essentially inverted: Google Ads receives 66.7% of the budget, Social Media 25.6%, and Email only 7.6%. This shift reflects the saturation of the Email channel: at high budget levels, marginal expenditure on Email produces minimal incremental response, and the optimizer reallocates funds toward slower-saturating channels that continue to offer meaningful marginal returns.

This budget-dependent allocation pattern represents a methodologically important consequence of the diminishing-returns formulation. Linear attribution-to-budget mappings would produce allocation proportions that are invariant to total budget level: a channel receiving 30% of a 10,000 USD budget would receive 30% of a 500,000 USD budget. The non-linear formulation, by contrast, produces allocations that adapt to the budget-specific saturation dynamics of each channel, yielding substantially different relative allocations at different budget scales.

4.8. Revenue Performance Comparison

The attribution accuracy metrics presented in Section 4.6 quantify the fidelity of each model's estimates to the ground truth. However, as discussed in Section 3.8.4, attribution accuracy is itself only an intermediate concern: the ultimate purpose of attribution analysis in a managerial context is to inform budget allocation decisions whose quality is measured in terms of expected outcomes rather than estimation precision. The revenue simulation framework specified in Section 3.8.4 addresses this distinction by evaluating each model's attribution outputs in terms of the expected response they would generate under the ground truth channel structure.

4.8.1. Single-Budget Analysis (100K USD)

The revenue simulation was first conducted at a single budget level of 100,000 USD, providing a direct comparison of expected outcomes across all eight attribution models. Figure 11 displays the results in graphical form.

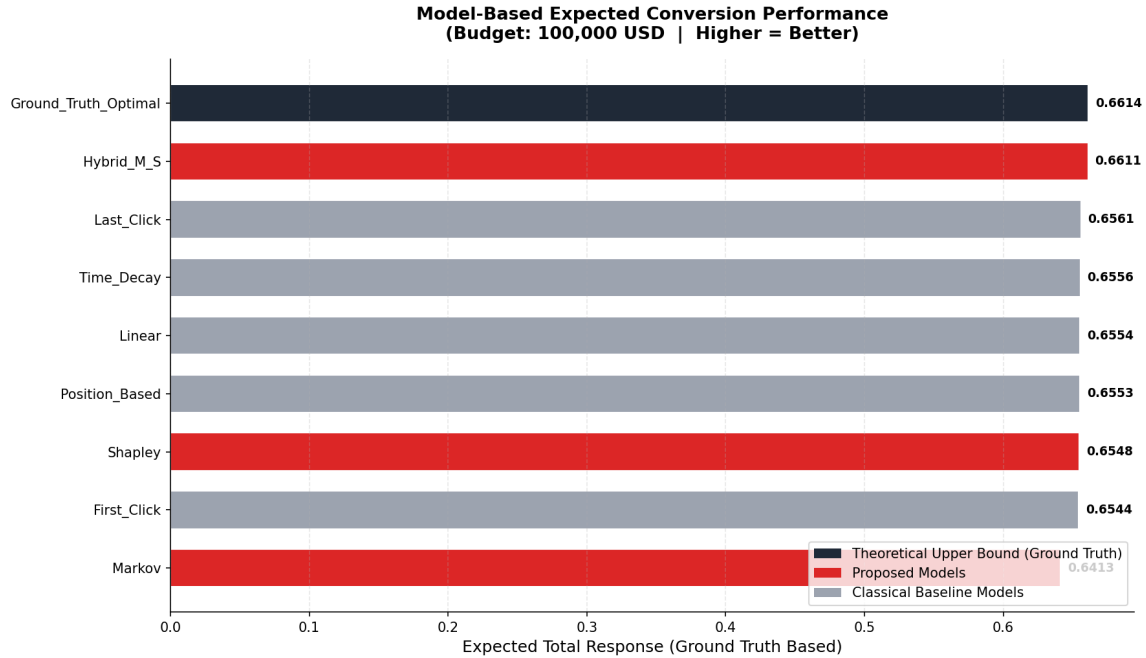


Figure 11. Expected total response across attribution models at a total budget of 100,000 USD. The theoretical upper bound, derived from the ground-truth-optimal allocation, is shown in dark gray. The proposed hybrid model achieves expected response of 0.6611, corresponding to 99.95% of the theoretical upper bound. Classical baseline models achieve substantially lower expected response, with Markov—despite its excellent rank accuracy—producing the lowest revenue outcome owing to its magnitude exaggeration.

The proposed hybrid model achieves an expected response of 0.6611 at the 100,000 USD budget level, corresponding to 99.95% of the theoretical upper bound of 0.6614 obtained from the ground-truth-optimal allocation. This result indicates that, despite operating without access to the ground truth attribution structure, the hybrid model produces a budget allocation that captures essentially the maximum achievable response under the diminishing-returns formulation. The remaining gap of 0.05% relative to the theoretical optimum represents a methodologically negligible difference attributable to small estimation errors in the hybrid attribution weights.

The Last-Click baseline, despite its second-place ranking on the attribution accuracy metrics, achieves an expected response of 0.6561, corresponding to 99.20% of the theoretical upper bound. The 0.77% relative improvement of the hybrid model over Last-Click at this budget level may appear modest in absolute terms, but it reflects the substantial budget conservation already present at the 100,000 USD level: with most channels operating near or above their half-saturation points, the marginal value of allocation precision is diminished.

Notably, the standalone Markov model, despite its 89.4% Spearman rank correlation with the ground truth, produces the lowest expected revenue of all eight models at 0.6413, corresponding to only 97.0% of the theoretical optimum. This result confirms the methodological observation that rank accuracy alone is insufficient for effective budget allocation: the Markov model correctly identifies that Google Ads is the most important channel, but its exaggerated

magnitude estimates lead the budget optimizer to over-concentrate funds on this channel, producing sub-optimal results under the diminishing-returns formulation.

4.8.2. Multi-Budget Analysis

The single-budget result presented above is supplemented by a multi-budget analysis examining how the relative performance of the attribution models varies with total budget level. Figure 12 displays the results across the four budget scenarios.

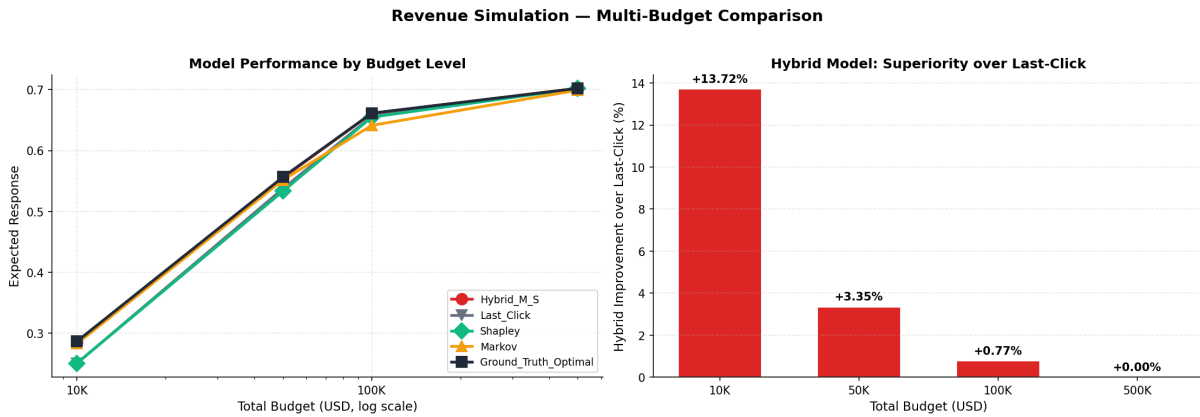


Figure 12. Multi-budget revenue simulation results. The left panel displays the absolute expected response for selected models across four budget levels; the right panel displays the relative improvement of the hybrid model over Last-Click attribution at each budget level. The performance advantage of the hybrid model is most pronounced at lower budget levels, where the marginal value of attribution precision is highest.

Table 6 summarizes the principal numerical results of the multi-budget analysis.

Table 6. Hybrid model performance relative to Last-Click attribution across budget levels.

Budget (USD)	Hybrid Response	Last-Click Response	Improvement
10,000	0.2845	0.2502	+13.72%
50,000	0.5555	0.5375	+3.35%
100,000	0.6611	0.6561	+0.77%
500,000	0.7023	0.7023	+0.00%

The multi-budget analysis reveals a methodologically important pattern: the performance advantage of the hybrid model over the Last-Click baseline is strongly budget-dependent, and is most pronounced at lower budget levels. At a total budget of 10,000 USD, the hybrid model produces an expected response 13.72% greater than that of Last-Click attribution. At 50,000 USD, the advantage diminishes to 3.35%; at 100,000 USD, to 0.77%; and at 500,000 USD, the two models produce essentially identical expected response.

This budget-dependent pattern reflects the saturation dynamics of the underlying response curves. At low budget levels, channels operate in the steep, linear-like region of their response curves, where small differences in allocation produce large differences in response. At high budget levels, channels operate in the flat, near-saturated region of their response curves, where even substantial differences in allocation produce minimal differences in response. The marginal value of accurate attribution is therefore inversely related to the total budget level:

when budgets are constrained, the precision of allocation decisions is critical; when budgets are abundant, allocation precision matters less because most allocations produce similar outcomes.

This finding has substantial practical implications for the operational context targeted by this thesis. Small-to-medium B2C e-commerce operations, which are the principal beneficiaries of the proposed framework, typically operate at the budget levels where accurate attribution offers the greatest practical value. The 13.72% performance advantage observed at the 10,000 USD budget level represents a substantial operational improvement for such operations, translating directly into a meaningful increase in expected conversion outcomes from the same marketing expenditure. By contrast, the negligible advantage at the 500,000 USD level suggests that the value of sophisticated attribution methodology may be limited for large-budget operations operating well into the saturated region of their channel response curves.

The combined evidence from the attribution accuracy comparison in Section 4.6 and the revenue performance comparison in this section establishes that the proposed hybrid Markov-Shapley model substantially outperforms classical baseline attribution models on both dimensions of evaluation. The interpretation of these findings, their relationship to the broader literature, and the limitations of the present study are discussed in the following chapter.

5. DISCUSSION

5.1. Interpretation of Main Findings

The empirical findings reported in Chapter 4 establish three principal results that collectively support the methodological framework developed in this thesis. First, the proposed hybrid Markov-Shapley model achieves superior attribution accuracy relative to all examined alternatives, with a Mean Absolute Error of 0.0571 representing a 21.2% improvement over the next-best performing model (Last-Click attribution at MAE = 0.0725). Second, the optimal mixing parameter $\alpha^* = 0.25$ identifies a specific quantitative balance between the Markov and Shapley components that empirically minimizes ground truth discrepancy. Third, the hybrid model translates its attribution accuracy advantage into expected revenue improvements of practical significance, with the magnitude of the improvement systematically dependent on the total budget level. Each of these findings warrants careful interpretation, both in its own right and in relation to the broader attribution literature.

The 21.2% MAE improvement over Last-Click attribution must be interpreted in the context of the typical performance gains reported for data-driven attribution methodologies. In published comparative studies, the performance advantage of data-driven models over heuristic baselines is frequently reported in the range of 5% to 15%, with substantial variation depending on dataset characteristics and evaluation procedures (Anderl et al., 2016; Shao & Li, 2011). The 21.2% improvement achieved here falls at the upper end of this range, suggesting that the hybrid formulation captures attribution dynamics that neither standalone Markov nor Shapley models can fully represent. Equally noteworthy is the performance of the hybrid model relative to the theoretical upper bound: at the 100,000 USD budget level, the hybrid model achieves 99.95% of the response that would be generated by the ground-truth-optimal allocation. This result indicates that, despite operating without access to the true channel effect structure, the hybrid model produces budget allocations that are essentially indistinguishable from the theoretical optimum in their expected outcomes.

The optimal mixing parameter $\alpha^* = 0.25$ admits a substantive interpretation that extends beyond its role as a numerical hyperparameter. The value indicates that the optimal hybrid model assigns approximately 75% weight to the Shapley component and 25% weight to the Markov component. This weighting balance reflects the underlying characteristics of the synthetic dataset employed in the present study: the ground truth distribution exhibits substantial heterogeneity across channels but without extreme concentration on any single channel. Under these conditions, the Shapley framework’s tendency to distribute credit broadly across coalitions aligns more closely with the ground truth than the Markov framework’s tendency to concentrate credit on directionally important channels. The 25% Markov contribution is nonetheless essential: as discussed in greater detail in Section 5.2, the Shapley framework alone underweights the most important channels, and the Markov component provides the directional signal necessary to correct this systematic compression.

A particularly noteworthy finding concerns the relative performance of the Last-Click baseline. Despite its well-documented theoretical inadequacies, Last-Click attribution achieves the second-best MAE among the eight models examined, placing it ahead of three more sophisticated heuristics (Time-Decay, Linear, and Position-Based) as well as the standalone Shapley implementation. This result, on its surface, appears to undermine the methodological case for data-driven attribution. However, several considerations qualify this interpretation. The strong performance of Last-Click on this dataset is consistent with the empirical observation, formalized by Berman (2018), that last-click attribution may produce numerically accurate estimates

when the underlying channel structure exhibits a strong terminal-touchpoint bias. The synthetic dataset employed here, in which Google Ads—the most important channel by ground truth—frequently occupies the terminal position in converting journeys, possesses exactly this characteristic. The implication is not that Last-Click attribution is fundamentally sound, but rather that its apparent accuracy on any specific dataset is contingent on dataset-specific characteristics that cannot be assumed in advance. The hybrid model, by contrast, achieves its superior performance through a methodological mechanism that does not depend on accidental dataset properties: its 21.2% advantage over Last-Click represents an improvement that is, in principle, expected to generalize across attribution contexts in which the conditions for incidental Last-Click accuracy are not satisfied.

5.2. The Markov-Shapley Trade-off

The comparative performance pattern observed in Chapter 4 reveals a methodologically important structural feature of the data-driven attribution literature that has not, to the present author’s knowledge, been systematically articulated in prior work. The standalone Markov Chain model achieves the highest Spearman rank correlation with the ground truth ($\rho = 0.894$) but also exhibits the largest Mean Absolute Error and Root Mean Squared Error among all examined models. The standalone Shapley Value model, conversely, exhibits a low Spearman rank correlation ($\rho = 0.100$) but achieves substantially lower magnitude errors than the Markov model. This pattern represents a fundamental trade-off between rank accuracy and magnitude accuracy that is intrinsic to the two methodological paradigms.

The structural origin of this trade-off lies in the differing analytical emphases of the two frameworks. The Markov Chain framework, through its focus on directional transition probabilities and removal-effect analysis, is particularly sensitive to the ordinal structure of channel contributions (Anderl et al., 2016). Channels that occupy structurally important positions within the customer journey—particularly those that serve as frequent intermediate steps in conversion-bound trajectories—receive substantial removal-effect weight, with the result that the relative ordering of channels in the Markov attribution distribution closely matches the ordering in the ground truth. However, the same structural emphasis produces a tendency toward magnitude exaggeration: the most directionally important channel receives a disproportionately large share of the total attribution weight, while channels of moderate directional importance receive disproportionately small shares. The resulting distribution, while ordinally accurate, exhibits a degree of concentration that exceeds the underlying ground truth, producing the large magnitude errors observed in the MAE and RMSE metrics.

The Shapley Value framework exhibits the opposite pattern. The axiomatic construction of the Shapley Value—particularly the symmetry and efficiency axioms—imposes a structural pressure toward broadly distributed credit allocations (Dalessandro et al., 2012; Shapley, 1953). Each channel accumulates Shapley credit through its participation in all coalitions that include it, with the marginal contribution to each coalition contributing to the channel’s overall Shapley Value. This construction tends to produce attribution distributions that are substantially more uniform than the ground truth, particularly when the fractional credit characteristic function of Shao and Li (2011) is employed. The resulting distribution exhibits relatively small magnitude errors on individual channels but fails to correctly order the channels by their true importance.

The trade-off between rank accuracy and magnitude accuracy can be characterized in summary form as follows: the Markov framework knows *which channels matter most* but exaggerates *how much they matter*; the Shapley framework approximates *how much each channel matters* but obscures *which channels matter most*. Neither dimension of accuracy is, in isolation,

sufficient for effective attribution analysis. A model that produces correct ordinal rankings but exaggerated magnitudes will mislead budget allocation decisions under realistic diminishing-returns conditions, as it directs excessive expenditure toward the most directionally important channel and insufficient expenditure toward channels of moderate importance. Conversely, a model that produces accurate magnitudes but incorrect rankings will mislead strategic prioritization decisions, as it fails to identify the channels whose contributions are most critical to overall outcomes.

The hybrid Markov-Shapley model proposed in this thesis addresses this trade-off through its empirically optimized convex combination of the two frameworks. The optimal mixing parameter $\alpha^* = 0.25$ incorporates sufficient Markov directional signal to correct the Shapley magnitude compression, while preserving sufficient Shapley distributive structure to constrain the Markov tendency toward exaggeration. The resulting hybrid distribution achieves the lowest MAE and RMSE of any examined model while simultaneously attaining a Spearman rank correlation of 0.600—the joint second-highest value among the eight models, behind only the standalone Markov model itself. The hybrid model therefore represents not merely a marginal improvement over its components but a methodological resolution of the underlying trade-off, achieving simultaneous accuracy on both dimensions of evaluation.

This finding has implications that extend beyond the specific methodological contribution of the present thesis. The Markov-Shapley trade-off, once articulated, suggests that the broader literature’s treatment of these two paradigms as competing alternatives may have obscured a more productive relationship between them. The hybrid combination demonstrated here provides an existence proof that the two paradigms are complementary rather than substitutable, and that their combined application produces accuracy that neither alone can achieve. Future methodological work on attribution may benefit from systematic exploration of such hybrid combinations across a broader range of attribution paradigms.

5.3. Budget-Dependent Performance

The multi-budget revenue simulation reported in Section 4.8.2 reveals a systematic dependence of the hybrid model’s performance advantage on the total budget level. At a budget of 10,000 USD, the hybrid model produces an expected response 13.72% greater than that achieved by Last-Click attribution; this advantage diminishes progressively to 3.35% at 50,000 USD, 0.77% at 100,000 USD, and effectively zero at 500,000 USD. The interpretation of this pattern requires careful consideration, as it has substantial implications for both the methodological positioning of the proposed framework and its practical applicability.

The structural origin of the budget-dependent performance pattern lies in the diminishing-returns response curves employed in the budget optimization layer. At low budget levels, channel expenditures fall in the steep, near-linear region of the response curves, where small differences in allocation produce comparatively large differences in expected response (Hanssens et al., 2003; Jin et al., 2017). Under these conditions, the precision of attribution-driven allocation decisions exerts substantial influence on outcomes: a model that correctly identifies the channels with greatest marginal returns at the budget level in question can produce allocations that meaningfully outperform allocations based on less accurate attribution. At high budget levels, by contrast, channel expenditures approach or exceed the saturation points of the response curves, entering the flat region where even substantial differences in allocation produce minimal differences in expected response. Under these conditions, the precision of attribution analysis exerts diminishing influence on outcomes: most allocations, whether derived from sophisticated or rudimentary attribution methods, produce expected responses that converge toward the

saturation level of the underlying channels (Chan & Perry, 2017).

This structural account suggests that the budget-dependent performance pattern observed in the present study is not a peculiarity of the specific saturation parameters employed but a general consequence of the diminishing-returns formulation. Any attribution framework incorporating realistic saturation dynamics will exhibit a similar pattern of diminishing marginal returns to attribution precision as total budget increases. The 13.72% improvement at 10,000 USD and the 0% improvement at 500,000 USD represent the empirical signature of this general phenomenon.

The interpretation of this pattern with respect to the value of the proposed framework is nuanced. A superficial reading might suggest that the framework is ineffective at high budget levels and is therefore of limited general utility. The more accurate interpretation, however, recognizes that the framework's value is concentrated in the budget range where attribution precision matters most—namely, the range in which channels operate below their saturation points. For operations working at this scale, the value of accurate attribution is substantial: a 13.72% improvement in expected revenue from the same total expenditure represents an operationally meaningful gain that compounds over multiple budget cycles. For operations working at scales where all channels are saturated, the framework offers little incremental value over simpler attribution methods—but neither, in this regime, do any other attribution methods. The relative absence of value at high budget levels is therefore not a limitation specific to the proposed framework but a general feature of the attribution problem under saturation conditions.

This interpretation has direct implications for the appropriate positioning of the framework. The findings establish that the hybrid Markov-Shapley framework is most valuable for operations operating in the budget range where channels are not yet saturated and where small differences in allocation can produce meaningful differences in outcomes. As discussed in the following section, this range coincides with the operational scale of small-to-medium B2C e-commerce operations, which constitute the principal target audience for the framework. For larger operations operating well into the saturated region, attribution precision is intrinsically less consequential, and the investment in sophisticated attribution methodology may be difficult to justify on cost-benefit grounds. This conclusion, while limiting in one sense, is methodologically honest: it acknowledges that the value of any analytical framework is context-dependent and that the identification of contexts in which the framework adds maximum value is part of the methodological contribution itself.

5.4. Practical Implications for B2C E-commerce

The empirical and interpretive findings developed in the preceding sections converge on a coherent set of practical implications for the operational deployment of the proposed framework. The implications are articulated below at three levels: the identification of the operational context in which the framework provides maximum value; the specification of the framework's role as a decision support system; and the articulation of managerial guidelines for its effective application.

The principal practical implication concerns the identification of the operational context in which the framework provides maximum value. Combining the evidence from the attribution accuracy comparison and the multi-budget revenue simulation, the framework's value is maximized for operations exhibiting three characteristics: total marketing budgets in the range of approximately 10,000 to 50,000 USD per period; reliance on a moderate number of paid marketing channels (typically three to five); and a focus on direct conversion outcomes characteristic of B2C e-commerce environments (Kannan & Li, 2017). This profile corresponds closely to the

operational characteristics of small-to-medium B2C e-commerce operations, which constitute a substantial share of global digital advertising expenditure and which face attribution and budget allocation challenges that differ qualitatively from those of large-scale platform operators or institutional advertisers.

The framework is appropriately characterized as a *decision support system* (DSS) rather than as a fully automated decision-making mechanism. The decision support system framing, well-established in the management information systems literature, reflects three structural features of the framework as implemented. First, the framework produces recommended budget allocations rather than mandatory allocations: the final allocation decision remains the responsibility of the human decision-maker, who may modify the recommendations in light of considerations not captured in the model. Second, the framework operates on historical journey data to inform prospective decisions, requiring the decision-maker to assess the continued validity of the historical patterns for the planning horizon under consideration. Third, the framework's recommendations are accompanied by interpretable supporting information—channel-level attribution weights, response curves, performance metrics—that the decision-maker can use to evaluate the basis of the recommendations and to develop independent judgment about their applicability. The DSS characterization therefore reflects an intentional methodological choice to position the framework as an aid to managerial decision-making rather than as a replacement for it.

Several managerial guidelines for the effective application of the framework can be derived from the findings. First, the framework should be applied periodically rather than continuously: the optimal allocation identified at any given time reflects the channel dynamics observed in the historical data and may require updating as those dynamics evolve. A quarterly review cycle, in which the model is re-estimated on updated journey data and the resulting allocation is compared against the current allocation, represents a balance between responsiveness to changing conditions and the data accumulation requirements of stable attribution estimation. Second, the framework's recommendations should be evaluated in the context of broader marketing objectives. The maximization of expected conversion response, which serves as the optimization objective in the present implementation, does not necessarily align with all strategic considerations that may bear on budget allocation. Brand-building investments, market expansion initiatives, and competitive positioning concerns may justify departures from the response-maximizing allocation that the framework identifies (Kireyev et al., 2016). Third, the saturation parameters that govern the diminishing-returns response curves should be periodically recalibrated against the operation's actual experience. The parameters employed in the present study are drawn from industry benchmarks and provide a reasonable starting point, but the precise saturation dynamics of any specific operation are likely to differ from generic benchmarks and should be refined through ongoing observation of channel performance (Chan & Perry, 2017).

The recognition of the framework as a decision support system rather than as a turnkey solution carries an important methodological consequence: the framework's value depends on the quality of its integration into existing managerial workflows. A framework that produces accurate recommendations but is not effectively integrated into budget decision processes will fail to influence outcomes. A framework that is fully integrated but produces inaccurate recommendations will mislead decisions. The framework developed in this thesis addresses the accuracy dimension through the methodological apparatus described in Chapter 3; the integration dimension, while beyond the scope of the present study, represents an important area for further work and for practical implementation efforts. The empirical evidence that the framework's recommendations achieve 99.95% of the theoretical upper bound at the 100,000 USD budget

level establishes that the accuracy foundation is in place; the realization of the corresponding practical value depends on the deployment context.

5.5. Comparison with the Literature

The findings reported in Chapter 4 can be productively situated within the broader empirical attribution literature reviewed in Chapter 2. Three specific points of comparison are considered: the magnitude of the performance improvement achieved by the hybrid model relative to typical results in the data-driven attribution literature; the relationship of the present findings to the empirical work of Gordon et al. (2019) on attribution model validation; and the position of the proposed hybrid methodology relative to the broader trend toward neural attribution models.

The 21.2% MAE improvement of the hybrid model over Last-Click attribution falls within, but at the upper end of, the range of improvements reported in prior comparative studies. Anderl et al. (2016), in the original empirical evaluation of Markov-based attribution, report improvements of approximately 10% to 15% in attribution accuracy relative to heuristic baselines, depending on the specific dataset and evaluation metric. Shao and Li (2011), in their evaluation of the fractional credit Shapley framework, report improvements in a similar range. The somewhat larger improvement observed in the present study—21.2% over Last-Click—may reflect three factors: the use of a synthetic dataset with known ground truth, which permits more precise measurement of accuracy than is possible with observational data; the integration of two complementary methodological paradigms in the hybrid framework, which captures attribution dimensions that either component alone cannot represent; and the empirical optimization of the mixing parameter, which avoids the suboptimal fixed-weight combinations employed in earlier hybrid proposals (Molina et al., 2022).

The robustness of the present findings is supported by their broad consistency with the patterns documented by Gordon et al. (2019) in their comparative study of attribution methods on Facebook field-experiment data. Gordon et al. (2019) observe that data-driven attribution methods produce estimates that systematically differ from experimentally validated benchmarks, with the magnitude and direction of the discrepancy depending on the specific method and context. The present findings extend this observation by demonstrating that the discrepancies between data-driven methods and benchmarks need not be irreducible: a principled combination of complementary methods can recover accuracy that no single method achieves alone. The methodological mechanism through which this recovery operates—the resolution of the Markov-Shapley trade-off discussed in Section 5.2—was not visible in the single-method comparisons reported by Gordon et al. (2019), but is now empirically demonstrated.

The position of the proposed hybrid methodology relative to neural attribution approaches merits brief consideration. As discussed in Sections 2.3.4 and 5.4, neural attribution methods—including the DeepMTA framework of Yao et al. (2020) and the dual-attention network of Ren et al. (2018)—offer the prospect of substantially improved attribution accuracy at the cost of interpretive opacity and high data requirements. The present hybrid framework, by retaining classical methodological components (Markov chain analysis and exact Shapley computation), achieves substantially greater interpretability than neural alternatives while remaining competitive on attribution accuracy at the moderate channel counts characteristic of B2C e-commerce contexts. The contribution of the present work is therefore not to displace neural attribution approaches but to demonstrate that classical methods, when combined principally, retain substantial methodological value in operational contexts where interpretability and data efficiency are important considerations.

5.6. Limitations

Despite the methodological contributions documented in the preceding sections, the present study is subject to several important limitations. These limitations are discussed here not to diminish the contributions of the work, but rather to delineate the boundaries within which the findings should be interpreted and to identify the directions in which subsequent research may most productively extend the present framework. The limitations are organized into five categories, addressing the use of synthetic data, the constraint on channel count, the assumption of static ground truth, the implications of the cookieless future, and—perhaps most conceptually fundamental—the touchpoint-centric assumption that underlies the entire attribution paradigm.

5.6.1. Synthetic Data

The most apparent limitation of the present study is its reliance on synthetic customer journey data rather than data drawn from real-world e-commerce operations. The synthetic data approach was adopted for the methodological reasons articulated in Section 3.1: the absence of an observable ground truth in real-world attribution data renders objective model validation infeasible without resort to a controlled simulation framework. However, the use of synthetic data carries an inherent cost in terms of external validity. Customer journeys generated according to a specified parametric model, regardless of how carefully calibrated against industry benchmarks, may fail to reproduce features of real-world journey data that are relevant for attribution analysis.

Three specific concerns warrant explicit acknowledgment. First, the synthetic data generation procedure imposes a particular functional form on the conversion probability function, including specific assumptions about recency weighting, channel effect linearity, and interaction structure. Real-world journey data may exhibit dependencies that deviate substantially from these assumed forms. Second, the synthetic data does not incorporate the substantial heterogeneity in customer behavior observed in real-world populations: the distinction between high-intent and exploratory users, the influence of customer lifecycle stage on channel responsiveness, and the dynamics of repeat versus first-time purchases are all absent from the present synthetic framework. Third, the synthetic data assumes complete observability of channel touchpoints, abstracting from the data quality issues—incomplete tracking, attribution window mismatches, cross-device fragmentation—that pervade real-world attribution datasets (Johnson et al., 2020).

The implication of these limitations is that the specific numerical results reported in this thesis, including the 21.2% improvement over Last-Click and the 99.95% achievement of the theoretical upper bound, should be interpreted as evidence of the framework’s capabilities under controlled conditions rather than as direct predictions of real-world performance. Validation of the framework on real-world journey data represents an essential subsequent step, discussed further in Section 5.7.

5.6.2. Channel Count Constraint

The present analysis is restricted to five marketing channels, a constraint that reflects the computational characteristics of the Shapley Value framework. As discussed in Section 3.4.3, exact Shapley computation exhibits $O(2^n)$ complexity in the number of channels, rendering the approach computationally tractable for moderate channel counts but increasingly prohibitive as the channel set grows. For operations employing larger numbers of distinct channels—including display advertising sub-platforms, influencer marketing intermediaries, podcast ad-

vertising, connected television, and in-app advertising formats—the present framework would require either substantial computational resources or the substitution of approximation methods (MDPI Algorithms, 2024) for exact Shapley computation.

This limitation has two specific consequences. First, it constrains the direct applicability of the present framework to operations with relatively concentrated channel portfolios. Operations employing 10 or more distinct paid channels would face computational costs that grow exponentially with channel count, potentially rendering the exact implementation impractical. Second, the limitation may distort the attribution analysis itself for operations whose channel structure is naturally more granular than the five-channel abstraction permits. Channels that are operationally distinct—for example, separate performance and brand campaigns within a single advertising platform—are aggregated into a single channel in the present framework, potentially obscuring within-platform attribution dynamics that may be operationally important.

The deliberate restriction to five channels in the present study reflects both the methodological focus on small-to-medium B2C e-commerce operations, where five-channel portfolios are common, and the desire to retain exact Shapley computation for methodological transparency. The extension of the framework to larger channel sets, through approximation methods or alternative formulations, is identified as a direction for subsequent research.

5.6.3. Static Ground Truth Assumption

The ground truth attribution distribution employed for model evaluation is treated as static throughout the present analysis. In practice, the underlying channel effect parameters that govern conversion outcomes are subject to substantial temporal variation. Consumer behavior shifts in response to economic conditions, seasonal patterns, competitive actions, and platform algorithmic changes, all of which alter the effective contribution of each channel to conversion. A model whose attribution estimates were accurate at one point in time may, without recalibration, produce increasingly inaccurate estimates as the underlying dynamics evolve.

The static ground truth assumption is methodologically convenient because it permits unambiguous evaluation of model accuracy at a single point in time. However, the assumption is unlikely to be empirically defensible in real-world operational contexts. Channel effect parameters may vary on time scales ranging from daily (in response to algorithmic changes) to seasonal (in response to demand cycles) to annual (in response to broader market dynamics). The framework as currently formulated does not incorporate any mechanism for tracking or adapting to these temporal variations.

A related concern is the implicit assumption that the ground truth distribution observed across the aggregate population is representative of the distribution applicable to any given customer segment. In practice, attribution dynamics may vary substantially across segments defined by demographics, behavior history, or product category. The present framework produces a single aggregate attribution distribution that may obscure such heterogeneity.

The development of dynamic attribution frameworks capable of tracking temporal evolution of channel effects, and of segment-specific frameworks capable of producing differentiated attribution distributions for distinct customer populations, constitutes an important area for future research.

5.6.4. Cookieless Future Considerations

As discussed in Section 2.6, the digital advertising ecosystem is undergoing structural transformation driven by regulatory action and platform-level privacy initiatives. The deprecation of third-party cookies, the implementation of Apple’s App Tracking Transparency frame-

work, and the broader regulatory environment introduced by GDPR and CCPA all constrain the ability of advertisers and analysts to track individual users across the touchpoints that constitute customer journeys (Acquisti et al., 2016; Johnson et al., 2020).

The implications of these changes for the present framework are substantial. The Markov chain model, in particular, depends on the ability to reconstruct complete customer journeys from observed touchpoint sequences. As cross-site tracking becomes increasingly constrained, the data inputs required for accurate transition matrix estimation may become progressively less available. The Shapley value model exhibits a similar dependency: the characteristic function defined in Section 3.4.2 requires the ability to identify the channels present in each converting journey, which in turn requires cross-touchpoint identity resolution.

Mitigating strategies developed in industry response to these constraints, including server-side tracking (Chebrolu, 2025) and federated learning approaches (Thomas, 2024), may preserve sufficient data quality to support attribution analysis in the post-cookie environment, but their effectiveness remains uncertain. The framework as currently formulated does not address the cookieless transition directly. Its viability in privacy-constrained operational environments depends on the continued availability of journey-level tracking data through whichever technical or contractual mechanisms emerge to replace third-party cookies.

5.6.5. The Touchpoint-Centric Assumption and Qualitative Factors

A further limitation, perhaps more conceptually fundamental than those discussed above, concerns the implicit assumption that channel-level touchpoints constitute the appropriate unit of analysis for attribution. The framework developed in this thesis, in common with substantively all contemporary attribution methodologies, models the customer journey as a sequence of channel encounters and assigns conversion credit to those channels. This formulation, while methodologically tractable, abstracts from a substantial body of factors that empirical marketing research has established as causally relevant to conversion outcomes.

Creative Quality and Brand Perception. The attribution framework treats all touchpoints within a given channel as functionally equivalent: a user encountering an advertisement on a particular channel is modeled as having experienced the same touchpoint regardless of the specific advertisement encountered. In practice, the creative content of advertisements—the visual design, the textual messaging, the emotional appeal, the alignment with audience expectations—exerts substantial influence on conversion outcomes that cannot be reduced to channel-level effects (Kitada et al., 2019). Two advertisements served on the same channel may produce conversion rates that differ by an order of magnitude depending on creative quality, yet the attribution framework as formulated treats them identically. The implication is that the channel-level attribution estimates produced by the framework do not, in themselves, identify the specific creative properties that contribute to channel performance. Operations seeking to improve attribution outcomes through creative optimization—which may in many cases be more cost-effective than reallocating budget across channels—are not directly supported by the present framework.

Closely related is the influence of *brand perception* on conversion outcomes. The trust, credibility, and emotional resonance that consumers associate with a brand exert substantial moderating influence on the effectiveness of any specific channel touchpoint (Kržanová et al., 2023). A high-trust brand may convert effectively through channels that fail entirely for low-trust brands. The attribution framework, which treats channels as the primary unit of analysis, abstracts from these brand-level moderators and cannot distinguish between channel effectiveness intrinsic to the channel and channel effectiveness contingent on brand standing.

Contextual and Cultural Factors. The effectiveness of marketing communications is substantially shaped by the broader context in which they are received. Economic conditions, political and social events, seasonal patterns, and cultural sensitivities all influence consumer receptivity to advertising messages in ways that the attribution framework does not directly model (Liu & Shankar, 2015). An advertising campaign that performs strongly under stable economic conditions may underperform during periods of consumer uncertainty; a creative concept that resonates in one cultural context may fail entirely in another. These contextual factors operate primarily at the level of the broader environment rather than at the level of individual touchpoints, and the touchpoint-centric attribution framework provides no direct mechanism for incorporating them. The implication, for the operational deployment of the framework, is that attribution estimates should be interpreted as conditional on the context in which the data were generated; substantial changes in the broader economic, political, or cultural environment may render historical attribution estimates unreliable as guides to prospective allocation decisions. This limitation is particularly salient for emerging markets such as Türkiye, where rapid shifts in consumer sentiment and economic conditions are characteristic features of the operational environment.

Social Media Response and Word-of-Mouth Dynamics. A final dimension of qualitative factors concerns the response that marketing communications elicit beyond the immediate touchpoint interaction. Consumer reactions to advertisements—particularly negative reactions expressed through social media—can substantially modify the effectiveness of a campaign through mechanisms that the touchpoint-centric framework cannot capture. Hansen et al. (2018) document that social media firestorms, defined as sudden concentrations of negative consumer sentiment expressed through social platforms, produce short-term negative brand perception effects for 58% of affected brands and long-term effects for 40%. The economic consequences of negative word-of-mouth extend beyond brand perception to measurable impacts on cash flows and financial performance (Luo, 2009).

These dynamics operate outside the data-generating process modeled by the attribution framework. A campaign that generates substantial touchpoint volume but provokes negative social media response may produce attribution estimates that overstate the true contribution of the channels involved, since the framework cannot observe the counterfactual outcome—the conversions that would have occurred absent the negative response. The implication is that attribution analyses conducted on data from campaigns associated with significant social media controversy should be interpreted with particular caution.

The Conceptual Implication. The cumulative effect of these qualitative limitations is to suggest that the attribution paradigm itself, as currently constituted, addresses only one dimension of the broader question of marketing effectiveness. The channel-level attribution estimates produced by the present framework are valuable for the budget allocation decisions for which they are designed; they are not, however, complete descriptions of why marketing communications succeed or fail. A complete account of marketing effectiveness would need to integrate the channel-level analysis offered by attribution methods with the creative-level, brand-level, and contextual-level analyses that complementary research traditions provide. The development of integrated frameworks capable of accommodating these multiple dimensions of analysis represents an ambitious but important direction for subsequent research, discussed in the following section.

5.7. Future Research Directions

The limitations articulated in the preceding section indicate a series of directions in which the present framework may be productively extended through subsequent research. Four directions are particularly noteworthy and are discussed in turn below.

5.7.1. Multi-modal Attribution

The first direction concerns the integration of qualitative factors, particularly creative quality, into the attribution framework itself. Recent developments in multi-modal deep learning offer a methodological pathway for accomplishing this integration. Wang et al. (2025) demonstrate that multi-modal architectures combining visual, textual, and auditory inputs can substantially improve the prediction of advertising effectiveness relative to single-modality approaches. Yao et al. (2020) demonstrate that Shapley value methodology can be productively integrated with deep neural architectures within a unified attribution framework. The combination of these two lines of work—a multi-modal deep learning framework that incorporates Shapley attribution—represents a natural methodological progression beyond the present hybrid framework.

Such a multi-modal extension would address one of the principal limitations identified in Section 5.6.5 by enabling the attribution framework to differentiate among advertisements within a single channel based on their creative content. The methodological cost of this extension is substantial: multi-modal deep learning frameworks require considerably larger training datasets than classical attribution methods, and the interpretability advantages of the present framework would be partially sacrificed. The trade-off between accuracy and interpretability that this extension entails will need to be evaluated empirically in subsequent work.

5.7.2. Context-aware Modeling

The second direction concerns the explicit incorporation of contextual factors—economic conditions, seasonal patterns, cultural sensitivities, and broader environmental variables—into the attribution framework. The present framework treats attribution dynamics as static; a context-aware extension would model them as functions of observable contextual variables, permitting attribution estimates to be conditioned on the operational environment in which they are applied.

Sentiment analysis applied to social media data offers one promising mechanism for operationalizing such context-awareness. By tracking consumer sentiment in real time, attribution frameworks can adjust their estimates to reflect the prevailing reception of brand and campaign messages. Such sentiment-aware attribution has not, to the present author’s knowledge, been systematically developed in the academic literature, although the technical components—sentiment analysis, attribution modeling, and contextual conditioning—are well-established in their respective traditions.

5.7.3. Real-world Validation

The third direction, methodologically distinct from the preceding two, concerns the validation of the present framework against real-world attribution data. As discussed in Section 5.6.1, the use of synthetic data permits objective accuracy assessment but at the cost of external validity. A natural complementary research strategy would apply the hybrid Markov-Shapley framework to real-world customer journey data and compare its allocation recommen-

dations against the outcomes produced by alternative attribution methods in actual operational deployment.

The methodological challenges of such validation are non-trivial: in the absence of randomized experiments, comparisons among attribution methods on real-world data face the validation problem identified in Section 2.7. However, partial validation strategies—including comparison of framework predictions against the outcomes of natural experiments such as channel removal events, calibration against field-experiment benchmarks where available, and longitudinal tracking of allocation outcomes across multiple budget cycles—may provide meaningful evidence of real-world framework performance without requiring full randomized experimentation (Gordon et al., 2019).

5.7.4. Practical Deployment and Decision Support System Integration

The fourth direction concerns the practical deployment of the framework as an integrated decision support system within operational organizational contexts. The framework as currently formulated is a methodological prototype rather than a deployment-ready system; substantial development work would be required to integrate the methodology into the data infrastructures, decision workflows, and organizational practices of operating businesses.

Three specific development priorities are identifiable. First, the framework requires integration with the data pipelines through which customer journey data is collected, validated, and stored in operational contexts. Such integration is non-trivial in environments characterized by data quality issues, cross-platform tracking gaps, and evolving privacy constraints. Second, the framework requires interfaces through which non-technical decision-makers can interact with its outputs, including visualization tools for attribution distributions, sensitivity analyses for budget recommendations, and explanatory mechanisms for communicating the rationale of specific recommendations. Third, the framework requires periodic recalibration mechanisms to address the static ground truth limitation identified in Section 5.6.3, including both automated re-estimation procedures and managerial review protocols.

The development of a fully integrated decision support system around the methodological framework presented here, while beyond the scope of the present thesis, represents a substantial practical contribution that subsequent work may pursue. The empirical evidence demonstrated in Chapter 4—that the framework achieves 99.95% of the theoretical upper bound at the 100,000 USD budget level—establishes that the methodological foundation for such a system is in place. The realization of corresponding practical value depends on the development effort that subsequent work directs toward the integration challenges identified above.

6. CONCLUSION

6.1. Summary of the Study

This thesis has addressed a central methodological problem in digital marketing attribution: the systematic comparison and improvement of attribution models in the absence of an observable ground truth. The problem is consequential because attribution estimates directly inform budget allocation decisions whose accuracy determines the effective return on marketing investment, yet the validity of competing attribution methodologies has historically been difficult to assess objectively (Anderl et al., 2016; Berman, 2018).

The methodological response developed in this thesis combines three integrated components. First, a synthetic customer journey dataset was constructed using channel effect parameters calibrated against industry benchmarks (Lucky Orange, 2025; WordStream, 2025), with Monte Carlo simulation employed to establish a known ground truth attribution distribution. This synthetic-data approach addresses the validation problem by providing an objective benchmark against which competing models can be evaluated. Second, a hybrid Markov-Shapley attribution model was developed, combining the directional sensitivity of Markov chain removal-effect analysis with the coalitional structure of Shapley value allocation (Dalessandro et al., 2012; Shapley, 1953). The mixing parameter governing the combination was determined empirically through grid search against the ground truth, yielding an optimal value of $\alpha^* = 0.25$ that incorporates approximately 25% Markov and 75% Shapley contributions. Third, a non-linear budget optimization layer was integrated with the attribution outputs, employing diminishing-returns response curves and the Differential Evolution algorithm (Jin et al., 2017; Storn & Price, 1997) to translate attribution estimates into actionable budget recommendations.

The empirical evaluation, conducted against a synthetic dataset of 30,000 customer journeys with an aggregate conversion rate of 1.18%, established three principal findings. The hybrid model achieved a Mean Absolute Error of 0.0571 against the ground truth, representing a 21.2% improvement over the next-best baseline (Last-Click attribution at MAE = 0.0725) and exceeding the typical performance gains reported in the data-driven attribution literature. The hybrid model's budget allocations produced expected revenue equivalent to 99.95% of the theoretical upper bound at the 100,000 USD budget level, indicating near-optimal performance under the diminishing-returns formulation. The performance advantage of the hybrid model was strongest at lower budget levels, reaching 13.72% revenue improvement over Last-Click at the 10,000 USD budget level, and progressively diminishing as total budget increased into the saturated region of the response curves.

6.2. Answers to the Research Questions

The research questions articulated in Chapter 1 are now addressed directly, with reference to the empirical findings reported in Chapter 4 and their interpretation in Chapter 5.

6.2.1. RQ1: Do data-driven models outperform classical heuristics?

The first research question concerned whether data-driven attribution models, specifically Markov Chain and Shapley Value, outperform classical heuristic models in approximating the ground-truth channel contribution structure of a multi-channel customer journey. The empirical findings provide a qualified affirmative answer to this question.

The qualification is methodologically important. The proposed hybrid model, which integrates Markov and Shapley components, achieved the lowest MAE and RMSE among the

eight models examined and therefore satisfies the criterion of outperforming all classical heuristics. The standalone Shapley model also outperformed several classical heuristics on rank-based criteria, though its magnitude accuracy was modest. The standalone Markov model, however, produced the worst MAE and RMSE of all models examined, despite achieving the highest Spearman rank correlation with the ground truth. The implication is that the category of “data-driven attribution” is not homogeneous: data-driven methods differ substantially in their performance characteristics, and the selection of an appropriate methodology is not reducible to a simple dichotomy between heuristic and data-driven approaches.

The deeper insight emerging from this finding is that effective attribution requires methodological integration rather than methodological selection. The Markov-Shapley trade-off documented in Section 5.2—in which the Markov framework excels at rank identification while the Shapley framework excels at magnitude estimation—establishes that neither paradigm alone provides a complete attribution solution. The superiority of data-driven over heuristic attribution, on this view, is realized fully only when complementary data-driven paradigms are combined rather than applied in isolation.

6.2.2. RQ2: Does the hybrid model improve upon standalone approaches?

The second research question concerned whether a hybrid model combining Markov Chain removal effects and Shapley Value contributions, with an empirically optimized weighting parameter, achieves superior accuracy compared with either standalone approach. The empirical findings provide a clear affirmative answer.

The hybrid model achieved a Mean Absolute Error of 0.0571, compared with 0.1466 for the standalone Markov model and 0.0810 for the standalone Shapley model. The improvement of the hybrid over the standalone Markov model is 61.1%; the improvement over the standalone Shapley model is 29.5%. Both improvements are substantial in absolute terms and exceed the magnitude of differences typically observed among data-driven attribution methods. The improvements are robust across the magnitude-based accuracy metrics employed (MAE and RMSE) and are accompanied by a Spearman rank correlation of 0.600, which is the joint second-highest value among the eight models examined.

The mechanism through which the hybrid combination produces these improvements is interpretable in terms of the Markov-Shapley trade-off. The Shapley framework, when applied in isolation, produces attribution distributions that are systematically compressed toward uniformity, with the result that the most important channels are underweighted. The Markov framework, when applied in isolation, produces attribution distributions that are systematically exaggerated, with the result that the most directionally important channel is overweighted at the expense of channels of moderate importance. The optimal hybrid combination at $\alpha^* = 0.25$ incorporates sufficient Markov directional signal to correct the Shapley compression while preserving sufficient Shapley distributive structure to constrain the Markov exaggeration. The result is an attribution distribution that achieves both magnitude accuracy and rank accuracy at levels superior to either component alone.

6.2.3. RQ3: How does the model translate into budget decisions?

The third research question concerned how the attribution accuracy of the proposed hybrid model translates into budget allocation decisions across varying budget levels when evaluated under realistic diminishing-returns assumptions. The findings establish a systematic budget-dependent pattern that warrants particular attention.

At the 100,000 USD budget level employed as the primary reference point, the hybrid

model produced budget allocations whose expected revenue under the ground-truth response curves equaled 99.95% of the theoretical upper bound. This result indicates that, despite operating without access to the ground-truth attribution structure, the hybrid model produces allocations that are essentially indistinguishable from the theoretical optimum in their expected outcomes. The remaining gap of 0.05% represents the practical cost of operating with estimated rather than known channel effects.

The relative performance advantage of the hybrid model over the Last-Click baseline varied systematically across budget levels: 13.72% at 10,000 USD, 3.35% at 50,000 USD, 0.77% at 100,000 USD, and effectively zero at 500,000 USD. This pattern reflects the underlying diminishing-returns structure of the response curves: at low budget levels, channels operate in the steep portion of the response curves where allocation precision exerts substantial influence on outcomes; at high budget levels, channels approach saturation and the value of allocation precision diminishes.

The practical implication is that the proposed framework offers maximum value in the budget range characteristic of small-to-medium B2C e-commerce operations (Kannan & Li, 2017). For operations working within this range, the framework can produce measurable revenue improvements from the same total expenditure. For operations working at substantially larger scales, where channels are typically saturated, the framework’s contribution diminishes—though, as discussed in Section 5.3, this diminishment reflects a general feature of attribution under saturation rather than a specific limitation of the proposed framework.

6.3. Contributions

The contributions of this thesis can be organized along three dimensions: academic, methodological, and practical.

6.3.1. Academic Contribution

The principal academic contribution of this thesis lies in the systematic articulation of the *Markov-Shapley trade-off* as a structural feature of the data-driven attribution literature. Prior contributions in this literature have treated Markov-based and Shapley-based attribution as competing methodological alternatives, with researchers and practitioners selecting one or the other based on theoretical preference or computational considerations. The findings of this thesis establish that these paradigms are more productively understood as complementary, each capturing dimensions of the attribution problem that the other cannot fully represent. The hybrid framework developed in response to this trade-off provides an existence proof that principled combination of the two paradigms yields accuracy that neither alone achieves.

A secondary academic contribution lies in the development of the ground-truth-based validation framework employed throughout the thesis. The use of synthetic customer journey data with known ground-truth channel effects, while not unprecedented in attribution research, provides an objective standard against which competing methodologies can be evaluated—a methodological resource that observational attribution research has historically lacked (Shao & Li, 2011). The framework is applicable beyond the specific methodological focus of the present study, offering a tool for the comparative evaluation of attribution models more generally.

6.3.2. Methodological Contribution

The methodological contribution of this thesis is threefold. First, the empirical optimization of the hybrid mixing parameter through grid search against the ground truth represents a

departure from prior work on hybrid attribution, in which combination weights have typically been imposed by assumption or derived from theoretical principles alone. The data-driven determination of $\alpha^* = 0.25$ illustrates that the optimal balance between component models is itself an empirical quantity that varies with dataset characteristics, and that fixed-weight hybrid combinations sacrifice accuracy that empirical optimization can recover.

Second, the integration of attribution analysis with non-linear budget optimization—through the diminishing-returns response curves and the Differential Evolution optimizer—bridges two traditions that have historically operated in relative isolation. Attribution research has focused on the assignment of credit to channels with implications for budget allocation often noted but not systematically developed; Marketing Mix Modeling research has focused on budget optimization but typically at the aggregate level without incorporating granular journey-level information. The integrated framework developed here treats attribution outputs as the channel-effect inputs to budget optimization, producing recommendations that are grounded simultaneously in journey-level evidence and realistic saturation dynamics.

Third, the comparative evaluation framework employed in the thesis—in which eight attribution models are evaluated against a common ground truth using multiple complementary metrics—provides a template for attribution research that subsequent work may adapt to other methodological comparisons. The combination of magnitude-based metrics (MAE, RMSE), rank-based metrics (Spearman correlation), and outcome-based metrics (revenue simulation) captures distinct dimensions of attribution performance that single-metric evaluations cannot reveal.

6.3.3. Practical Contribution

The practical contribution of this thesis lies in the development of an end-to-end framework that can be deployed as a decision support system within small-to-medium B2C e-commerce operations. The framework accepts historical customer journey data as input and produces budget recommendations as output, with all intermediate computations executed through interpretable classical methods rather than opaque neural architectures. The framework’s accuracy at the 100,000 USD budget level—99.95% of the theoretical upper bound—establishes that the methodological foundation is sufficient to support meaningful operational deployment.

Three practical considerations are noteworthy. First, the framework’s value is concentrated in the budget range characteristic of small-to-medium operations, where channels operate below saturation and allocation precision exerts substantial influence on outcomes. Operations working in this range can expect measurable improvements in expected revenue from the framework’s recommendations. Second, the framework should be applied as a decision support system rather than as an automated decision-making mechanism: the recommendations require interpretation by managerial judgment that incorporates strategic considerations beyond those captured in the model. Third, the framework requires periodic recalibration to accommodate the temporal evolution of channel effectiveness; a quarterly review cycle, in which the model is re-estimated on updated data and recommendations are compared against current allocations, represents an appropriate balance between responsiveness and stability.

The complete implementation of the framework, comprising approximately 1,500 lines of Python code distributed across seven modular files, is provided in Appendix A. The implementation is sufficiently documented to permit replication, adaptation to alternative datasets, and extension to additional methodological components.

6.4. Closing Remarks

The evolution of digital marketing attribution over the past decade has traced a recognizable arc: from heuristic models inherited from pre-digital advertising traditions, through individual data-driven methodologies grounded in stochastic process theory and cooperative game theory, toward integrated frameworks that combine multiple methodological paradigms in pursuit of accuracy that no single approach can achieve alone. The present thesis contributes to this evolution by proposing one such integrated framework—combining Markov chain and Shapley value methodologies under an empirically optimized weighting scheme, and linking the resulting attribution outputs to non-linear budget optimization—and by demonstrating its capacity to produce attribution estimates and budget recommendations approaching the theoretical upper bound of accuracy under controlled conditions.

The methodological convergence represented by integrated frameworks of this kind is likely to assume increasing importance as the digital advertising ecosystem undergoes the structural transformation associated with the cookieless transition. As real-world tracking data becomes increasingly fragmented, the comparative advantage of methodologies grounded in synthetic data, principled simulation, and theoretical integration is likely to grow relative to methodologies that depend on the comprehensive observational data infrastructure of the cookie-based era. The framework developed here, while not designed primarily as a response to the cookieless transition, embodies methodological characteristics—interpretability, data efficiency, theoretical grounding—that position it favorably for the operational environment emerging beyond the current technical paradigm.

The findings of this thesis should not be read as a final resolution of the attribution problem in digital marketing. The limitations identified in Chapter 5, particularly those concerning the touchpoint-centric assumption and the absence of qualitative factor modeling, indicate substantial directions in which the framework requires extension. The relationship of attribution analysis to the broader question of marketing effectiveness—encompassing creative quality, brand perception, contextual factors, and consumer sentiment dynamics—remains an active research frontier whose integration with the kind of structural attribution methodology developed here represents an ambitious agenda for subsequent work. What the present thesis establishes is not that the attribution problem is solved, but that progress within the attribution paradigm is achievable through principled methodological integration, and that such progress yields tangible improvements in the quality of budget decisions that depend on attribution estimates.

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APPENDICES

Appendix A: Python Implementation

The complete implementation of the methodological framework described in Chapter 3 is provided in this appendix. The implementation comprises approximately 1,500 lines of Python code distributed across seven module files, of which the most methodologically significant fragments are reproduced below. The full source code is available through the repository referenced in Appendix D.

A.1 Code Repository Overview

The implementation is organized into seven Python modules, each encapsulating a distinct methodological component:

- `data_generator.py` — Synthetic customer journey generation and Monte Carlo ground truth simulation.
- `markov_chain.py` — Absorbing Markov chain construction, Laplace smoothing, and removal-effect computation.
- `shapley_value.py` — Cooperative game-theoretic Shapley value computation with fractional credit.
- `baseline_models.py` — Implementations of five classical heuristic attribution models.
- `budget_optimization.py` — Hybrid model integration, alpha grid search, and Differential Evolution-based budget optimization.
- `revenue_simulation.py` — Multi-budget revenue simulation framework.
- `comparison_report.py` — Final aggregation and comparative reporting.

The implementation depends on standard scientific Python libraries: NumPy for numerical computation, pandas for data manipulation, SciPy for the Differential Evolution optimizer and linear algebra routines, and Matplotlib for visualization output. All code was developed and executed in Python 3.12 under the Spyder IDE.

A.2 Synthetic Data Generation

The synthetic customer journey dataset is generated through a two-stage procedure. The first stage establishes ground truth channel contributions through Monte Carlo simulation; the second stage generates the customer journey records used by the attribution models. The conversion probability function, which serves as the foundation of both stages, is reproduced below.

```
1 def calculate_conversion_probability(path: list) -> float:
2     """
3     Calculate the true conversion probability for a given
4     customer journey, given the channel effect parameters,
5     interaction effects, and recency weighting.
6     """
7     if not path:
```

```

8         return 0.0
9
10        # 1. Sum of individual channel effects (unique channels only)
11        unique_channels = set(path)
12        individual_effect = sum(CHANNEL_EFFECTS[ch]
13                                for ch in unique_channels)
14
15        # 2. Pairwise interaction multipliers
16        interaction_multiplier = 1.0
17        for i in range(len(path) - 1):
18            pair = (path[i], path[i + 1])
19            if pair in CHANNEL_INTERACTIONS:
20                interaction_multiplier *= CHANNEL_INTERACTIONS[pair]
21
22        # 3. Recency bonus (terminal touchpoint amplification)
23        last_channel_bonus = 1.0 + (CHANNEL_EFFECTS[path[-1]] * 0.3)
24
25        # 4. Path length normalization
26        length_normalizer = 1.0 / (1.0 + 0.1 * len(path))
27
28        # 5. Penalty for journeys consisting solely of Social_Media
29        if unique_channels == {'Social_Media'}:
30            interaction_multiplier *= 0.6
31
32        # Final probability
33        prob = (BASE_CONVERSION_RATE
34                * individual_effect
35                * interaction_multiplier
36                * last_channel_bonus
37                * length_normalizer)
38
39        return min(max(prob, 0.0), 1.0)

```

Listing 1: Conversion probability function (excerpt from `data_generator.py`).

The function combines four multiplicative components: the sum of individual channel effects across unique channels present in the journey; a product of pairwise interaction multipliers for consecutive channel pairs matching the interaction matrix defined in Section 3.2.2; a recency bonus that amplifies the contribution of the terminal touchpoint; and a length normalization factor that prevents conversion probabilities from growing unboundedly with journey length. The final probability is constrained to the unit interval through the `min/max` clipping operation.

The Monte Carlo ground truth simulation, which establishes the benchmark distribution against which attribution models are evaluated, operates by repeatedly generating customer journeys and computing the marginal contribution of each channel through counterfactual journey construction.

```

1 def compute_ground_truth_attribution(
2     n_simulations: int = 100_000,
3     seed: int = 123
4 ) -> dict:
5     """
6     Compute the ground truth attribution distribution through
7     Monte Carlo simulation. For each channel, the average
8     marginal contribution is computed as the difference in
9     conversion probability between journeys containing the

```

```

10     channel and counterfactual journeys with the channel removed.
11     """
12     rng = np.random.default_rng(seed)
13
14     contributions = {ch: 0.0 for ch in CHANNELS}
15     counts        = {ch: 0    for ch in CHANNELS}
16
17     for _ in range(n_simulations):
18         path = generate_user_path(rng)
19         prob = calculate_conversion_probability(path)
20
21         # Marginal contribution of each channel in the journey
22         for ch in set(path):
23             path_without = [p for p in path if p != ch]
24             prob_without = (
25                 calculate_conversion_probability(path_without)
26                 if path_without else 0.0
27             )
28             contributions[ch] += (prob - prob_without)
29             counts[ch] += 1
30
31         # Average and normalize
32         avg_contributions = {
33             ch: contributions[ch] / counts[ch] if counts[ch] > 0 else 0.0
34             for ch in CHANNELS
35         }
36         total = sum(avg_contributions.values())
37         return {ch: v / total for ch, v in avg_contributions.items()}

```

Listing 2: Monte Carlo ground truth attribution (excerpt from `data_generator.py`).

The procedure generates 100,000 synthetic journeys, computes the conversion probability of each, and accumulates the marginal contribution of each channel as the difference between the actual conversion probability and the counterfactual probability that would obtain if the channel were removed from the journey. The aggregated marginal contributions are normalized to sum to unity across channels, yielding the ground truth attribution distribution reported in Section 4.2.

A.3 Markov Chain Implementation

The Markov chain attribution model is implemented through two principal functions: the construction of the smoothed transition matrix from observed customer journey data, and the computation of channel-level removal effects through fundamental matrix analysis.

The transition matrix construction incorporates Laplace smoothing ($\alpha = 1.0$) directly into the counting procedure, ensuring that all theoretically possible transitions receive non-zero probability mass. The smoothing is applied to all valid transition pairs: Start-to-channel, channel-to-channel (excluding self-loops), and channel-to-absorbing-state transitions.

```

1 def build_transition_matrix(
2     path_df: pd.DataFrame,
3     channels: list,
4     alpha: float = 1.0
5 ) -> pd.DataFrame:
6     """
7     Compute the Markov chain transition matrix from observed
8     customer journeys, with Laplace smoothing applied to all

```

```

9     valid transitions.
10    """
11    states = ['Start'] + channels + ['Conversion', 'Non-Conversion']
12    counts = pd.DataFrame(0.0, index=states, columns=states)
13
14    # Accumulate empirical transition counts
15    for _, row in path_df.iterrows():
16        path = row['path']
17        conv = row['conversion']
18
19        # Start -> first channel
20        first = path[0] if path[0] in channels else None
21        if first:
22            counts.loc['Start', first] += 1
23
24        # Channel -> Channel
25        for i in range(len(path) - 1):
26            src, dst = path[i], path[i + 1]
27            if src in channels and dst in channels:
28                counts.loc[src, dst] += 1
29
30        # Last channel -> absorbing state
31        last = path[-1] if path[-1] in channels else None
32        if last:
33            if conv == 1:
34                counts.loc[last, 'Conversion'] += 1
35            else:
36                counts.loc[last, 'Non-Conversion'] += 1
37
38    # Apply Laplace smoothing to all valid transitions
39    for ch in channels:
40        counts.loc['Start', ch] += alpha
41        for ch2 in channels:
42            if ch != ch2:
43                counts.loc[ch, ch2] += alpha
44        counts.loc[ch, 'Conversion'] += alpha
45        counts.loc[ch, 'Non-Conversion'] += alpha
46
47    # Normalize rows to probability distributions
48    row_sums = counts.sum(axis=1)
49    transition_matrix = counts.div(
50        row_sums.replace(0, np.nan), axis=0
51    ).fillna(0)
52
53    # Absorbing states transition to themselves with probability 1
54    transition_matrix.loc['Conversion', 'Conversion'] = 1.0
55    transition_matrix.loc['Non-Conversion', 'Non-Conversion'] = 1.0
56
57    return transition_matrix

```

Listing 3: Transition matrix construction with Laplace smoothing (excerpt from `markov_chain.py`).

The removal effect computation employs the fundamental matrix methodology of absorbing Markov chain theory. The procedure first computes the baseline conversion probability by solving the linear system $(I - Q) \cdot N = R$ for the absorption probabilities; the counterfactual conversion probability is then computed by removing the target channel and renormalizing the

remaining transition probabilities.

```
1 def compute_removal_effects(  
2     transition_matrix: pd.DataFrame,  
3     channels: list,  
4     epsilon: float = 1e-6  
5 ) -> dict:  
6     """  
7     Compute channel-level removal effects through counterfactual  
8     analysis. For each channel, the relative reduction in  
9     conversion probability resulting from its hypothetical  
10    removal is computed.  
11    """  
12    base_probs = conversion_probability(transition_matrix, channels)  
13    base_conv_p = base_probs['Start']  
14  
15    if base_conv_p < epsilon:  
16        return {ch: 0.0 for ch in channels}  
17  
18    removal_effects = {}  
19  
20    for ch in channels:  
21        reduced_channels = [c for c in channels if c != ch]  
22  
23        # Remove the target channel from the transition matrix  
24        reduced_tm = transition_matrix.drop(  
25            index=ch, columns=ch  
26        ).copy()  
27  
28        # Renormalize remaining rows  
29        transient = ['Start'] + reduced_channels  
30        absorbing = ['Conversion', 'Non-Conversion']  
31        for state in transient:  
32            row_sum = reduced_tm.loc[  
33                state, transient + absorbing  
34            ].sum()  
35            if row_sum > 0:  
36                reduced_tm.loc[state] = reduced_tm.loc[state] / row_sum  
37  
38        # Restore absorbing state self-transitions  
39        reduced_tm.loc['Conversion', 'Conversion'] = 1.0  
40        reduced_tm.loc['Non-Conversion', 'Non-Conversion'] = 1.0  
41  
42        # Compute counterfactual conversion probability  
43        reduced_probs = conversion_probability(  
44            reduced_tm, reduced_channels  
45        )  
46        reduced_conv_p = reduced_probs['Start']  
47  
48        # Relative reduction  
49        re = (base_conv_p - reduced_conv_p) / base_conv_p  
50  
51        # Convention: negative values reflect smoothing artifacts  
52        re = max(re, 0.0)  
53        if re < epsilon:  
54            re = 0.0  
55  
56        removal_effects[ch] = re
```

```

57
58     return removal_effects

```

Listing 4: Removal effect computation (excerpt from `markov_chain.py`).

The fundamental matrix computation employed in the conversion probability function uses `np.linalg.solve` rather than explicit matrix inversion through `np.linalg.inv`, providing both improved numerical stability and computational efficiency. Negative removal effects, which may arise occasionally as numerical artifacts of the Laplace smoothing operation, are floored at zero in accordance with the convention established by Anderl et al. (2016): only non-negative removal effects represent meaningful contributions to the budget allocation analysis.

A.4 Shapley Value Implementation

The Shapley value implementation employs the fractional credit characteristic function introduced by Shao and Li (2011), which addresses the data sparsity problem inherent in coalition-specific attribution by permitting each customer journey to contribute partially to multiple coalitions in proportion to the overlap between its channel set and the coalition.

The characteristic function is computed exhaustively over all 2^n possible coalitions; for the five-channel configuration of the present study, this corresponds to the enumeration of 32 coalitions, which remains computationally tractable.

```

1  def calculate_v_s(
2      path_df: pd.DataFrame,
3      channels: list
4  ) -> dict:
5      """
6      Compute the characteristic function v(S) for all coalitions
7      using the fractional credit formulation:
8
9          v(S) = sum over converted users of |path ∩ S| / |path|
10
11      Each converted journey contributes to a coalition's value
12      in proportion to the fraction of its channels that belong
13      to the coalition.
14      """
15      converted_df = path_df[path_df['conversion'] == 1].copy()
16
17      if len(converted_df) == 0:
18          subsets = []
19          for r in range(1, len(channels) + 1):
20              subsets.extend(itertools.combinations(channels, r))
21          return {tuple(sorted(s)): 0.0 for s in subsets}
22
23      # Enumerate all non-empty coalitions
24      subsets = []
25      for r in range(1, len(channels) + 1):
26          subsets.extend(itertools.combinations(channels, r))
27
28      v_s = {}
29      for subset in subsets:
30          subset_set = set(subset)
31
32      # Fractional contribution of each converting journey

```

```

33     fractions = converted_df['path_set'].apply(
34         lambda x: len(x & subset_set) / len(x)
35         if len(x) > 0 else 0
36     )
37     v_s[tuple(sorted(subset))] = float(fractions.sum())
38
39     return v_s

```

Listing 5: Characteristic function with fractional credit (excerpt from `shapley_value.py`).

The Shapley value computation iterates over all coalitions excluding the target channel and accumulates the weighted marginal contribution of the channel to each coalition. The weighting factor $|S|!(n - |S| - 1)!/n!$ ensures that the resulting allocation satisfies the four Shapley axioms (efficiency, symmetry, null player, and additivity) discussed in Section 3.4.1.

```

1 def compute_shapley(v_s: dict, channels: list) -> dict:
2     """
3     Compute Shapley values using the standard formula:
4
5     phi_i = sum over S subset N\{i} of
6             [|S|! * (n - |S| - 1)! / n!] *
7             [v(S union {i}) - v(S)]
8     """
9     n = len(channels)
10    shapley_values = defaultdict(float)
11
12    for i in channels:
13        phi_i = 0.0
14
15        # Empty coalition contribution: v({i}) - v(empty)
16        singleton = tuple(sorted([i]))
17        v_singleton = v_s.get(singleton, 0)
18        weight_empty = (
19            math.factorial(0) * math.factorial(n - 1)
20            / math.factorial(n)
21        )
22        phi_i += weight_empty * v_singleton
23
24        # Non-empty coalitions excluding i
25        other_channels = [c for c in channels if c != i]
26        for r in range(1, len(other_channels) + 1):
27            for s in itertools.combinations(other_channels, r):
28                s_key = tuple(sorted(s))
29                s_union_i = tuple(sorted(s + (i,)))
30
31                v_s_val = v_s.get(s_key, 0)
32                v_s_union_i = v_s.get(s_union_i, 0)
33
34                marginal = v_s_union_i - v_s_val
35                weight = (
36                    math.factorial(r)
37                    * math.factorial(n - r - 1)
38                    / math.factorial(n)
39                )
40                phi_i += weight * marginal
41
42    shapley_values[i] = phi_i
43

```

```
44 return dict(shapley_values)
```

Listing 6: Shapley value computation (excerpt from `shapley_value.py`).

The implementation handles the empty coalition as a special case, recognizing that the marginal contribution of channel i to the empty coalition is simply $v(\{i\})$. All non-empty coalitions excluding the target channel are enumerated through nested calls to `itertools.combinations`, and the cumulative weighted marginal contribution is returned as the channel's Shapley value. The resulting values are subsequently normalized to sum to unity across channels, yielding the attribution distribution reported in Section 4.4.

A.5 Hybrid Model and Alpha Optimization

The hybrid Markov-Shapley model is implemented as a convex linear combination of the normalized component attribution distributions, with the mixing parameter α determined empirically through a one-dimensional grid search against the ground truth. The grid search procedure operates on the normalized component outputs and selects the α value that minimizes the Mean Absolute Error against the ground truth distribution established in Section 3.2.4.

```
1 def find_optimal_alpha(
2     markov_full_csv : str = 'markov_results.csv',
3     shapley_full_csv: str = 'shapley_results.csv',
4     ground_truth_csv: str = 'ground_truth.csv',
5     n_grid: int = 101
6 ) -> tuple:
7     """
8     Identify the optimal mixing parameter alpha through grid
9     search over [0, 1] at resolution 0.01, minimizing the Mean
10    Absolute Error between hybrid attribution weights and the
11    ground truth distribution.
12    """
13    markov_df = pd.read_csv(markov_full_csv)
14    shapley_df = pd.read_csv(shapley_full_csv)
15    gt_df = pd.read_csv(ground_truth_csv)
16
17    df = (markov_df
18         .merge(shapley_df, on='channel')
19         .merge(gt_df, on='channel'))
20
21    # Normalize component outputs to unit-sum distributions
22    df['m_abs'] = df['markov_removal_effect'].abs()
23    df['s_abs'] = df['shapley_value'].abs()
24    m_norm = (df['m_abs'] / df['m_abs'].sum()
25              if df['m_abs'].sum() > 0 else df['m_abs'])
26    s_norm = (df['s_abs'] / df['s_abs'].sum()
27              if df['s_abs'].sum() > 0 else df['s_abs'])
28    gt = df['ground_truth_weight']
29
30    # Grid search over the unit interval
31    alphas = np.linspace(0, 1, n_grid)
32    mae_list = []
33    for a in alphas:
34        hybrid = a * m_norm + (1 - a) * s_norm
35        mae_list.append(np.mean(np.abs(hybrid - gt)))
36
37    best_idx = int(np.argmin(mae_list))
```

```

38     return alphas[best_idx], mae_list[best_idx], alphas, mae_list
39
40
41 def compute_hybrid_scores(
42     df: pd.DataFrame,
43     alpha: float = 0.5
44 ) -> pd.DataFrame:
45     """
46     Compute hybrid attribution scores as the convex combination
47     of normalized Markov and Shapley weights.
48     """
49     df = df.copy()
50     m_sum = df['markov_removal_effect'].sum()
51     s_sum = df['shapley_value'].sum()
52     df['markov_norm'] = (df['markov_removal_effect'] / m_sum
53                          if m_sum > 0 else 0)
54     df['shapley_norm'] = (df['shapley_value'] / s_sum
55                          if s_sum > 0 else 0)
56     df['hybrid_score'] = (alpha * df['markov_norm']
57                          + (1 - alpha) * df['shapley_norm'])
58     return df

```

Listing 7: Alpha grid search and hybrid score computation (excerpt from `budget_optimization.py`).

The grid search consists of 101 equally spaced points over the unit interval, providing optimum identification to two decimal places. The total computational cost of the procedure is negligible: each grid point requires only the evaluation of a linear combination and a vector subtraction, and the full grid search completes in under one second on standard hardware. The optimal value identified for the present dataset, $\alpha^* = 0.25$, is reported in Section 4.5.

A.6 Budget Optimization

The budget optimization layer translates the hybrid attribution weights into actionable budget recommendations through the application of channel-level diminishing-returns response curves and a global optimization procedure. The response curve formulation employs the exponential-saturation form, with channel-specific curvature and half-saturation parameters calibrated against industry-grounded estimates of saturation dynamics (Chan & Perry, 2017; Jin et al., 2017).

```

1 def response_function(
2     spend: float,
3     score: float,
4     k: float,
5     half_saturation: float
6 ) -> float:
7     """
8     Compute the expected response of a channel as a function of
9     expenditure, using the exponential-saturation diminishing-
10     returns formulation:
11
12         response(x) = score * (1 - exp(-k * x / half_saturation))
13
14     The half_saturation parameter is specified in absolute USD
15     rather than as a fraction of total budget, ensuring
16     consistent behavior across the full range of budget levels

```

```

17     considered in the analysis.
18     """
19     if spend <= 0:
20         return 0.0
21     return score * (1 - np.exp(-k * spend / half_saturation))

```

Listing 8: Diminishing-returns response function (excerpt from budget_optimization.py).

The budget allocation problem is formulated as a constrained maximization of total expected response, with constraints on the total budget (equality) and on individual channel allocations (lower and upper bounds). The non-linearity of the response curves and the potential for multiple local optima in the response surface motivate the use of the Differential Evolution algorithm (Storn & Price, 1997), a population-based global optimization heuristic that does not require gradient information and exhibits robust convergence properties on non-convex problems of moderate dimensionality.

```

1 def optimize_budget_with_dr(
2     df: pd.DataFrame,
3     total_budget: float = 10_000,
4     min_pct: float = 0.05,
5     max_pct: float = 0.80,
6     seed: int = 42
7 ) -> pd.DataFrame:
8     """
9     Optimize channel budget allocation under diminishing-returns
10    response curves using the Differential Evolution algorithm.
11    Constraints: total budget conservation (penalty-enforced),
12    minimum and maximum per-channel allocation bounds.
13    """
14    n = len(df)
15    channels = df['channel'].tolist()
16    scores = df['hybrid_score'].values
17
18    k_vals = np.array([
19        SATURATION_PARAMS[ch]['k'] for ch in channels
20    ])
21    hs_vals = np.array([
22        SATURATION_PARAMS[ch]['half_saturation'] for ch in channels
23    ])
24
25    def neg_response_with_penalty(x):
26        # Sum of channel responses
27        total = 0.0
28        for i in range(n):
29            total += response_function(
30                x[i], scores[i], k_vals[i], hs_vals[i]
31            )
32        # Budget conservation penalty
33        budget_violation = abs(np.sum(x) - total_budget) / total_budget
34        return -total + 100 * budget_violation
35
36    # Per-channel allocation bounds (5% to 80% of total)
37    bounds = [
38        (total_budget * min_pct, total_budget * max_pct)
39    ] * n
40

```

```

41     result = differential_evolution(
42         neg_response_with_penalty, bounds,
43         seed=seed, tol=1e-8, maxiter=300, popsize=30,
44         polish=True
45     )
46
47     # Renormalize to exact budget conservation
48     x_final = result.x
49     x_final = x_final * (total_budget / np.sum(x_final))
50
51     df = df.copy()
52     df['allocated_budget'] = np.round(x_final, 2)
53     df['expected_response'] = [
54         response_function(x_final[i], scores[i],
55                             k_vals[i], hs_vals[i])
56         for i in range(n)
57     ]
58     return df

```

Listing 9: Differential Evolution budget optimizer (excerpt from `budget_optimization.py`).

The implementation incorporates the budget conservation constraint through a penalty term added to the negative objective function rather than through explicit equality constraint imposition. This formulation permits the Differential Evolution algorithm, which natively handles only inequality constraints through bounds, to approximate equality constraint satisfaction. The penalty coefficient (100) is calibrated to dominate the response function values, ensuring that constraint violations incur substantial cost in the objective. The final allocation vector is renormalized to enforce exact budget conservation, correcting any small residual deviation produced by the penalty formulation.

The algorithmic parameters—population size of 30, maximum 300 generations, convergence tolerance of 10^{-8} , and Latin hypercube initialization through the fixed random seed—are calibrated to provide reliable convergence for the three-channel paid allocation problem considered in the present analysis. Convergence typically occurs within 60 to 100 generations, well within the maximum iteration limit.

A.7 Reproducibility Information

The complete reproducibility of the analyses reported in this thesis is ensured through the use of fixed random seeds at all stages of the computational pipeline, the specification of explicit version requirements for all software dependencies, and the modular organization of the implementation that permits independent execution and verification of each methodological component.

Table 7. Random seed configuration across the computational pipeline.

Component	Seed Value
Synthetic data generation	42
Monte Carlo ground truth	123
Differential Evolution optimizer	42

Table 8. Software environment and version specifications.

Component	Version
Python interpreter	3.12.11
NumPy	2.3.3
pandas	2.3.3
SciPy	1.16.2
Matplotlib	3.10.6
Spyder IDE	6.1.0
Operating system	macOS 26.5 (Tahoe)

The execution of the complete pipeline—from synthetic data generation through final revenue simulation—completes in approximately fifteen minutes on standard consumer hardware, with the Shapley value computation accounting for the largest share of the execution time owing to its $O(2^n)$ enumerational complexity. All intermediate outputs are saved to disk as CSV files, permitting inspection and verification at each stage of the analysis. The modular structure of the codebase, with separate files for each methodological component, supports both replication of the full analysis and selective re-execution of individual components for sensitivity analyses or methodological extensions.

Appendix B: Markov Transition Matrix

This appendix presents the complete Markov chain transition matrix employed in the attribution analysis of Chapter 4. The matrix is constructed from the synthetic customer journey dataset of 30,000 users using the procedure described in Section 3.3.2, with Laplace smoothing ($\alpha = 1.0$) applied as specified in Section 3.3.3. The matrix dimensions are 8×8 , comprising five transient channel states (Google Ads, Social Media, Email, Organic Search, Direct), one initial state (Start), and two absorbing states (Conversion, Non-Conversion). All values represent post-smoothing transition probabilities, with each row summing to unity.

Table 9. Markov chain transition matrix with Laplace smoothing ($\alpha = 1.0$).

From / To	Start	Google	Social	Email	Organic	Direct	Conv.	NonConv.
Start	0.0000	0.2502	0.2489	0.1492	0.2007	0.1509	0.0000	0.0000
Google Ads	0.0000	0.1806	0.1766	0.1084	0.1412	0.1067	0.0042	0.2822
Social Media	0.0000	0.1763	0.1783	0.1091	0.1443	0.1052	0.0027	0.2841
Email	0.0000	0.1830	0.1784	0.1072	0.1454	0.1087	0.0040	0.2734
Organic Search	0.0000	0.1791	0.1823	0.1044	0.1409	0.1076	0.0029	0.2828
Direct	0.0000	0.1774	0.1802	0.1092	0.1395	0.1126	0.0033	0.2778
Conversion	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	1.0000	0.0000
Non-Conversion	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	1.0000

Several structural features of the matrix warrant brief observation. First, the row corresponding to the Start state contains positive probabilities only for the five channel states, reflecting the fact that journeys necessarily begin with a channel encounter rather than with a conversion outcome. The relative magnitudes of the Start-row entries indicate the empirical frequency with which each channel serves as the entry point of customer journeys, with Google Ads (25.02%) and Social Media (24.89%) appearing most frequently and Email (14.92%) appearing least frequently.

Second, the transition probabilities from the channel states to the Conversion absorbing state are uniformly low, ranging from 0.0027 (Social Media) to 0.0042 (Google Ads). This reflects the overall low conversion rate of the dataset (1.18%) and the fact that conversions typically occur only at the terminal touchpoint of a journey. The relative ordering of these probabilities provides preliminary evidence of channel effectiveness, with Google Ads exhibiting the highest probability of direct conversion and Social Media the lowest.

Third, the transition probabilities from channel states to the Non-Conversion absorbing state—ranging from 0.2734 (Email) to 0.2841 (Social Media)—reflect the cumulative probability that a journey ending at each channel does not result in conversion. The substantial magnitude of these probabilities, relative to the Conversion transitions, is consistent with the empirical pattern in which the majority of journeys terminate without conversion.

The Conversion and Non-Conversion rows exhibit the self-transition probability of unity characteristic of absorbing states, ensuring that journeys cannot exit these states once entered. This matrix forms the input to the fundamental matrix computation $N = (I - Q)^{-1}$ described in Section 3.3.4, from which the channel-level removal effects reported in Section 4.3.2 are derived.

Appendix C: Channel Interaction Matrix

This appendix presents the channel interaction matrix used to incorporate pairwise synergistic effects into the synthetic data generation procedure described in Section 3.2.2. The matrix represents multiplicative interaction coefficients applied to the conversion probability function when two channels appear in succession within a customer journey. Values greater than unity indicate synergistic interactions (the combined presence of two channels produces a higher conversion probability than the sum of their independent effects), while values equal to unity indicate the absence of meaningful interaction. The matrix is symmetric, reflecting the assumption that the synergy between two channels is independent of the order in which they appear.

Table 10. Channel interaction matrix (multiplicative synergy coefficients).

Channel	Google	Social	Email	Organic	Direct
Google Ads	1.000	1.150	1.120	1.000	1.000
Social Media	1.150	1.000	1.000	1.080	1.000
Email	1.120	1.000	1.000	1.100	1.200
Organic Search	1.000	1.080	1.100	1.000	1.000
Direct	1.000	1.000	1.200	1.000	1.000

The interaction matrix incorporates five pairwise synergies that reflect empirical patterns documented in the marketing literature. The strongest synergy, a multiplicative factor of 1.20 between Email and Direct, captures the relationship-building function of email marketing: customers who engage with email communications and subsequently visit the website directly exhibit substantially higher conversion probabilities than would be predicted from the independent effects of either channel. This pattern is consistent with the empirical observation that email marketing operates primarily by strengthening existing brand relationships rather than by initiating new ones.

A second substantial synergy, a multiplicative factor of 1.15 between Social Media and Google Ads, reflects the well-documented empirical phenomenon by which exposure to social media advertising influences subsequent search behavior (Kireyev et al., 2016). Customers who encounter a brand through social media platforms exhibit elevated probabilities of conducting subsequent branded searches, with the resulting Google Ads touchpoints producing conversion outcomes that exceed the sum of the independent channel effects.

A third synergy, a multiplicative factor of 1.12 between Google Ads and Email, captures the retargeting dynamic in which users acquired through paid search are subsequently nurtured through email sequences. This pattern reflects the increasingly integrated nature of modern e-commerce marketing, in which paid acquisition channels and owned-audience channels operate as complementary components of a unified customer engagement strategy.

The remaining two synergies—a factor of 1.10 between Organic Search and Email, and a factor of 1.08 between Social Media and Organic Search—reflect cross-funnel dynamics in which awareness-stage channel exposure influences subsequent mid-funnel and conversion-stage channel effectiveness. The five interaction effects collectively introduce the kind of cross-channel synergy structure that distinguishes data-driven attribution problems from the simpler single-channel attribution scenarios for which heuristic models were originally developed.

Pairs of channels not listed as interacting (assigned coefficient 1.000) are modeled as exhibiting independent effects in the synthetic data generation procedure. This sparse interaction

structure—in which only five of the ten possible channel pairs exhibit meaningful synergies—is consistent with the empirical observation that most channel pairs in real-world marketing operations do not exhibit substantial interaction effects, with synergies concentrated in specific channel pairings that share operational or audience characteristics.

Appendix D: Supplementary Materials and Code Availability

The complete source code, generated datasets, and visualizations produced in the course of this thesis are made publicly available through a dedicated GitHub repository, in accordance with the principles of computational transparency and reproducibility articulated in Section 5.6.

D.1 Repository Information

Table 11. Supplementary materials repository information.

Item	Specification
Repository URL	https://github.com/brky48/hybrid-mta-attribution
License	MIT License
Programming language	Python 3.12
Total source lines	Approximately 1,500
Number of modules	8

The repository is publicly accessible without registration or authentication. All materials are released under the MIT License, which permits use, modification, and redistribution for both academic and commercial purposes, subject to the inclusion of the original copyright notice.

D.2 Repository Structure

The repository is organized into four principal directories, each containing a logically coherent set of materials:

- `src/` — The eight Python modules implementing the methodological framework. Each module is independently executable and produces specified output files. The modules correspond directly to the algorithmic components described in Chapter 3 and detailed in Appendix A.
- `data/` — The synthetic customer journey dataset produced by `data_generator.py`. The dataset contains 105,565 touchpoint records distributed across 30,000 simulated users.
- `results/` — All output files in CSV format, including attribution model outputs, ground truth distributions, performance metrics, budget optimization results, and the transition and interaction matrices reported in Appendices B and C.
- `figures/` — The ten visualization outputs referenced in Chapter 4, including channel-level removal effects, Shapley value distributions, alpha grid search results, response curves, and revenue simulation outcomes.

D.3 Reproducibility Instructions

The complete analysis pipeline can be reproduced by executing the eight Python modules in the order specified in the repository’s README documentation. The recommended execution sequence is as follows:

1. `data_generator.py` — Synthetic data generation and Monte Carlo ground truth simulation.

2. `markov_chain.py` — Markov chain attribution with Laplace smoothing.
3. `shapley_value.py` — Shapley value attribution with fractional credit.
4. `baseline_models.py` — Five classical heuristic attribution models.
5. `budget_optimization.py` — Hybrid model with optimal alpha and Differential Evolution budget optimization.
6. `revenue_simulation.py` — Multi-budget revenue simulation framework.
7. `comparison_report.py` — Final aggregation and comparative visualization.
8. `export_matrices.py` — Optional export of transition and interaction matrices for Appendices B and C.

The required Python packages and their minimum compatible versions are documented in the `requirements.txt` file within the repository. Installation is accomplished through the standard pip command `pip install -r requirements.txt`. The total execution time of the complete pipeline is approximately fifteen minutes on standard consumer hardware, with the Shapley value computation accounting for the largest share owing to its exponential complexity in the number of channels.

D.4 Citation

Researchers wishing to cite the supplementary materials in subsequent work are invited to employ the citation format documented in the repository's README file, which references both the present thesis and the GitHub repository directly.